

From Comparative Benchmarking to Realization: Designing and Prototyping a Next-Generation Social Football Prediction App for Enhanced User Experience

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Abstract

Social football prediction platforms occupy a neglected corner of sports technology. The dominant product in the German-speaking market, Kicktipp, retains its position not through design quality but through network effects: users stay because their friends are already there. At the other end of the spectrum, commercial sports betting applications like Tipico have invested heavily in engagement design, but through dark nudge architectures that suppress reflection and exploit impulsive behaviour. No platform currently occupies the space between these two extremes, one that takes motivation seriously without exploiting it.

This thesis addresses that gap through a three-phase Design Science Research methodology. Phase I benchmarked four market-relevant applications, Kicktipp, Tackle, Teamtip, and Tipico, against the Gameful Design Heuristics framework across 28 motivational dimensions. The results identified the categories Social and Community (cross-application mean 2.10) and Immersion and Narrative (2.25) as the most severely underserved ones across the evaluated applications. Phase II translated these findings into a functional mobile application called PreMatch, designed around Self-Determination Theory and Positive Friction. The central design intervention is a hold-to-confirm submission gesture paired with a prediction rationale prompt, intended to encourage deliberate reflection at the moment of tip submission. Phase III evaluated the prototype with $N = 10$ participants using System Usability Scale (SUS), AttrakDiff, Intuitive Interaction Scale (INTUI), and Net Promoter Score (NPS), supplemented by a semi-structured qualitative debrief.

The results of the evaluation are broadly consistent with the central hypothesis. SUS produced a mean of 90.0, placing the prototype in the excellent range. AttrakDiff revealed a hedonic-dominant profile with HQ-I at 2.03, suggesting participants experienced the application as identity-affirming rather than merely functional. All four INTUI subscales exceeded the scale midpoint, with Gut Feeling at 5.30, indicating that the friction intervention does not substantially damage intuitive interaction quality. NPS reached 70 with zero Detractors. Together, these results tentatively support the argument that Positive Friction can elevate hedonic quality in a social prediction context without undermining usability or intuitiveness of the application.

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List of Acronyms

DSR	Design Science Research
HCI	Human-Computer Interaction
ICR	Inter-Coder Reliability
INTUI	Intuitive Interaction Scale
MVP	Minimum Viable Product
PENS	Player Experience of Need Satisfaction
SDT	Self-Determination Theory
SRQ	Sub-Research Question
SUS	System Usability Scale
UCD	User-Centered Design
UX	User Experience
METUX	Motivation, Engagement and Thriving in User Experience
UI	User Interface
WCAG	Web Content Accessibility Guidelines
AI	Artificial Intelligence
NPS	Net Promoter Score
PQ	Pragmatic Quality

1 Introduction

1.1 Football, Prediction, and the Problem Nobody Is Solving

Football is the most watched sport on the planet, generating billions in broadcast revenue and occupying a disproportionate share of attention in the daily lives of fans worldwide (Funk, Alexandris, & McDonald, 2016), and the digital infrastructure built around that attention has expanded accordingly. Real-time data platforms, fantasy leagues, social media communities, and statistics aggregators have turned passive spectatorship into something more interactive (Billings & Ruihley, 2013). Within this ecosystem, a quieter sub-genre has taken root in the form of the social prediction platform, apps where groups of friends, colleagues, and acquaintances predict match outcomes against each other within private leagues. By design, the winner at the end of the season earns bragging rights, the platform gains engagement, and what remains is a low-stakes social game built on genuine football interest.

The market leader in the German-speaking world is Kicktipp (Kicktipp GmbH, n.d.), a platform on which millions of users have created private leagues, invited friends, and entered score predictions before each matchday over the course of multiple seasons. From a User-Centered Design (UCD) perspective (*ISO 9241-210: Ergonomics of Human-System Interaction – Part 210: Human-Centred Design for Interactive Systems*, 2019), however, the application exhibits shallow gamification that rarely extends beyond point accumulation, inconsistent interface patterns, and interaction design that offers limited adaptability to varying usage contexts. Yet Kicktipp retains its dominance, not because of design quality but because the people you want to play against are already on it. The value of a network product increases with the number of participants, which means technically inferior platforms can maintain dominance as long as their user base is large enough to make switching costly (Shapiro & Varian, 1998). This dynamic is the starting point for this thesis.

1.2 The Problem: Stagnation, Dark Nudges, and the Gap Between Them

Two forces define the current landscape of digital sports prediction, and they pull in opposite directions.

On one side, platforms like Kicktipp are motivationally inert. They satisfy the minimum requirements for prediction and ranking, and they benefit from network effects strong enough to protect their market position regardless of design quality (Eisenmann, Parker, & Van Alstyne, 2006). Users tolerate the interface because the social cost of switching, persuading an entire friend group to migrate to a new platform together, outweighs the potential benefit of a more capable app (Shapiro & Varian, 1998). The consequences are visible in the overall design. Existing platforms implement game elements superficially without addressing the underlying psychological needs that they are intended to satisfy, resulting in engagement that is habitual rather than intrinsic (Tondello et al., 2016). Users return because of their social connections on the platform rather than because the platform itself is intrinsically motivating (Deterding, Dixon, Khaled, & Nacke, 2011).

By contrast, commercial sports betting applications have invested heavily in engagement design, though oriented towards suppressing reflection rather than enriching it. Schüll (2012) documents how these platforms deploy what she calls the machine zone, an architecture of frictionless, continuous engagement designed to suppress reflection and maximise impulsive action. The specific design patterns behind this have been catalogued by Newall (2019), who points to features like pre-filled stakes, one-click confirmations, and continuous scroll feeds as the primary mechanisms. These are not incidental features of convenience, they are deliberate mechanisms for keeping users in fast, automatic System 1 processing (Kahneman, 2011). Hing, Smith, et al. (2022) confirm in interview data that this architecture measurably increases impulsive betting and loss-chasing among vulnerable users, a finding consistent with the broader nudge literature's documentation of how design shapes decision-making (Thaler & Sunstein, 2008).

The gap between these two extremes forms the focus of this thesis. Tondello et al. (2016) identify exactly this polarisation as a failure of gameful design at the market level, with systems at one pole implementing game elements without motivational grounding while systems at the other exploit motivational vulnerabilities rather than satisfying them. Mo-

tivationally complete design remains the exception even in the broader app landscape, with only a quarter of reviewed behaviour change applications supporting all three basic psychological needs of autonomy, competence, and relatedness simultaneously (Villalobos-Zúñiga & Cherubini, 2020). As the heuristic benchmarking in Chapter 5.1 shows, this gap is even more pronounced in social prediction platforms specifically, where no existing application helps users make better predictions rather than faster ones and none treats the moment of commitment as something worth pausing for, which is precisely the gap this thesis sets out to fill.

1.3 Theoretical Grounding and Design Approach

Two theoretical frameworks shape the design of the proposed platform.

The first is Self-Determination Theory (SDT), developed by Ryan and Deci (2000), which identifies autonomy, competence, and relatedness as three innate psychological needs whose satisfaction determines whether motivation is sustained or collapses. These are universal psychological needs rather than preferences, and when they go unmet, motivation fades, intrinsic interest erodes, and continuous frustration tends to affect overall well-being (Ryan & Deci, 2000). Applied to digital product design through the Player Experience of Need Satisfaction (PENS) model (Ryan, Rigby, & Przybylski, 2006), SDT provides a vocabulary for evaluating whether an app genuinely supports the user's motivational needs or merely mimics doing so. Peters, Calvo, and Ryan (2018) formalise this in their Motivation, Engagement and Thriving in User Experience (METUX) model, arguing that digital products should be held accountable for whether they satisfy or frustrate psychological needs at every layer of the user experience. Points and badges can satisfy competence if they reflect genuine skill. They undermine it when they are designed to manufacture compulsion, a mechanism first demonstrated empirically by Lepper, Greene, and Nisbett (1973) and confirmed in digital contexts by Deci, Koestner, and Ryan (1999).

The second theoretical framework is Positive Friction, developed by Cox, Gould, Cecchinato, Iacovides, and Renfree (2016), which refers to deliberate interaction hurdles that slow users down at key decision points, nudging them away from fast and impulsive reactions towards more considered, reflective thinking (Kahneman, 2011). Laschke, Hassenzahl, and Diefenbach (2015) describe this in terms of the pleasurable troublemaker, building on the

slow technology paradigm of Hallnäs and Redström (2001), which argued that designing for reflection rather than efficiency is a distinct and legitimate design position. Mecke et al. (2019) demonstrate empirically that small motor-effort hurdles, a sustained press rather than a single tap, are sufficient to interrupt automatic behaviour without registering as usability obstacles. In a football prediction context, the moment that warrants this kind of pause is the instant before a tip is submitted. A user who briefly justifies their reasoning before committing to a score has made a different kind of decision than one who tapped a number and moved on.

These two frameworks are not in tension. Ryan and Deci (2000) are clear that autonomy satisfaction depends on the felt quality of volitional engagement, not just the presence of choice, meaning a prediction made impulsively does not satisfy autonomy even if nothing technically prevented it. Positive Friction, implemented well, therefore does not obstruct the autonomy principle but rather gives it a more concrete expression in interaction design (Cox et al., 2016).

1.4 Research Questions and Objectives

The central research question is:

How can design principles based on SDT and Positive Friction be embedded in a social football prediction application to enhance user experience beyond the current market standard?

Three Sub-Research Question (SRQ)s operationalise this across the three phases of the study.

SRQ 1 (Benchmarking): What quantifiable deficiencies exist in established social prediction applications regarding autonomy support and ethical design, as assessed against the Gameful Design Heuristics (Tondello et al., 2016)?

SRQ 2 (Conceptualisation): How must a Minimum Viable Product (MVP) be designed to productively balance gameful design principles with Positive Friction interventions (Cox et al., 2016; Deterding et al., 2011)?

SRQ 3 (Evaluation): How does the developed prototype perform in terms of pragmatic and hedonic quality, intuitive interaction, and system usability, as measured by AttrakDiff

(Hassenzahl et al., 2003), Intuitive Interaction Scale (INTUI) (Ullrich & Diefenbach, 2010a), System Usability Scale (SUS) (Brooke, 1996), and Net Promoter Score (NPS) (Reichheld, 2003)?

From these three research questions, the following objectives are derived. The first is to conduct a reliable and empirically grounded gap analysis of the current market. Building on that, a functional prototype that addresses identified gaps through a theoretically coherent design, following the Design Science Research (DSR) paradigm as formulated by Hevner, March, Park, and Ram (2004). Finally, empirical evidence gathered through a mixed-methods user study with validated psychometric instruments (Sauro & Lewis, 2012) that the prototype achieves its intended effect.

1.5 Contribution to Existing Research

This thesis sits at the intersection of three research areas that have rarely been brought together in the same study. SDT-informed mobile application design, Positive Friction as an interaction pattern, and the empirical evaluation of social sports prediction applications. To the author's knowledge, no prior work has combined all three in a single design and evaluation study. Each area has been addressed in isolation, but the intersection remains largely unexplored.

While the literature on Positive Friction provides a strong theoretical foundation, examples of its practical application remain comparatively scarce. Cox et al. (2016), Laschke et al. (2015), and Mecke et al. (2019) establish the conceptual case for deliberate interaction hurdles. Terpelle et al. (2023) survey the state of friction research in Human-Computer Interaction (HCI) and identify the absence of applied evaluation studies as a primary gap. This thesis provides an applied example through the implementation of a friction mechanism in a mobile application, tested with real users performing real-world tasks and evaluated using validated instruments.

The gamification literature has produced extensive evidence that superficial reward mechanics undermine intrinsic motivation over time (Deterding et al., 2011; Hamari, Koivisto, & Sarsa, 2014; Seaborn & Fels, 2015). Less attention has been paid to what ethical gamification looks like in practice as a measurable outcome rather than a design principle. The combination of AttrakDiff (Hassenzahl et al., 2003), INTUI (Ullrich & Diefenbach,

2010a), SUS (Brooke, 1996), and NPS (Reichheld, 2003) used in this thesis produces a multidimensional evaluation profile of a prototype explicitly designed to support motivation rather than exploit it, following the instrument logic established by Hinderks, Schrepp, Mayo, and Thomaschewski (2019) for User Experience (UX) key performance indicators.

A third gap concerns the product category itself. To the author's knowledge, social prediction platforms are almost entirely absent from the academic literature, and applications like Kicktipp, Tackle, and Teamtip have not been the subject of peer-reviewed UX evaluation. The heuristic benchmarking in Phase I therefore represents the first systematic application of the Gameful Design Heuristics (Tondello et al., 2016; Tondello, Kappen, Ganaba, & Nacke, 2019) to this product category, and the gap analysis it produces the first empirical documentation of what this market has and has not built.

1.6 Structure of the Thesis

The thesis proceeds across six chapters, each addressing a distinct stage of the research. The theoretical foundation is established in Chapter 2, covering SDT and the PENS model (Ryan & Deci, 2000; Ryan et al., 2006), Gameful Design and the overjustification risk (Deterding et al., 2011; Lepper et al., 1973), Positive Friction and its roots in Dual Process Theory (Cox et al., 2016; Kahneman, 2011), and the psychometric instruments used to evaluate the prototype (Hassenzahl et al., 2003; Ullrich & Diefenbach, 2010a; Brooke, 1996; Reichheld, 2003). In the following section, Chapter 3 sets out the DSR paradigm (Hevner et al., 2004; March & Smith, 1995), the research gap, and the research questions. The three-phase methodology is then detailed in Chapter 4, encompassing the heuristic benchmarking procedure, the artefact design and prototyping process, and the mixed-methods user study design. Findings from all three phases, including the benchmarking gap analysis, the design decisions and prototype documentation, and the user study results, are presented in Chapter 5. The subsequent discussion in Chapter 6 interprets those findings in relation to the research questions and draws out the theoretical and practical implications. This master's thesis concludes in Chapter 7 with a summary of contributions, limitations, and directions for future research.

2 Theoretical Background

This section establishes the conceptual foundation of this thesis, opening with SDT and its operationalisation through the PENS model, which together provide the motivational framework underlying all design decisions. From there, the focus shifts to Gameful Design Heuristics and how game design elements can be applied to non-game contexts without falling into the trap of superficial reward mechanics. Positive Friction and its roots in Dual Process Theory follow, explaining why deliberate interaction hurdles can enhance rather than damage the user experience. The measurement instruments used to evaluate the prototype are introduced afterwards, before the section closes with a conceptual synthesis that translates all of these theoretical inputs into the concrete feature set of the proposed platform.

2.1 Self-Determination Theory and the PENS Model

Self-Determination Theory

SDT is a macro-theory of human motivation developed by Ryan and Deci (2000). Its central claim is that people are not passive recipients of rewards and punishments but have innate psychological needs, and whether an environment satisfies those needs determines whether motivation holds or collapses. Ryan and Deci (2000) identify autonomy, competence, and relatedness as the three needs at the core of this framework.

Autonomy refers to the experience of acting from genuine volition, not independence in the sense of acting alone, but the feeling that one's actions are self-endorsed rather than pressured from outside. Competence is the need to feel effective and to take on challenges and master them, while relatedness describes the desire to feel meaningfully connected to others. Ryan and Deci (2000) are clear that these are universal psychological needs rather than preferences or personality traits, and that frustrating them leads to disengagement, loss of intrinsic interest, and in continuous cases, broader ill-being.

The intrinsic and extrinsic distinction sits at the heart of SDT. Intrinsic motivation refers to engagement in an activity for its own sake. Extrinsic motivation covers everything from external rewards to internalised values. What matters, Ryan and Deci (2000) argue, is not whether a motivation is technically extrinsic but to which extent a person has internalised

it. Regulations that feel chosen produce engagement qualitatively similar to intrinsic motivation in comparison to regulations that feel imposed. For app design this has a direct consequence, as points and badges can support motivation when experienced as informational and affirming, such as feedback that reflects genuine skill or progress, but undermine it when they feel controlling or contingent on performance in ways that shift the user's focus away from the activity itself (Peters et al., 2018; Deci et al., 1999). The robustness of these mechanisms across a wide range of contexts, including digital and interactive environments, has since been confirmed through large-scale reviews of SDT research (Ryan et al., 2022).

The PENS Model

SDT was developed in educational and clinical psychology. Ryan et al. (2006) applied it systematically to digital interactive media through the PENS model, running a series of studies on video game motivation and finding that the three SDT needs independently predicted enjoyment, preference, and sustained play. This applied to different game genres, which suggests the model captures something fundamental about human engagement with interactive systems rather than something genre-specific.

The PENS model translates these needs into concrete design criteria. Autonomy is supported through meaningful choices and informational rather than controlling feedback. Competence requires that challenge scales with the player's growing skill, staying difficult enough to feel meaningful but not so demanding as to produce frustration (Przybylski, Rigby, & Ryan, 2010). Relatedness requires social infrastructure that goes beyond passive ranking toward genuine shared goals and communal interaction (Rigby & Ryan, 2011).

Przybylski et al. (2010) extend this work by distinguishing presence, intuitive controls, and need satisfaction as separate predictors of positive experience. Competence satisfaction is the strongest predictor of return engagement. Relatedness becomes increasingly important in social and multiplayer contexts, a direct implication for a platform where users return weekly across an entire football season.

What makes SDT and PENS useful here is that they are not just descriptive, but also prescriptive. If autonomy, competence, and relatedness are the conditions under which intrinsic motivation thrives, a well-designed app can be evaluated against exactly those

criteria. The framework used to build the platform is also the framework used to critique it. Peters et al. (2018) formalise this in their METUX model, arguing that digital products should be held accountable for whether they satisfy or frustrate psychological needs at every layer of the user experience, from first contact through to long-term behaviour change. The principles underlying this argument are elaborated further by Ryan (2023), who provides a comprehensive account of SDT's application across contemporary digital contexts.

2.2 Gameful Design: Beyond Pointification

Gamification is one of the most overused and least precisely defined concepts in digital product design, most commonly reduced to points, badges, and leaderboards with a hope that engagement follows, though it rarely does, at least not sustainably. Deterding et al. (2011) drew a principled distinction between this surface-level reward layering and what they call Gameful Design, a distinction subsequent surveys have consistently confirmed (Seaborn & Fels, 2015).

The term *pointification* comes from this critique, capturing the idea that reducing game design to a scoring mechanism borrows the least interesting part of what makes games engaging. What actually drives engagement, Deterding et al. (2011) argue, is the underlying motivational structure in which the points sit, specifically clear goals, meaningful feedback, a sense of progress, and the feeling that skill is being developed and recognised. Without that structure, reward mechanics produce superficial compliance rather than genuine motivation, a distinction with direct consequences for long-term engagement.

Nacke and Deterding (2017) note that the field has matured enough to move past debating whether gamification works toward asking under what conditions and for whom (Hamari et al., 2014), and the answer connects back to SDT in that game elements support engagement when they satisfy psychological needs and undermine it when they substitute for those satisfactions rather than producing them. A badge marking genuine skill development carries motivational weight, whereas a badge for logging in three days in a row does not, and users tend to sense the difference quickly. Recent empirical work on mobile applications confirms this pattern at scale, with Oliveira, Carvalho, and Rodrigues (2024) finding in a systematic review across app categories that social features drive community and

belonging while achievement elements address competence and autonomy, consistent with SDT predictions. Paschmann, Bruno, van Heerde, Völckner, and Klein (2025) similarly demonstrate using individual-level field data that unlocking game levels produces a sense of competence and mastery that stimulates intrinsic motivation, though over-reliance on reward mechanics can erode engagement once the novelty wears off.

Gameful Design Heuristics

The Gameful Design Heuristics framework developed by Tondello et al. (2016) is one of the most practically useful contributions to this area. Rather than prescribing a fixed set of game elements, the framework offers 28 evaluative criteria organised around motivational affordances. It focuses not on which game features a system includes, but on whether those features genuinely support intrinsic motivation across the three needs of SDT. Tondello et al. (2019) validated this in a study with UX professionals, finding that evaluators using heuristics identified significantly more motivational design issues than those working without them.

The framework serves a dual function in this thesis. In Phase I, it is used as the primary instrument for heuristic benchmarking of existing prediction platforms, identifying which motivational dimensions are neglected by apps like Kicktipp and producing a gap analysis the design phase can address directly. In Phase II, the same heuristics then inform the design of the prototype itself, so that using them as both evaluation tool and design reference creates a traceable connection between the problems identified in benchmarking and the solutions embedded in the artefact.

The Overjustification Risk in Practice

Villalobos-Zúñiga and Cherubini (2020) examined 208 applications from the App Store and found that the most common SDT-aligned design features were also the most frequently misimplemented, with progress indicators and reward systems widespread but typically designed to maximise session length rather than genuinely support competence, leaving users feeling manipulated rather than motivated. This finding sits within a broader pattern identified by Koivisto and Hamari (2019), who documented widespread misalignment between gamification implementation and motivational theory across digital products.

For a football prediction platform, the risk is concrete. Predicting match outcomes is already intrinsically engaging for football fans. Aggressive reward mechanics layered on top can convert genuine interest into a transactional loop, exactly the pattern Schüll (2012) documents in gambling products. The design challenge is not to add as many game elements as possible but to select only those that extend the motivational value of prediction itself. Grasse, Kreminski, Wardrip-Fruin, Mateas, and Melcer (2022) make the same point in educational games. Elements perceived as integrated into the core activity produced significantly better motivational outcomes than those added as external incentives.

2.3 Positive Friction and Dual Process Theory

Most interaction design is based on the implicit assumption that faster is better, stripping out every obstacle, optimising every flow, reducing every tap. As a general principle this is not wrong, but it becomes a problem when the decision itself is worth slowing down for, and football prediction is exactly that kind of activity. A well-considered tip requires thinking about the current form and head-to-head record of the competing teams as well as the injury situation of both squads, none of which a one-tap submission used in applications like Kicktipp encourages.

Dual Process Theory

The psychological foundation here is Dual Process Theory, articulated by Kahneman (2011), which distinguishes between two modes of human cognitive processing. System 1 operates fast, automatically, and intuitively, handling most quick judgments and habitual behaviours without conscious effort. System 2 by contrast is slow, deliberate, and effortful, engaging only when a task genuinely demands reasoning and reflection. These are not measurement values but conceptual categories describing how the mind shifts between automatic and controlled processing depending on the demands of a given situation.

The vast majority of digital interaction is designed for System 1, built around gestures like swiping, tapping, and scrolling that allow users to move through an interface without conscious deliberation. This works well for many tasks, but it creates a vulnerability in that users can complete actions without genuinely deciding to. Kahneman (2011) calls this *cognitive ease*, a comfortable state of frictionless processing that feels natural in the

moment but is not always appropriate for decisions the user will care about afterward.

The commercial betting industry has learned to exploit this systematically, with Newall (2019) documenting how betting platforms deploy dark nudges, design patterns that eliminate any pause between impulse and action through mechanisms like pre-filled bet amounts, one-click confirmation, and continuous scroll feeds. Hing, Smith, et al. (2022) confirm in interview data that this architecture directly increases impulsive betting and loss-chasing among vulnerable users, making clear that the speed and ease of these interactions are not incidental features of convenience but the intended product. The broader nudge literature (Thaler & Sunstein, 2008) provides the conceptual scaffolding for understanding why these patterns are so effective and why deliberate counter-design is necessary, a case that has since been reinforced at the public health level by Wardle et al. (2024), who identify platform dark patterns as a systemic threat requiring regulatory response and explicitly connect frictionless betting architecture to measurable health harm.

Positive Friction

Positive Friction, developed by Cox et al. (2016), offers a principled design response to exactly this dynamic by introducing deliberate interaction hurdles that interrupt System 1 processing at key decision points and redirect attention toward more considered, System 2 deliberation. The qualifier *positive* is significant in that the goal is not to frustrate users or erect arbitrary barriers, but to create a pause that feels meaningful rather than obstructive precisely because it arrives at a moment when reflection genuinely adds value.

Laschke et al. (2015) describe a related concept, the *pleasurable troublemaker*, building on slow technology as articulated by Hallnäs and Redström (2001), which argued that designing for reflection rather than efficiency is a distinct and legitimate design paradigm. Their research shows that when friction is tied to the content of the decision rather than imposed as a technical barrier, users typically do not experience it as annoying. They experience the outcome as more personally owned.

Mecke et al. (2019) take an empirical approach to what they call thresholds for intentional interaction, finding that small motor-effort hurdles such as a sustained button press rather than a single tap are sufficient to interrupt automatic behaviour without registering as usability obstacles. Friction mechanisms tied to real informational content also resist

habituation, since their value is renewed each time new data becomes available rather than presenting the same information repeatedly. A prompt asking users to consider the latest injury news about both squads before confirming a tip, for instance, carries different weight every matchday depending on what has changed since the last round, which is precisely what prevents it from fading into background noise the way an arbitrary delay would. More recent work extends this logic into Artificial Intelligence (AI)-mediated contexts, with Chen and Schmidt (2024) developing a behavioural model of positive friction in human-AI interaction and arguing that deliberate delays and reflection prompts prevent users from defaulting to automatic, potentially biased responses, a mechanism they find generalises across interaction types where reflective engagement is desirable.

Friction as an Ethical Design Position

Cox et al. (2016) frame Positive Friction explicitly as an ethical design stance, arguing that in a landscape where the dominant commercial logic maximises engagement time through frictionless compulsion, choosing to slow users down at certain moments is a statement about what the platform values, namely that the quality of a user's decision matters more than the speed of their transaction.

This ethical dimension connects directly to SDT, since Ryan and Deci (2000) are clear that autonomy is not simply about offering choices but about the felt quality of those choices. A user who submits a prediction impulsively and later regrets it has not had their autonomy satisfied, even though technically nothing prevented them from doing so, which means that Positive Friction, implemented well, does not conflict with the autonomy principle but rather gives it a more concrete form in interaction design.

Calibration remains the central challenge, as Terpelle et al. (2023) note that too little friction collapses the design back into the compulsive patterns it was initially meant to counter, while too much produces frustration rather than reflection and damages both usability and the very sense of autonomy the friction was intended to protect.

2.4 UX Measurement: AttrakDiff, INTUI, SUS, and NPS

Designing a prototype is only part of the process, while the other part involves determining whether it actually achieves its intended function. To answer this, the thesis evaluates

the proposed platform using four instruments, three of which are validated psychometric scales covering usability, hedonic quality, and intuitiveness, while NPS is included as a lightweight supplementary indicator of recommendation intent. Taken together, AttrakDiff, INTUI, SUS, and NPS form a complementary evaluation framework that captures multiple dimensions of the user experience rather than reducing quality to a single score.

AttrakDiff

AttrakDiff was developed by Hassenzahl et al. (2003) in response to a limitation in traditional usability measurement. Most usability instruments focus on efficiency and effectiveness, what Hassenzahl (2003) calls *pragmatic quality*. This captures how well a product supports goal achievement. But it misses whether the product is enjoyable, stimulating, or identity-affirming. Hassenzahl (2003) calls this *hedonic quality*, and argues it is at least as important as pragmatic quality for understanding long-term engagement.

AttrakDiff operationalises this through a semantic differential questionnaire in which users rate the product on opposing adjective pairs across four dimensions covering pragmatic quality, hedonic quality stimulation, hedonic quality identity, and overall attractiveness. The result is a two-dimensional portfolio diagram that maps where a product sits relative to pragmatic and hedonic expectations, with hedonic quality on the vertical axis and pragmatic quality on the horizontal (Hassenzahl et al., 2003), an example of which is presented in Chapter 5 alongside the study results. Hinderks et al. (2019) draw on it as a foundational framework for UX key performance indicators, confirming its continued validity and sensitivity across product categories.

This thesis uses AttrakDiff as the primary instrument for evaluating whether the Positive Friction interventions achieve their intended effect. The hypothesis is that slowing users down at key moments will increase hedonic quality, making interactions feel more meaningful and deliberate, without significantly damaging pragmatic quality. A prototype scoring well on hedonic stimulation and identity while maintaining acceptable pragmatic ratings would be consistent with this hypothesis. One that sacrifices pragmatic quality for hedonic gains would suggest the friction was too heavy. AttrakDiff is precise enough to distinguish between these outcomes (Hassenzahl, 2013).

INTUI

INTUI was developed by Ullrich and Diefenbach (2010a) to measure intuitive interaction, a construct widely discussed in HCI but rarely operationalised with precision, capturing four components of effortlessness, gut feeling, magical experience, and verbalisability. The last of these is particularly notable, as it measures whether users can articulate why an interaction felt intuitive, something that tends to score low precisely because intuitive interactions are processed by System 1 without conscious reflection (Kahneman, 2011). Diefenbach and Ullrich (2013) later applied the instrument across a broader range of product contexts, finding stable four-factor structures and confirming the scale is sensitive enough to detect meaningful differences in perceived intuitiveness between interaction sequences within the same product.

The relevance to this thesis follows directly from the nature of the Positive Friction interventions, which are designed to interrupt automatic System 1 responses at specific moments and therefore carry the risk of damaging the overall sense of intuitive flow that modern mobile users expect. Ullrich and Diefenbach (2010b) demonstrate that intuitiveness is not binary and that products can score high on effortlessness in some interaction sequences while scoring lower in others, which is precisely the sensitivity needed here since INTUI scores that remain high despite the friction interventions would suggest the hurdles were experienced as natural parts of the interaction rather than as disruptive obstacles.

SUS

The SUS is the most widely used usability questionnaire in applied UX research, developed by Brooke (1996) as a ten-item instrument in which respondents rate statements about a product on a five-point scale ranging from strongly disagree to strongly agree, producing a single composite score between zero and 100. Despite its age, it has proven remarkably robust across decades of use and has been validated against more complex instruments with consistently strong results (Lewis, 2018). Bangor et al. (2008) established the widely used interpretive benchmarks, according to which a value above 68 is considered above average while a value above 80 indicates good usability.

In this thesis, SUS serves a baseline function, since before asking whether the prototype

achieves its hedonic and intuitiveness goals it is worth confirming that it meets basic usability standards. A prototype that scores poorly on SUS has a usability problem that needs to be addressed before any hedonic finding can be meaningfully interpreted, which is why the instrument is applied here as a first-pass check rather than a primary evaluative lens, with the full results presented in Section 5.3. Brooke (1996) himself conceived of the scale as a lightweight screening tool, and that is precisely the role it plays in this evaluation.

NPS

The NPS was introduced by Reichheld (2003) as a single-item measure of customer loyalty and recommendation intent. Respondents rate on a zero-to-ten scale how likely they would recommend a product to someone they know. Scores of nine or ten are classified as Promoters, seven or eight as Passives, and zero to six as Detractors. The NPS is calculated by subtracting the percentage of Detractors from the percentage of Promoters, producing a score between -100 and $+100$.

While NPS is primarily a business metric rather than a psychometric instrument, it has been adopted in product evaluation research as a lightweight indicator of overall satisfaction and perceived value (Reichheld, 2003), and is included here as a supplementary measure that captures the degree to which participants would voluntarily advocate for an application of this kind in their social environment, which is particularly relevant for a platform whose value proposition depends on social adoption. Given the small sample size of $N = 10$, the NPS result is treated as a directional indicator rather than a statistically generalisable metric.

Motivation for Combining the Four Evaluation Instruments

Each instrument addresses a different layer of evaluation, with SUS confirming whether the prototype meets basic usability standards, while AttrakDiff tests whether the design goes beyond usability to produce a genuinely engaging and identity-affirming experience. INTUI then addresses the most novel and theoretically contested part of the design, namely whether the friction interventions are experienced as natural parts of the interaction or as intrusive obstacles. NPS adds a practical perspective by asking whether participants would actually recommend the platform to someone in their social circle, independent of what the other instruments measure. No single instrument could answer the thesis's core question

on its own, but the four in combination provide a multidimensional profile that can.

2.5 Conceptual Synthesis: From Psychological Needs to Artifact Features

The instruments discussed above only become useful when translated into concrete decisions. SDT, Gameful Design, and Positive Friction each make specific claims about the motivation of people. This section maps those claims onto the features of the proposed platform, not to justify design choices after the fact, but to show that the MVP rests on a coherent motivational logic from the ground up.

Autonomy

Autonomy tends to be the most underemphasised of the three SDT needs in app design, with most platforms reducing it to little more than a set of toggle settings (Villalobos-Zúñiga & Cherubini, 2020), whereas Ryan and Deci (2000) place considerably higher demands on what genuine autonomy support requires. Satisfaction of this need depends on the overall felt quality of volitional engagement, meaning every structural decision in the interface either supports or quietly erodes it.

The most visible autonomy features are personalisation and profile editing, since allowing users to select their supported football club, set a profile picture, and customise notifications signals from the first session that the platform adapts to the user rather than demanding the user adapt to it (Villalobos-Zúñiga & Cherubini, 2020). Less obvious but equally important is the ability to edit a submitted tip before the kick-off of the respective match, as once a prediction feels locked in the interaction stops feeling voluntary and starts to resemble an obligation (Rigby & Ryan, 2011).

Wellbeing settings and the opt-out path deserve particular attention, as Schüll (2012) shows how commercial gambling products systematically bury exit affordances to keep users inside the loop, a pattern Hing, Russell, et al. (2022) confirm extends into sports betting. A platform that claims to respect user autonomy while obscuring the means to disengage is contradicting its own design principles. The same logic applies to the rules section, which explains how points are awarded for correct predictions and how the seasonal ranking is calculated. Keeping this reachable at all times ensures users always have clarity about

what they have agreed to and how their performance is being evaluated (Tondello et al., 2016).

Competence

Przybylski et al. (2010) find competence to be the strongest independent predictor of return engagement across digital systems, more than social features and rewards. The MVP developed in this thesis reflects this finding through several design decisions targeting competence support specifically.

Onboarding marks the starting point of the competence journey within the MVP. A well-designed first-run experience is not primarily about tutorial pop-ups, which Nielsen's tenth usability heuristic identifies as a sign of poor design when relied upon too heavily (Nielsen, 1994b), but about calibrating the initial challenge so users feel capable before they feel tested (Deterding et al., 2011). The home screen follows the same logic, being designed around one clear action and avoiding navigational overhead in line with the eighth heuristic of minimalist design (Nielsen, 1994b), since users who cannot locate the primary task quickly tend to attribute that difficulty to their own failure rather than to a design problem (Hassenzahl, 2003; Norman, 2013).

Match statistics, head-to-head records, and form tables give users the information to make a reasoned prediction rather than a guess (Przybylski et al., 2010), while a transparent scoring system ensures that users continue to attribute outcomes to their own skill rather than chance, since opacity in this area tends to break down the competence loop entirely (Rigby & Ryan, 2011). Feedback closes this loop through a running accuracy score, a browsable archive of past matchdays, and a streak counter that makes progress visible over time and creates a self-imposed performance standard renewing motivation after each matchday (Ryan et al., 2006; Deterding et al., 2011). Official club logos throughout the match display further reduce the cognitive effort needed to parse a fixture list, freeing attention for the prediction itself (Kahneman, 2011).

Relatedness

Tondello et al. (2016) describe relatedness as the most consistently underserved need in gameful applications. A leaderboard is a start, but it only satisfies relatedness when scoped

to people the user actually knows, since anonymous global rankings tend to produce anxiety rather than belonging (Festinger, 1954). All rankings are therefore scoped to private leagues and friend groups by default.

Two features extend the social layer of the MVP beyond the passive consumption of a standings table. Community chat creates a shared space in which genuine community can form around the prediction activity itself (Ryan & Deci, 2000), while showing friends' tips from the previous matchday turns what would otherwise be a solitary scoring activity into a comparative and social one, giving users a reason to return after results come in rather than only before kick-off (Ryan & Deci, 2000). Both features depend on the friend invitation flow as their entry point, since without it the private league remains empty and the relatedness features have no one to act on.

Positive Friction

The friction features are concentrated at the natural decision point of tip submission, which Cox et al. (2016) identify as precisely where friction is most effective, not as a barrier that blocks the user but as a pause that arrives at a moment when reflection genuinely adds value.

Two mechanisms work in parallel at the moment of submission. The hold-to-confirm gesture replaces a single tap with a short sustained press, which Mecke et al. (2019) demonstrate is sufficient to interrupt an automatic System 1 response without registering as a usability obstacle. Alongside this motor-effort hurdle, a rationale prompt asks users to select a short tag explaining the reasoning behind their prediction, drawing on contextual factors such as the team's recent performance, injury news, or head-to-head record. Together these two mechanisms operationalise the *pleasurable troublemaker* concept of Laschke et al. (2015), translating a theoretical design position into two concrete interaction patterns embedded at the point where reflective engagement matters most.

The reflection setting extends this logic into a user-controlled opt-in layer, allowing users to decide how much deliberation they want built into their tipping flow rather than having friction imposed on them uniformly. This directly satisfies the autonomy principle, since Ryan and Deci (2000) emphasise that felt volition matters and a friction mechanism the user has chosen to activate is experienced very differently from one they cannot switch off.

Unlike arbitrary delays, friction tied to real informational content also resists habituation, as Mecke et al. (2019) find its value is renewed each time new data becomes available.

Ethical Design and Accessibility

Accessibility is included in the MVP not as a legal obligation but as a direct implication of the autonomy principle, since Peters et al. (2018) argue that need satisfaction has to be considered at the level of physical access first, before any engagement design becomes meaningful. A platform built around respecting user autonomy that is structurally unusable for people with visual impairments contradicts its own foundational premise. The accessibility implementation in the MVP is therefore oriented around the Web Content Accessibility Guidelines (World Wide Web Consortium, 2018), which provide the internationally recognised standard for inclusive digital design. Gray, Kou, Battles, Hoggatt, and Toombs (2018) reinforce the ethical dimension of this decision by arguing that inaccessible design constitutes a dark pattern in its own right, since the harm may be unintentional but the effect remains the same in that it removes the user's ability to engage on their own terms. Screen reader compatibility and a high contrast mode are therefore treated as non-negotiable features of the MVP rather than optional additions.

3 Research Design

3.1 Research Paradigm: Design Science Research

This thesis is grounded in the DSR paradigm, a research approach that positions the creation of an artefact as a legitimate and rigorous form of knowledge production. Building on foundational work by March and Smith (1995), Hevner et al. (2004) define DSR as research that seeks to extend the boundaries of human and organisational capabilities by creating new and innovative artefacts. The artefact is therefore not just a deliverable but the primary means through which the research question is addressed. Designing and evaluating it counts as a contribution to knowledge, as long as the design decisions are theoretically grounded and the evaluation is carried out in a systematic way.

DSR differs from behavioural science research in a fundamental way, in that behavioural research seeks to understand and explain existing phenomena while DSR seeks to create solutions to identified problems and evaluate whether those solutions work (Hevner et al., 2004). This distinction matters for the present thesis, as the goal is not to study football prediction behaviour in the abstract but to design a platform that improves it and to produce evidence that the improvement is real.

Hevner et al. (2004) articulate seven guidelines that characterise rigorous DSR, of which five are directly relevant to this thesis. The first, design as an artefact, is satisfied by the MVP itself, a fully functional mobile application developed in Flutter, an open-source cross-platform framework by Google (Google LLC, 2024b), which produces a testable and demonstrable output rather than a purely theoretical proposition. Problem relevance is addressed through the heuristic benchmarking in Phase I, which empirically documents the UX stagnation and motivational poverty of existing prediction platforms and thereby establishes the design problem on evidential rather than assumed grounds. The design evaluation guideline is met through the mixed-methods user study in Phase III, which combines validated psychometric instruments with qualitative interview data to assess whether the MVP achieves its intended effect. Research rigour is maintained throughout by grounding every design decision in SDT and Positive Friction theory, creating a traceable connection between theoretical premises and interaction patterns. Finally, the research contribution lies in demonstrating how these two frameworks can be jointly operationalised in a working interactive system, extending the applied friction literature that Terpelle et al.

(2023) identify as underdeveloped.

The DSR Process Model

Peppers, Tuunanen, Rothenberger, and Chatterjee (2007) formalise DSR into a six-step process model encompassing problem identification and motivation, definition of objectives, design and development, demonstration, evaluation, and communication, though they emphasise that this sequence is not strictly linear but iterative, with evaluation findings feeding back into earlier stages across successive cycles. This iterative character is central to the present thesis, as the heuristic benchmarking of Phase I directly shapes the design decisions of Phase II, and the user study results of Phase III are interpreted against the theoretical framework established in Chapter 2.

The three phases of the methodology map directly onto the process model of Peppers et al. (2007). Phase I addresses problem identification and the definition of objectives by grounding both in empirical benchmarking data rather than assumption, while Phase II covers design and development by translating the resulting gap analysis and theoretical framework into a concrete MVP. Phase III then completes the cycle through demonstration and evaluation, testing the MVP with real users and measuring outcomes against validated psychometric instruments, thereby constituting what Peppers et al. (2007) would recognise as a complete DSR iteration.

Why Design Science Research Is a Suitable Framework for This Study

The choice of DSR is not incidental, as the research problem here is that existing social prediction platforms fail to satisfy basic psychological needs and that no empirically validated alternative exists, which is precisely the kind of problem DSR is designed to address. The gap cannot be closed by studying the phenomenon more carefully but only by building something better and demonstrating that it works, an approach Hevner et al. (2004) describe through the relevance cycle, according to which research should be motivated by a genuine problem in an application environment and produce outputs that are useful to that environment. A functional, theoretically grounded prototype that outperforms existing platforms on hedonic quality and intuitiveness represents exactly that kind of output (Gregor & Hevner, 2013).

3.2 Research Gap and Problem Formulation

As established in Chapter 1, the current digital sports prediction landscape is defined by two separate but related problems. The first is the UX stagnation of social prediction platforms, where network effects create structural lock-in that suppresses competitive pressure even when users are demonstrably dissatisfied with the incumbent product (Shapiro & Varian, 1998; Eisenmann et al., 2006). The consequences are visible in the design of platforms like Kicktipp, which exhibit inconsistent interface patterns, shallow gamification that rarely extends beyond point accumulation, and interaction design that offers little adaptability to different usage contexts. Tondello et al. (2016) identify exactly these shortcomings as failures of gameful design, producing engagement that is habitual rather than intrinsic and driven by social obligation rather than genuine motivation.

The second problem is the design polarisation of the wider sports prediction market between motivationally inert platforms on one side and commercially driven betting applications on the other, the latter of which deploy frictionless engagement architectures that suppress reflection and exploit motivational vulnerabilities rather than satisfying them (Schüll, 2012; Newall, 2019; Hing, Russell, et al., 2022). The gap between these two extremes, a platform that combines social infrastructure with genuine psychological need satisfaction and ethical interaction design, is the research space this thesis occupies and the contribution it sets out to make.

3.3 Research Questions and Study Design

Building on the problem formulation above, the following research questions define the scope and phased structure of the study.

How can design principles based on SDT and Positive Friction be embedded in a social football prediction application to enhance user experience beyond the current market standard?

Three SRQs operationalise this across the three phases of the study, each building on the previous to form a logical chain from diagnosis through design to evaluation.

SRQ 1 (Benchmarking): What quantifiable deficiencies exist in established social prediction applications regarding autonomy support and ethical design, as assessed against the

Gameful Design Heuristics?

SRQ 2 (Conceptualisation): How must a MVP be designed to productively balance gameful design principles with Positive Friction interventions (Cox et al., 2016; Deterding et al., 2011)?

SRQ 3 (Evaluation): How does the developed prototype perform in terms of pragmatic and hedonic quality, intuitive interaction, and system usability, as measured by AttrakDiff (Hassenzahl et al., 2003), INTUI (Ullrich & Diefenbach, 2010a), SUS (Brooke, 1996), and NPS (Reichheld, 2003)?

Taken together, the three SRQs define the scope and sequence of the thesis. The study moves from an empirical diagnosis of what the current market lacks, through a theoretically grounded design response, to a psychometric evaluation of whether that response achieves its intended effect, constituting a complete DSR cycle as described by Hevner et al. (2004).

4 Methodology

Grounding the methodology in the DSR paradigm as formulated by Hevner et al. (2004), this thesis treats research as the iterative design, construction, and evaluation of an artefact that addresses a real-world problem. Phase I diagnoses that problem through heuristic benchmarking, Phase II constructs the artefact in response to the identified gaps, and Phase III evaluates whether it achieves its intended effect.

4.1 Phase I: Heuristic Benchmarking and Inter-Coder Reliability

The first phase serves the purpose of establishing an empirical baseline by documenting, through structured means, what existing platforms lack rather than relying on informal observation. To this end, a heuristic evaluation of four market-relevant applications is conducted using the Gameful Design Heuristics framework (Tondello et al., 2016), producing a quantifiable gap analysis that directly informs the design decisions of Phase II.

Selection of Applications

Four applications were selected for evaluation, namely Kicktipp, Tackle, Teamtip, and Tipico, following a deliberate logic. Kicktipp represents the dominant platform in the German-speaking market and is the primary competitive reference for this thesis (Kicktipp GmbH, n.d.), while Tackle and Teamtip represent direct competitors in the social prediction space, offering an equivalent core feature set at varying levels of design maturity. Tipico is included as a representative of the commercial sports betting category, and its inclusion is intentional in that benchmarking a gambling platform alongside social prediction apps allows the evaluation to highlight the design patterns this thesis explicitly distances itself from, particularly the dark nudge architectures documented by Newall (2019). Together the four applications span the relevant design landscape from utilitarian prediction tools to engagement-maximising betting products, and the selection criteria are summarised in Table 1.

Table 1: Overview of Selected Applications and Rationale for Inclusion

Application	Type	Market	Reason for Inclusion
Kicktipp	Social prediction	DACH	Market leader; primary competitive reference (Kicktipp GmbH, n.d.).
Tackle	Social prediction	DACH	Direct competitor with comparable core feature set.
Teamtip	Social prediction	DACH	Direct competitor at a different level of design maturity.
Tipico	Commercial betting	DACH / EU	Representative of dark-nudge design patterns used as negative reference (Newall, 2019).

Evaluation Instrument

The primary instrument is the Gameful Design Heuristics framework developed by Tondello et al. (2016) and later validated as an inspection tool by Tondello et al. (2019). The framework deconstructs the user experience into motivational dimensions drawn directly from SDT, asking for each dimension whether the application actively supports, neglects, or undermines the relevant psychological need. This makes it particularly well suited to the present thesis, since the same theoretical vocabulary used to build the MVP is also used to evaluate the competition, creating a direct and traceable connection between the benchmarking findings and the design responses.

Each application is rated on a standardised five-point Likert-scale rubric across the heuristic dimensions, where the value 1 indicates the need is severely neglected and the value 5 indicates it is actively and effectively supported. The rubric is shown in Table 2. The scores are then visualised as a radar diagram overlaying all four applications, allowing the motivational profile of each platform to be compared at a glance. Dimensions where all four applications score consistently low represent the most significant gaps in the current market and the strongest candidates for design intervention in Phase II.

Table 2: Five-Point Likert-Scale Rubric for Heuristic Evaluation

Score	Label	Description
1	Not present	The heuristic is completely absent. No evidence of this design element anywhere in the app.
2	Barely present	There is a minimal or token attempt at implementing this heuristic, but it is incomplete or very poorly executed.
3	Partially implemented	The heuristic is recognisably present but only in some areas or with noticeable gaps. A mixed implementation.
4	Well implemented	The heuristic is clearly and consistently present throughout the app. Only minor improvements could be made.
5	Fully & excellently implemented	The heuristic is implemented to a high standard across the entire app. It stands as a best-practice example.

Inter-Coder Reliability

A central methodological challenge in heuristic evaluation is evaluator bias. A single evaluator's ratings reflect their prior knowledge, assumptions, and aesthetic preferences as much as the properties of the system under review. Nielsen (1994a) established that using multiple independent evaluators significantly improves the validity of heuristic findings, as systematic biases from one evaluator are unlikely to be replicated by another.

To address this, the evaluation follows a two-coder design in which the same heuristic rubric was handed to a second evaluator with relevant professional qualifications, namely a UX designer and software developer with a background in UX design and professional experience in software testing and benchmarking. The two evaluations were conducted independently, with no communication between coders during the rating process.

Following independent scoring, the two sets of ratings are compared to assess Inter-Coder Reliability (ICR), which O'Connor and Joffe (2020) describe as the degree to which

independent coders arrive at the same conclusions when applying the same framework to the same material, and which they argue is a prerequisite for treating qualitative evaluation data as reliable evidence rather than subjective opinion. Where the two ratings agree, the score is accepted directly, while significant divergences are resolved through a negotiated agreement process in which the two evaluators discuss the outlier, examine the specific heuristic criterion and the application behaviour in question, and arrive at a consensus score. The nature of each disagreement is noted, as O'Connor and Joffe (2020) observe that divergences are themselves informative about the limits of the instrument and the ambiguity of the design patterns under evaluation. The resulting scores and ICR coefficients are reported in Section 5.1.

4.2 Phase II: Artefact Design, Prototyping, and Implementation

The second phase translates the diagnostic findings from Phase I into a tangible artefact, which following the DSR logic articulated by Hevner et al. (2004) is not merely a deliverable but the primary vehicle through which the theoretical claims of this thesis are tested. Two parallel tracks structure this phase, combining the iterative design of a high-fidelity prototype in Figma (Figma, Inc., 2024) with the subsequent implementation of a functional mobile application.

Prototype Design in Figma

The design process begins with the gap analysis produced by Phase I, with each dimension in which the benchmarked applications scored consistently low treated as a design requirement representing a motivational need that the current market fails to satisfy and that the MVP is explicitly tasked with addressing. This creates a direct and traceable line from the heuristic findings to the design decisions, which is a prerequisite for the DSR paradigm to function as intended (Hevner et al., 2004).

The prototype is developed in Figma (Figma, Inc., 2024) as a high-fidelity, interactive artefact covering the full user journey from onboarding through social interaction, with the design process structured iteratively so that each round of internal testing refines the interaction patterns in response to observed issues. Screens are designed to operationalise the four theoretical constructs identified in the conceptual framework, providing autonomy

support through an opt-in reflection layer, competence support through contextual match data visualisation, relatedness support through shared group features, and Positive Friction through a deliberate hold-to-confirm mechanism that interrupts the prediction gesture and prompts the user to briefly reason their tip before it is committed.

The hold-to-confirm interaction is the most consequential design decision in this phase, replacing a single tap to submit a prediction with a sustained press followed by a brief rationale prompt before the tip is saved. This mechanic operationalises the theoretical argument advanced by Cox et al. (2016) that deliberate, well-placed interaction hurdles can interrupt automated System 1 behaviour and invite a moment of System 2 reflection without necessarily degrading the experience, while the design challenge of calibrating the threshold at which friction becomes productive rather than obstructive, identified by Mecke et al. (2019), was addressed through internal testing across multiple prototype iterations before the artefact proceeded to Phase III evaluation.

Technical Implementation

In parallel with the Figma (Figma, Inc., 2024) prototype, the MVP is implemented as a fully functional, coded mobile application. This step goes beyond what a purely prototypical evaluation would require, but serves two purposes within the DSR logic. It ensures that the evaluated artefact reflects genuine interaction behaviour rather than the idealised frictionlessness that Figma (Figma, Inc., 2024) prototypes can inadvertently produce, and it validates that the design decisions are technically feasible within the constraints of a real production environment.

The application is built using Flutter and the Dart programming language (Google LLC, 2024b), developed within Antigravity (Google LLC, 2025), a development environment by Google. Choosing Flutter is motivated by its cross-platform capability, allowing a single codebase to target both iOS and Android, which is relevant for the heterogeneous device landscape of the target demographic. The development workflow is supported by Claude Code (Anthropic, 2025) as an AI-assisted terminal-based coding tool, which is used to accelerate implementation while maintaining full developer control over architectural decisions and interaction logic.

Real-time sports data is sourced through the API-Sports service (API-Sports, 2024), which

provides live and historical fixture data, team form, and league standings for the German league Bundesliga and other competitions included in the MVP. User authentication, league management, group data, and prediction records are handled through Firebase and Firestore (Google LLC, 2024a), providing a scalable, real-time database backend that supports the social features central to the application's value proposition. Firestore's real-time listeners enable live leaderboard and group updates without manual refresh, ensuring that the social infrastructure behaves consistently with the relatedness argument advanced in the theoretical framework.

The social architecture of the app is organised around private leagues. Users create or join a league via an invite code, submit predictions ahead of each matchday, and receive points based on the accuracy of their tips after results are confirmed. The leaderboard, group chat, shared achievement features, and friends' tip comparison are built on top of this foundation. The combination of live sports data and real-time social infrastructure means that the artefact evaluated in Phase III reflects a genuinely functional system rather than a Figma (Figma, Inc., 2024) prototype, a distinction that is methodologically significant for the validity of the evaluation results.

4.3 Phase III: Mixed-Methods User Study

The third phase evaluates whether the artefact achieves its intended effect by combining quantitative psychometric instruments with a semi-structured qualitative debrief, designed to answer SRQ 3 directly in terms of how the developed prototype performs across pragmatic and hedonic quality and whether the Positive Friction intervention preserves the intuitiveness of the interaction. Both strands of data are collected within the same session, with the quantitative instruments providing structured measurement and the debrief providing contextual depth and participant voice.

Sample and Recruitment

The target sample consists of $N = 10$ participants, a size consistent with qualitative saturation principles for semi-structured interviews within homogeneous samples (Guest, Bunce, & Johnson, 2006) and with established guidance on formative usability evaluation in HCI (Nielsen, 1994a). Given that the qualitative debrief serves a triangulation function

within a mixed-methods DSR evaluation study rather than as a primary evidence source, this sample size is considered sufficient while maintaining feasibility within the academic timeframe. Participants are recruited through purposive sampling via social media channels, targeting adults with an active interest in football who are open to using a social prediction platform of this kind. This profile reflects the intended primary user group of the MVP and ensures that responses to both the functional features and the friction interventions are grounded in contextually relevant prior expectations. Inclusion criteria require smartphone ownership, basic digital literacy, and willingness to participate in a moderated session of approximately 25 to 55 minutes. Participants with professional UX or usability research backgrounds are excluded to avoid evaluator effects.

Study Design and Procedure

Each session follows a structured protocol consisting of eight task-based scenarios that cover the full feature set of the MVP, sequenced to reflect a natural first-use journey. Participants move through onboarding and league joining before encountering the core prediction flow, which includes fixture browsing and the embedded Positive Friction intervention, followed by league standing review, the opt-in reflection layer, friends' tip comparison, group chat, and the friend invitation flow. The friction intervention is placed in the second scenario rather than isolated at the end, so that its effect is observed in a natural usage context rather than under artificial spotlight conditions.

Before the task phase began, each participant received a standardised information sheet explaining the study purpose and completed an informed consent form. Both documents are provided in Appendix A. This ensured that all participants understood the scope of data collection and provided consent prior to any interaction with the prototype.

All sessions were conducted in person. Participants were asked to think aloud throughout the task phase (Ericsson & Simon, 1993), narrating their expectations and reactions as they interacted with the application. Sessions were audio-recorded using Apple Voice Memos on a secondary device. Transcripts were generated automatically and subsequently translated into English using Claude (Anthropic, 2025) for thematic analysis.

Each session proceeded in four sequential stages, beginning with the eight task-based scenarios completed under think-aloud conditions. Once the tasks were finished, the three

psychometric instruments were administered in order, covering AttrakDiff (Hassenzahl et al., 2003), INTUI (Ullrich & Diefenbach, 2010a), and SUS (Brooke, 1996), followed by the single NPS item (Reichheld, 2003). The session closed with a semi-structured debrief during which demographic data was also gathered. Throughout the task phase, facilitator observations paid particular attention to verbal and emotional reactions at the moment the hold-to-confirm gesture activated and the rationale prompt appeared.

Instruments

Following the task phase, three validated psychometric instruments and one supplementary measure are administered sequentially. The SUS, developed by Brooke (1996) and validated against adjective-based benchmarks by Bangor et al. (2008), provides a global usability score on a zero-to-one-hundred scale and serves as a baseline indicator of whether the artefact meets industry-standard usability thresholds. The instrument consists of ten alternating positive and negative items rated on a five-point Likert scale, with an accepted benchmark for good usability set at 68 (Bangor et al., 2008).

The AttrakDiff questionnaire (Hassenzahl et al., 2003) extends the usability measure into the hedonic domain by splitting the user experience into four dimensions covering pragmatic quality, hedonic stimulation, hedonic identity, and overall attractiveness. Pragmatic quality reflects usability and task efficiency, while the two hedonic dimensions capture novelty and creative appeal on the one hand and the degree to which the product supports the user's sense of self on the other. For this thesis, the hedonic dimensions are of primary theoretical interest, as the central hypothesis is that the Positive Friction features will produce elevated hedonic quality scores relative to the pragmatic baseline established by SUS, precisely because slowing the interaction and inviting reflection is more likely to produce a sense of meaningfulness than of efficiency. This would be consistent with Hassenzahl (2013)'s argument that meaningful interaction is not reducible to task performance.

The INTUI scale (Ullrich & Diefenbach, 2010a) completes the psychometric battery by measuring four facets of intuitive interaction, namely Gut Feeling, Magical Experience, Verbalisability, and Confidence. Its inclusion is motivated by a specific theoretical risk that the deliberate friction introduced by the hold-to-confirm mechanism will be experienced as a disruption to the effortless, automatic quality that modern mobile users expect, with the Gut Feeling and Confidence sub-scales testing whether this risk materialises in practice.

The Verbalisability sub-scale is of particular interest in the friction scenario, as a score above the midpoint would suggest that the rationale prompt is functioning as intended by making the prediction process more reflective without making it opaque (Ullrich & Diefenbach, 2010a).

Additionally, a NPS item is administered at the close of each session (Reichheld, 2003), asking participants to rate on a zero-to-ten scale how likely they would be to recommend an application of this kind to a friend, followed by an open field for the primary reason behind the rating. Given the sample size of $N = 10$, the NPS is treated as a directional indicator rather than a statistically generalisable metric.

Qualitative Debrief

Following the psychometric instruments, each session concludes with a semi-structured debrief of approximately eight to twelve minutes. Seven open questions, labelled O1 through O7 and listed in Table 3, are posed in sequence, each anchored to a specific theoretical construct from the framework developed in Chapter 2. The questions are designed to triangulate the psychometric instrument scores with qualitative evidence, moving from an unanchored overall impression through targeted probing of the Positive Friction interventions, the three SDT needs, and a direct comparison with existing platforms, before closing with an open invitation for any observations not covered by the preceding questions. Debrief responses are referenced throughout Chapter 5 using the format (P $_{xx}$, O $_n$), where xx denotes the participant number and n the question identifier.

Table 3: Semi-structured debrief questions with theoretical grounding.

ID	Question	Construct
O1	<i>Looking back at the whole session: what is your overall impression of the app? What stands out most, positively or negatively?</i>	Overall impression (Ericsson & Simon, 1993)

Table 3 continued

ID	Question	Construct
O2	<i>There were moments where the app deliberately slowed you down, asked you to briefly justify a prediction, or confirm a choice. Did those moments feel meaningful, or just annoying? Did they change anything about how you interacted?</i>	Positive Friction (Cox et al., 2016; Mecke et al., 2019)
O3	<i>Did you feel in control of your own experience throughout? Were there moments where the app felt like it was pushing you in a direction you didn't choose?</i>	Autonomy (Ryan & Deci, 2000)
O4	<i>Did the match data and statistics make you feel more capable and well-informed when making your predictions, or did they feel like extra noise?</i>	Competence (Ryan & Deci, 2000; Przybylski et al., 2010)
O5	<i>Did the social features, the leaderboard or chat, feel meaningful? Did they make you feel more connected to the other players, or were they just functional extras?</i>	Relatedness (Ryan & Deci, 2000; Rigby & Ryan, 2011)
O6	<i>How does this compare to apps like Kicktipp or any other prediction tools you've used? What feels genuinely different, better or worse?</i>	Market comparison (Tondello et al., 2016)
O7	<i>Is there anything else you'd like to share, anything about the app, about today's experience, or anything that came to mind during the session that we haven't talked about yet?</i>	Open invitation

Analysis

The quantitative data from SUS, AttrakDiff, INTUI, and NPS are analysed descriptively, with means and standard deviations reported for each instrument dimension and interpreted against published benchmarks where available. Given the sample size of $N = 10$, inferential statistics are not applied and the study is treated as exploratory rather than confirmatory (Sauro & Lewis, 2012), which is consistent with the DSR paradigm's focus on evaluating

whether a designed artefact works in practice rather than on producing statistically generalisable findings (Hevner et al., 2004). The qualitative debrief responses are analysed thematically, with themes derived from the SDT needs and the Positive Friction framework, and selected participant quotes are used throughout Chapter 5 to contextualise patterns observed in the quantitative data.

5 Results

5.1 Phase I: Benchmarking Findings and Gap Analysis

Inter-Coder Reliability

Before presenting the substantive findings, the reliability of the evaluation data must be established. The two evaluators (the primary researcher and a secondary coder with professional experience in UX design and software testing) rated each of the 28 heuristics independently across all four applications, producing 112 rating pairs in total. Following the procedure described in the methodology, scores were compared and divergences of two or more points on the five-point scale were resolved through a negotiated agreement process between the two evaluators.

Of the 112 pairs, 42 (37.5%) were in exact agreement, with a further 51 pairs differing by exactly one point, bringing the within-one-point agreement rate to 83.0%, a level considered acceptable for heuristic evaluation using Likert-scale instruments (O'Connor & Joffe, 2020). It should be noted that percentage agreement does not account for chance agreement in the way that formal statistical coefficients do, with Cohen's Kappa (Cohen, 1960) and Krippendorff's Alpha (Krippendorff, 2004) being the most widely used alternatives, both correcting for the probability that two coders agree by chance alone. However, heuristic evaluation is an inspection method rather than a coding study, and the negotiated agreement process serves as a structured qualitative validity check that substitutes for a statistical coefficient in established applications of this framework (Tondello et al., 2019; Nielsen, 1994a). The remaining 19 rating pairs (17.0%) that diverged by two or more points were resolved through this process, and the full agreement statistics are summarised in Table 4.

Table 4: Inter-Coder Reliability Summary across 112 Rating Pairs (4 Apps $\times$ 28 Heuristics)

Agreement Level	Count	Percentage
Exact agreement (diff = 0)	42	37.5%
Agreement within 1 point (diff $\leq$ 1)	93	83.0%
Divergence $\geq$ 2 points (negotiated agreement)	19	17.0%
Total rating pairs	112	100%

The 19 negotiated agreement cases were distributed across all four applications and revealed two recurring sources of disagreement. Heuristic IDs follow the format used in Table 5, where the prefix denotes the category and the number identifies the individual heuristic within it. The first concerned the interpretation of partially implemented features, with the primary evaluator rating a feature as absent or minimal based on its practical invisibility to users in several instances, while the secondary coder rated it higher based on its technical existence within the app. This pattern was most pronounced for Kicktipp's GC4 (Rules are clear and transparent), where the secondary coder credited the technical presence of a rules section while the primary evaluator weighted its poor discoverability, a position partially accepted in the consensus, moving the score from divergent ratings of B=2 and L=5 to a consensus of 4. The same dynamic arose for Kicktipp's FR5 (Progress indicators), where the primary evaluator's assessment prevailed given the absence of any meaningful visual progression system, resulting in a consensus of 2 from initial ratings of B=1 and L=4. The second source of disagreement concerned the scope of the Tipico evaluation, as the two coders disagreed on whether features present in Tipico's broader platform but absent from the football-specific interface should be credited. The negotiated agreement resolved in favour of evaluating only the in-app experience, leading to downward revisions for SC1, SC2, and SC3, all of which were adjusted to 2. All consensus scores are documented in Table 5.

Table 5: Final Consensus Heuristic Scores by Category of All Four Applications

ID	Heuristic	Kicktipp	Tackle	Teamtip	Tipico
<i>Goals & Challenges</i>					
GC1	Provide clear and meaningful goals	3	4	4	4
GC2	Provide a variety of challenges	2	1	2	5
GC3	Challenges scaled to skill level	1	1	3	4
GC4	Rules are clear and transparent	4	1	5	4
<i>Feedback & Rewards</i>					
FR1	Timely and meaningful feedback	3	2	4	5
FR2	Visible progress and score	2	4	5	4

ID	Heuristic	Kicktipp	Tackle	Teamtip	Tipico
FR3	Rewards acknowledge effort	1	2	3	3
FR4	Unexpected or variable rewards	1	4	3	4
FR5	Progress indicators are meaningful	2	3	4	4
<i>Freedom & Creativity</i>					
FC1	Freedom of choice in participation	4	1	5	4
FC2	Creative self-expression is possible	3	4	3	3
FC3	Exploration and discovery encouraged	3	3	3	5
<i>Social & Community</i>					
SC1	Social comparison and leaderboards	4	4	4	2
SC2	Cooperation and team play	1	1	1	2
SC3	Social sharing and external visibility	1	1	1	2
SC4	Communication and community features	3	1	4	1
SC5	Community help and peer information	1	3	1	4
<i>Immersion & Narrative</i>					
IN1	Narrative context and storytelling	1	2	1	1
IN2	Thematic visual consistency	1	3	3	4
IN3	Avatar or persistent identity	2	4	3	3
IN4	Atmosphere and sensory engagement	1	2	1	4
<i>Engagement Loops</i>					
EL1	Onboarding introduces system gradually	2	2	4	3
EL2	Regular engagement triggers	3	3	4	4
EL3	Streaks and consistency rewards	1	4	3	4
EL4	Urgency and loss aversion mechanics	2	2	4	4

ID	Heuristic	Kicktipp	Tackle	Teamtip	Tipico
<i>Ease & Accessibility</i>					
EA1	Core interaction is simple and immediate	3	4	5	5
EA2	Error prevention and recovery	3	4	4	4
EA3	Accessibility and inclusive design	2	2	2	2
Total (max 140)		60	72	89	98
Mean (max 5.00)		2.14	2.57	3.18	3.50

Overall Scores and Application Profiles

Table 5 presents the final consensus scores across all 28 heuristics for each of the four applications. The motivational profiles are visualised as a radar diagram in Figure 1, allowing category-level patterns to be compared across all four applications at a glance.

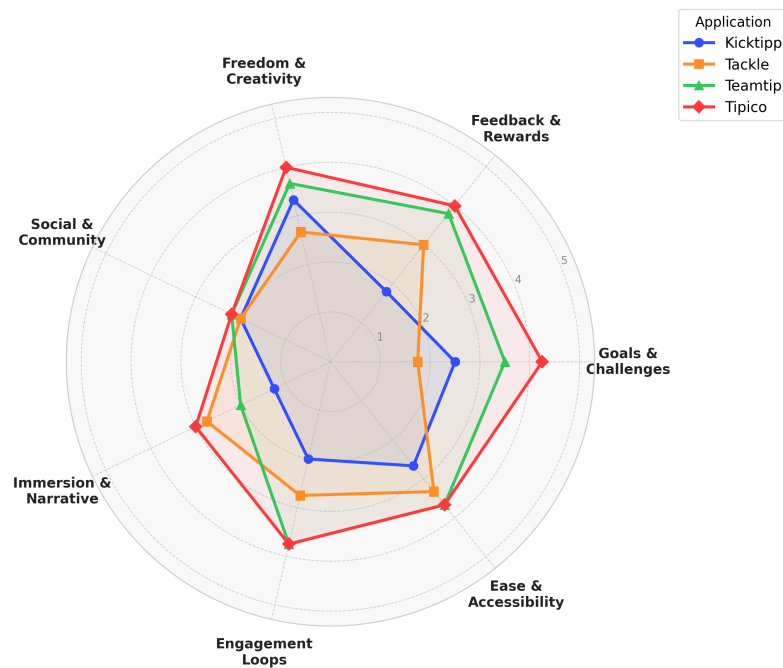


Figure 1: Gameful Design Heuristics Radar: Category-level motivational profiles of the four evaluated applications. Higher scores indicate stronger need satisfaction across each dimension (Tondello et al., 2016). Social & Community was the consistently weakest category across all four apps (range: 2.00–2.20).

The total scores range from 60 out of a maximum of 140 (Kicktipp, mean 2.14) to 98 (Tipico, mean 3.50), with Tackle (72, mean 2.57) and Teamtip (89, mean 3.18) occupying intermediate positions. Although Teamtip achieves the highest aggregate score among the social prediction applications, its market share remains marginal, a finding consistent with the network effect dynamics documented by Shapiro and Varian (1998), which suggest that technical superiority alone is insufficient to overcome the social capital embedded in an incumbent platform. That Tipico, the commercial betting application, produces the highest aggregate score overall is not an endorsement of its design philosophy. On the contrary, it reflects the significant investment gambling platforms make in engagement mechanics, precisely the dark nudge architecture that this thesis distances itself from (Newall, 2019; Schüll, 2012). Its high scores in Feedback & Rewards (4.00), Engagement Loops (3.75), and Ease & Accessibility (3.67) are consistent with a product engineered to maximise compulsive return rather than meaningful participation.

Gap Analysis by Category

Table 6 aggregates the 28 heuristic scores into the seven framework categories and presents cross-application means, making the structural gaps visible at a higher level of abstraction.

Table 6: Category-Level Mean Scores across All Four Applications

Category	Kicktipp	Tackle	Teamtip	Tipico	Cross-app mean
Goals & Challenges	2.50	1.75	3.50	4.25	3.00
Feedback & Rewards	1.80	3.00	3.80	4.00	3.15
Freedom & Creativity	3.33	2.67	3.67	4.00	3.42
Social & Community	2.00	2.00	2.20	2.20	2.10
Immersion & Narrative	1.25	2.75	2.00	3.00	2.25
Engagement Loops	2.00	2.75	3.75	3.75	3.06
Ease & Accessibility	2.67	3.33	3.67	3.67	3.34

Three findings from this analysis directly motivate the design decisions of Phase II and are discussed in turn.

Social & Community is the weakest category across the entire market. With a cross-application mean of 2.10 and a range of only 2.00 to 2.20, Social & Community produced the lowest and least variable scores of any category. No application scored higher than 2.20, meaning that even the strongest performer in this category achieved only a minimal implementation. The individual heuristics driving this floor are particularly stark, with SC2 (Cooperation and team play) and SC3 (Social sharing and external visibility) both producing a cross-application mean of 1.25, suggesting these features are effectively absent across the entire market. The leaderboard (SC1, mean 3.50) was the sole social feature to perform consistently, suggesting that competition is addressed while cooperation, community, and social expression are almost entirely neglected. For a category of applications whose entire value proposition rests on social prediction, this may represent a foundational design gap that also maps directly onto SDT's relatedness construct, since none of the evaluated platforms meaningfully support the psychological need for connection beyond a ranked list (Ryan & Deci, 2000; Rigby & Ryan, 2011).

Immersion & Narrative is the second most neglected category, with Kicktipp performing worst. The cross-application mean of 2.25 masks a particularly low result for Kicktipp (1.25), which scored the minimum possible on three of the four Immersion heuristics, namely IN1 (Narrative context, 1), IN2 (Thematic visual consistency, 1), and IN4 (Atmosphere and sensory engagement, 1). The evaluator notes for IN2 describe an application that relies exclusively on plain text, with no football-specific visual language, club imagery, or thematic design of any kind. This finding carries additional weight given that Kicktipp is the market leader (Kicktipp GmbH, n.d.), and its continued dominance despite near-total absence of hedonic design investment is consistent with the lock-in effect described by Shapiro and Varian (1998), where users may remain not because the experience is strong but because their social networks are already there. The practical implication for this thesis is that raising the hedonic baseline through thematic coherence and contextual data visualisation could represent a meaningful differentiator.

Feedback & Rewards is structurally absent in Kicktipp and mechanically present but ethically problematic in Tipico. Kicktipp's Feedback & Rewards mean of 1.80 reflects an almost complete absence of reward mechanics, with FR3 (Rewards acknowledge effort, 1) and FR4 (Unexpected or variable rewards, 1) both receiving the minimum score and

FR5 (Progress indicators, 2) faring only marginally better. There are no badges, no streaks, no progress visualisations, and no milestone acknowledgements of any kind, which is consistent with the overjustification concern in reverse in that the platform has no extrinsic layer at all, leaving intrinsic motivation without any structural reinforcement (Lepper et al., 1973; Deterding et al., 2011). Tipico, by contrast, scored 4.00 in this category, though through mechanics that appear to exploit urgency, loss aversion, and variable reward schedules in ways documented as harmful (Newall, 2019; Hing, Smith, et al., 2022). The benchmarking data therefore points to a design space the current market has not addressed, namely feedback structures that could support intrinsic motivation without exploiting it.

Lowest-Performing Individual Heuristics

Beyond the category level, five individual heuristics merit particular attention as the most severely neglected across all four applications, as shown in Table 7.

Table 7: Five Lowest-Scoring Heuristics by Cross-Application Mean

ID	Heuristic	KT	TCL	TT	TPC	Mean
IN1	Narrative context and storytelling	1	2	1	1	1.25
SC2	Cooperation and team play	1	1	1	2	1.25
SC3	Social sharing and external visibility	1	1	1	2	1.25
EA3	Accessibility and inclusive design	2	2	2	2	2.00
IN4	Atmosphere and sensory engagement	1	2	1	4	2.00

Most notable among the Phase I findings is the convergence of IN1, SC2, and SC3 at a cross-application mean of 1.25. Across all four evaluated applications, including the commercial betting platform, there is no meaningful investment in narrative framing, cooperative mechanics, or social sharing, representing motivational territory that is simultaneously the most neglected by the market and the most directly addressed by the design principles of this thesis through relatedness-oriented social architecture and hedonic quality through contextual meaning-making. Equally consistent across all four applications is the uniform score of 2.00 for EA3 (Accessibility and inclusive design), suggesting that accessibility has been treated as an afterthought across the entire evaluated landscape. While the MVP does not claim full Web Content Accessibility Guidelines (WCAG) compliance within the

constraints of a Master's thesis prototype, it addresses this gap through contrast-compliant colour usage and standard platform accessibility features, establishing a meaningfully higher baseline than any of the benchmarked applications.

Summary: Design Implications for Phase II

The benchmarking findings converge on three actionable design gaps that directly inform the feature priorities of Phase II. The near-total absence of cooperative social features and community mechanics across all four applications points to the relatedness dimension of SDT as an underserved design space (Ryan & Deci, 2000), which the MVP addresses through a group chat and a friends tip comparison feature. The consistent neglect of narrative and thematic immersion, most severely in Kicktipp, identifies hedonic quality as a potential differentiating opportunity that the MVP addresses through contextual match data visualisation and a thematically coherent interface. Finally, the polarisation between Kicktipp's reward vacuum and Tipico's manipulative engagement mechanics points to the need for an ethical middle ground, meaning feedback structures that support intrinsic motivation without exploiting variable reward schedules or loss aversion (Cox et al., 2016; Deterding et al., 2011), and this tension forms the theoretical and practical foundation for the Positive Friction intervention developed in Phase II.

5.2 Phase II: Design Decisions and Prototype Documentation

The second phase translates the empirical gap analysis of Phase I into a concrete artefact, unfolding across three distinct stages that move from initial low-fidelity sketching to establish the structural logic, through high-fidelity prototyping in Figma to define the visual language and interaction design, and finally to technical implementation as a functional Flutter application. Each stage is documented in turn.

Feature Selection and SDT Mapping

Before any design work began, the benchmarking findings were translated into a prioritised feature set, with each feature mapped to one of four motivational categories derived from the theoretical framework, namely the three basic psychological needs of SDT covering autonomy, competence, and relatedness, and Positive Friction as the thesis's primary design intervention. This mapping ensured that every design decision carried a

traceable theoretical justification and that the features most neglected by the benchmarked applications received the highest implementation priority.

Table 8 presents the complete feature set of the MVP, organised by SDT dimension.

Table 8: PreMatch MVP Feature Set Mapped to Theoretical Framework

SDT Dimension	Feature	Priority
Autonomy	Club selection and personalisation	High
	Editable profile and profile picture	High
	Edit submitted tip (pre-deadline)	High
	Easy opt-out (leave league)	Medium
	Persistently accessible league rules	Medium
Competence	Onboarding flow	High
	One-click home screen prediction	High
	Countdown timer to tip deadline	High
	Deep statistics and data visualisation	High
	Lineup prediction tool	High
	Prediction accuracy and history browser	Medium
	Streak counter	Medium
	Club logos in fixtures	Low
Relatedness	Leaderboard (friend-scoped)	High
	Community chat	Medium
	Prediction percentage distribution	Medium
	Badges	Low
Positive Friction	Hold-to-confirm submission	High
	Prediction rationale prompt	High
	Push notifications (reminders, lineups, injuries)	Medium
Ethical Design / Accessibility	Screen reader support and bold font option	Medium

The feature priorities were assigned in direct response to the benchmarking findings. Features addressing the three lowest-scoring heuristic domains (Social & Community,

cross-app mean 2.10; Immersion & Narrative, 2.25; and Feedback & Rewards for Kicktipp, 1.80) received the highest weighting. The two positive friction features, hold-to-confirm and the prediction rationale prompt, were both assigned high priority as the central design intervention of this thesis, directly operationalising the theoretical argument of Cox et al. (2016) that deliberate interaction hurdles can support reflective decision-making without degrading the user experience.

Low-Fidelity Sketching

The design process began with hand-drawn low-fidelity sketches on paper, a medium chosen deliberately to constrain the initial ideation phase to structural logic and information hierarchy rather than visual detail. Each sketch represents a single screen's content model, capturing the information present on that screen, its order, and the structure of the primary interaction, without committing to colour, typography, or spacing decisions that would be resolved in the subsequent high-fidelity phase.

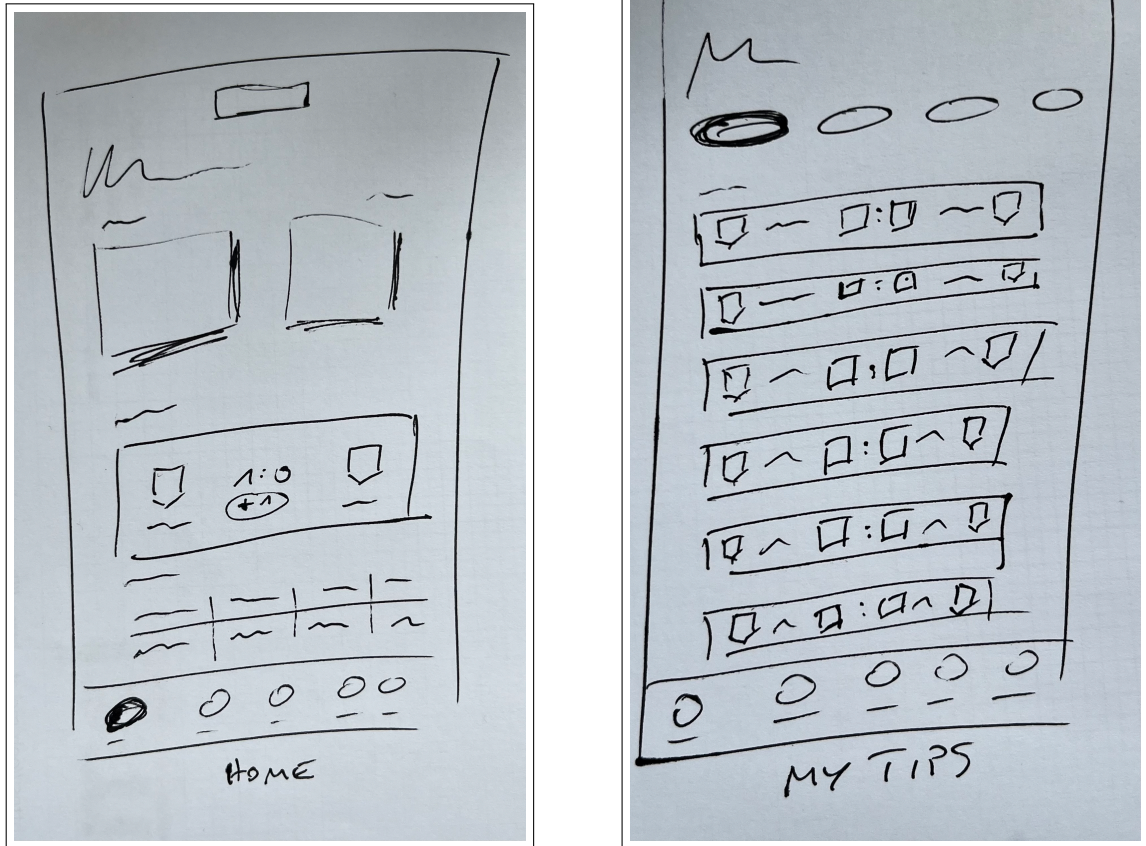


Figure 2: Low-fidelity hand-drawn sketches for the two primary screens of PreMatch, the social football prediction application developed in this thesis. Left: the Home screen with a card-based layout featuring a live match module, prediction summary, and leaderboard. Right: the My Tips screen with a matchday tab selector and scrollable match list for entering predictions.

The sketch of the Home screen (Figure 2, left) establishes the core information architecture of the application through a card-based layout grouping matchday performance data, a live match module showing the current score alongside the user's prediction, and a leaderboard table below. This structure directly addresses Kicktipp's identified weaknesses in GC1 (providing a clear primary goal) and FR2 (visible progress and score) by making the user's current standing and active prediction status the dominant content of the first screen.

The sketch of the My Tips screen (Figure 2, right) establishes the fixture list, representing the scheduled matches for a given matchday, as the core of the prediction flow, with a matchday tab selector at the top and individual match rows below for entering score

predictions.

Both sketches include a five-icon bottom navigation bar, though the exact destinations were not yet defined at this stage and were finalised during the subsequent high-fidelity phase.

Brand Identity and Visual Language

Before proceeding to high-fidelity design, a visual identity was established for the application under the name **PreMatch**, chosen to reflect the app's primary interaction moment, namely the period before a match when predictions are formed and submitted, and to position the product clearly within the football context without adopting the visual language of commercial betting.

The pattern library is built on the foundations shown in Figure 3. The primary typeface is Inter, selected for its high legibility at small sizes on mobile screens and its neutral, modern character that allows the football content such as club colours, match data, and league information to carry the visual weight of each screen rather than the interface itself. The colour palette is built around a primary blue (#304FFF), black (#000000), and white (FFFFFF) as the core triad, complemented by four semantic colours covering error red (#FF383C), success green (#34C759), warning orange (#FF8D28), and alert yellow (FFCC00). These semantic colours play a functional role in the prediction feedback system, with correctly predicted results surfaced in success green and incorrect ones in error red, giving users immediate visual confirmation of their performance without requiring them to read score text. To support accessibility, all colour-coded states are additionally distinguished by icons sourced from the Phosphor Icon Library (Phosphor Icons, 2024), ensuring that the feedback system remains interpretable for users with colour vision deficiencies. The consistent stroke weight and legibility of the Phosphor icons at small sizes also contributes to the overall visual coherence of the system, which further defines core UI components including buttons, tags, switches, and achievement badges to ensure consistency across the full feature set.

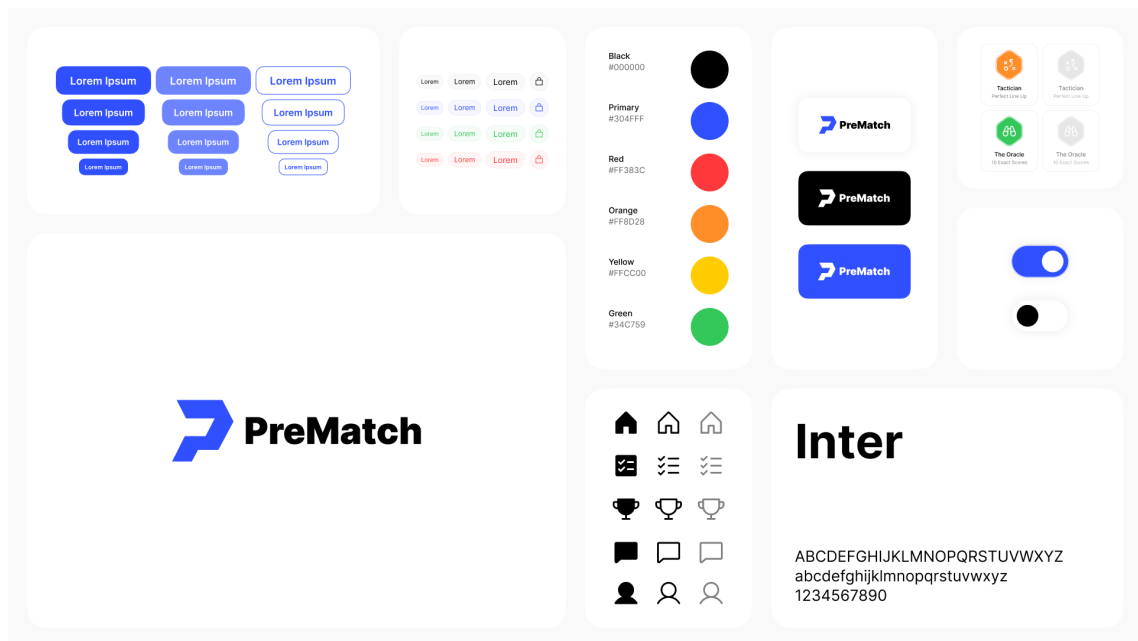


Figure 3: PreMatch design system overview, including primary typeface (Inter), colour palette (#304FFF blue, semantic error, success, warning, and alert colours), core UI components (buttons, tags, switches, achievement badges), and iconography from the Phosphor Icon Library (Phosphor Icons, 2024). The full interactive design file is provided in Appendix A.

High-Fidelity Prototype in Figma

The high-fidelity prototype was developed in Figma (Figma, Inc., 2024), translating the structural decisions of the low-fidelity sketches into a fully specified interactive design covering the complete user journey across five navigation destinations, namely Home, My Tips, My League, Chat, and Profile. All screens use the PreMatch design system and are connected with interactive transitions that reflect the intended navigation behaviour of the implemented application. The prototype was initially designed using Bundesliga fixtures and club logos as placeholder content, giving the screens a realistic football context during the design phase. Not all screens were fully specified in Figma prior to implementation, as several interaction details including the rationale prompt were designed directly during the implementation phase in response to technical and timing constraints. Minor visual differences between the Figma prototype and the implemented application therefore reflect this iterative process rather than inconsistencies in the design intent. The full interactive

design file is provided in Appendix A.

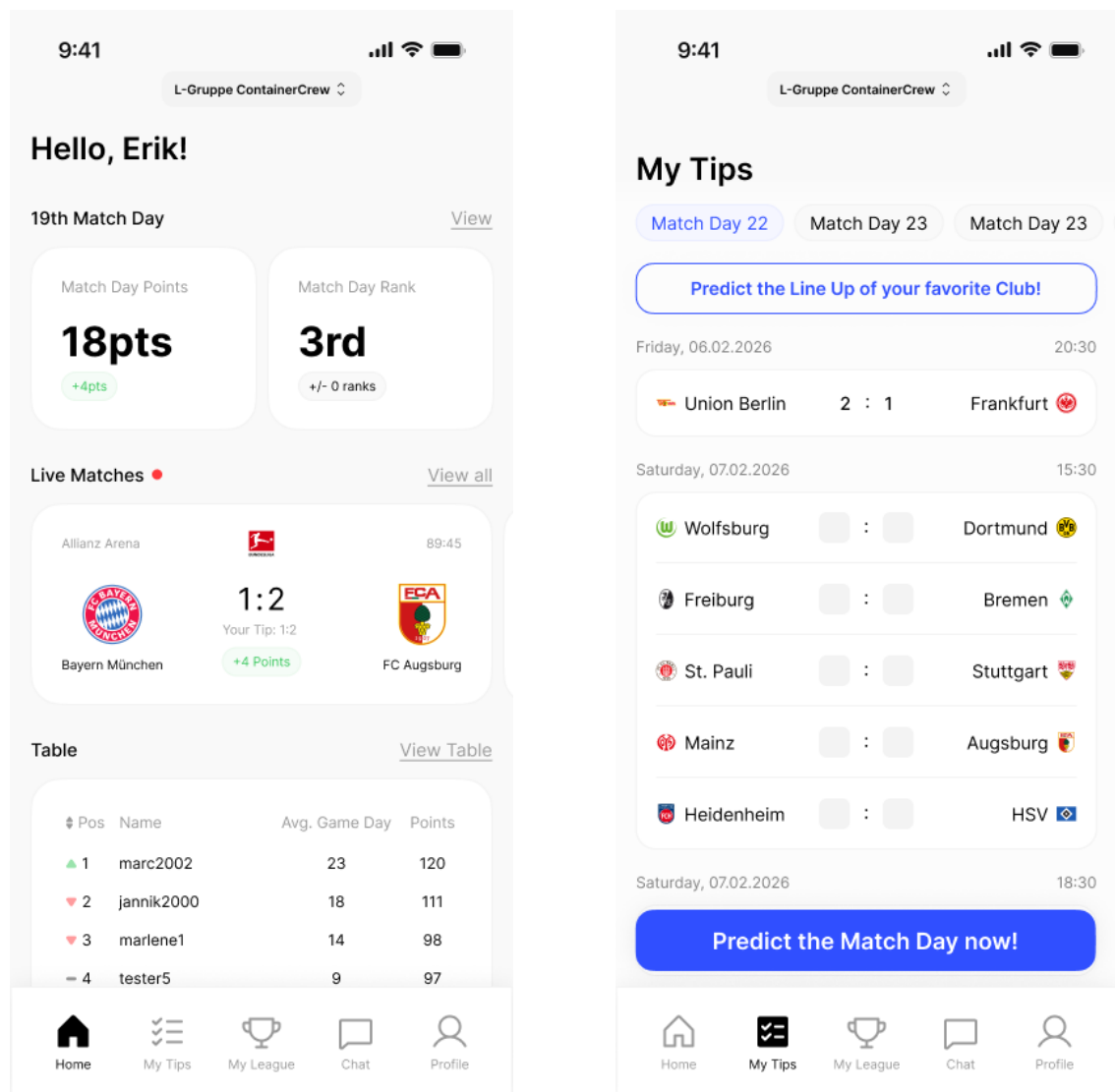


Figure 4: High-fidelity Figma prototype screens for PreMatch. Left: the Home screen displaying matchday performance cards (points and rank), a live match module with real-time score, the user's submitted tip, and the resulting points awarded, followed by a friend-scoped leaderboard. Right: the My Tips screen showing the matchday tab selector, fixture rows with club logos, score input fields, and a completed result (Union Berlin 2:1 Frankfurt) with the primary call-to-action button anchored at the bottom. Both screens use Bundesliga placeholder content inserted during the design phase.

The comparison between the low-fidelity sketches (Figure 2) and the high-fidelity screens (Figure 4) illustrates how the structural decisions made on paper were carried through into the designed artefact. The information hierarchy established in the sketches is preserved

throughout, with the Home screen leading with a performance summary, moving to live context, and closing with social comparison, while the My Tips screen maintains the tab-first navigation, the fixture row pattern, and the bottom-anchored primary action. The high-fidelity layer adds the PreMatch visual identity, club logo integration, semantic colour feedback, and the interaction density required for a functional mobile application.

Several design decisions visible in Figure 4 relate directly to the benchmarking findings. The live match module on the Home screen surfaces the user's own tip alongside the current score and communicates the points outcome in real time, addressing FR1 (Timely and meaningful feedback), which scored 3.0 for Kicktipp and was identified as a structural weakness across the social prediction category. The leaderboard on the Home screen is friend-scoped rather than global, showing only the user's immediate league as a deliberate relatedness design decision, since the competitive reference point being the user's social circle rather than an anonymous global ranking is what Rigby and Ryan (2011) identify as a stronger driver of sustained engagement. Club logos are present on every fixture row in the My Tips screen, addressing IN2 (Thematic visual consistency), which scored 1.0 for Kicktipp and represents one of the most visible differences between PreMatch and the benchmarked market leader.

Technical Implementation

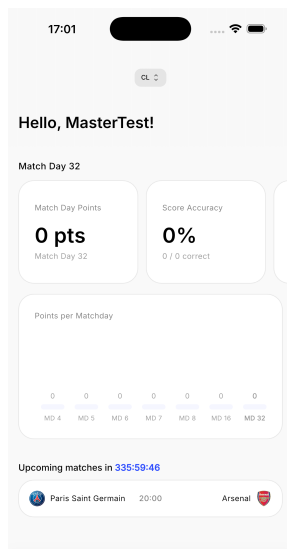
Following the completion of the Figma prototype, the MVP was implemented as a functional mobile application using Flutter and the Dart programming language (Google LLC, 2024b), developed within Antigravity (Google LLC, 2025), a development environment by Google. The implementation was supported by Claude Code (Anthropic, 2025) as an AI-assisted terminal-based coding tool, which accelerated development of boilerplate and integration logic while full architectural and interaction design decisions remained under direct developer control. The full application source code and setup instructions are available in Appendix A.

The application is structured around three external services. Live and historical football data (fixture schedules, match results, club logos, team form, head-to-head statistics, and league standings) are sourced from the API-Sports service (API-Sports, 2024), which provides a comprehensive RESTful data layer covering multiple competitions. The application was designed using Bundesliga content, but as the Bundesliga season concluded before the user

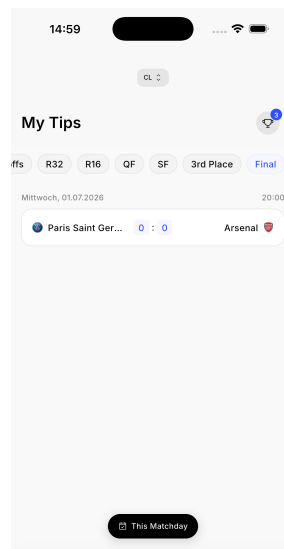
study took place, the implementation switched to the UEFA Champions League, which was still ongoing at the time of data collection. User authentication, league membership, prediction records, leaderboard data, chat messages, and badge assignments are managed through Firebase Authentication and Firestore (Google LLC, 2024a). Firestore's real-time document listeners are used for the live leaderboard and community chat features, enabling instant updates across all connected clients without manual refresh, a technical requirement for the relatedness features to feel genuinely social rather than asynchronous. Push notifications for tip deadlines, lineup releases, and injury alerts are handled through Firebase Cloud Messaging, directly addressing EL2 (Regular engagement triggers), which scored 3.0 for Kicktipp and was identified in Phase I as an underdeveloped mechanism for maintaining habitual engagement.

The hold-to-confirm interaction is implemented at the Flutter widget level as a long-press gesture recogniser with a configurable duration threshold, after which the rationale prompt sheet is presented modally before the prediction is written to Firestore. This architecture means the friction intervention is not bypassable by rapid tapping, a deliberate constraint that ensures the System 2 pause described by Kahneman (2011) is structurally enforced rather than merely suggested by the interface. The duration threshold was calibrated through internal testing to a value that felt purposeful without feeling punitive, consistent with the design guidance of Mecke et al. (2019) on calibrating friction thresholds.

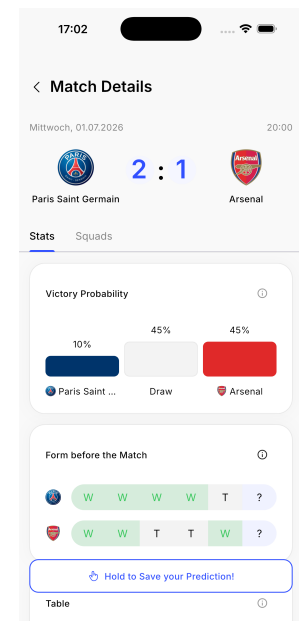
Figure 5 and Figure 6 present the key screens of the implemented application as used during the user study.



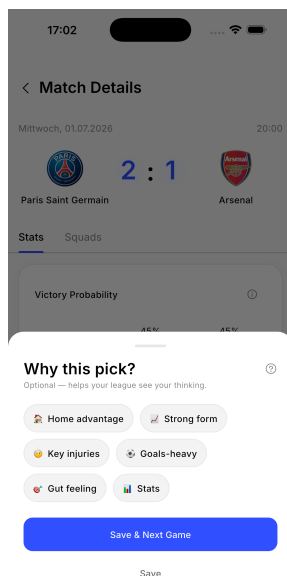
Home



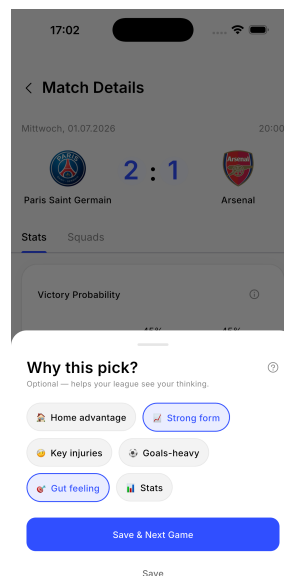
My Tips



Game Detail

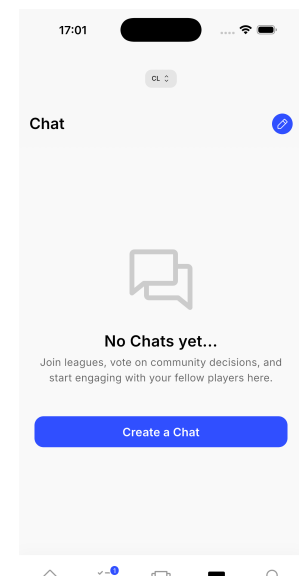


Rationale Prompt



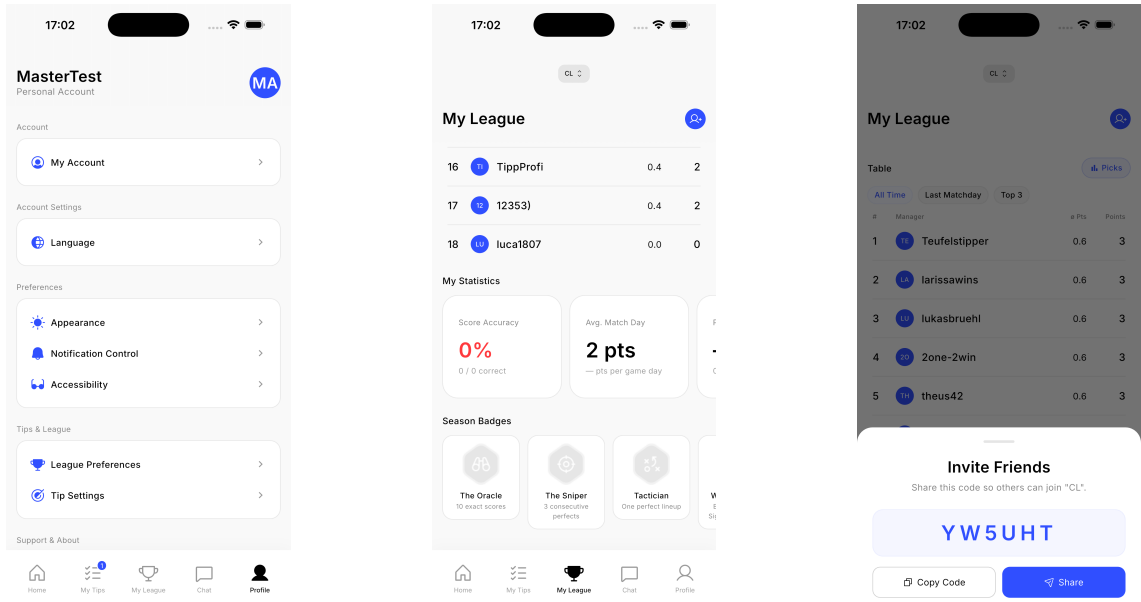
Rationale Prompt

Selected



Chat

Figure 5: Key screens of the PreMatch application, part 1 of 2 (Champions League content). Row 1, left to right: Home (matchday summary cards, live match module, friend leaderboard), My Tips (fixture list with club logos and score input fields), and Game Detail (match statistics and head-to-head data). Row 2: Rationale Prompt Selection, Rationale Prompt Selected (reasoning tag selection before the tip is saved), and Chat (group messaging). Continued in Figure 6.



Settings

My League

Friend Invitation

Figure 6: Key screens of the PreMatch application, part 2 of 2. Left to right: Settings (reflection layer toggle and notification preferences), My League (leaderboard and league overview), and Friend Invitation (invite code sharing flow). Screens designed directly during the implementation phase, such as the Rationale Prompt, may differ slightly from the Figma prototype. Source code and setup instructions are available in Appendix A.

5.3 Phase III: Mixed-Methods Evaluation Results

Ten participants completed the study between the 16th of May and the 7th of June 2026. Sessions were conducted in person, each lasting between 28 and 51 minutes. All participants interacted with the functional Flutter application on the researcher’s device, working through the eight task-based scenarios before completing the psychometric instruments and the semi-structured debrief. One participant (P01) encountered a technical issue with the audio recording. Their session data was reconstructed from facilitator notes and written questionnaire responses. All other sessions were audio-recorded and transcribed using Apple Voice Memos, with transcripts subsequently translated into English using Claude (Anthropic, 2025). Full transcripts, individual participant response sheets, and the aggregated results file are provided in Appendix A. All results are reported descriptively, consistent with the exploratory framing of a DSR evaluation study at $N = 10$ (Hevner et al., 2004; Sauro & Lewis, 2012).

System Usability Scale

Every participant scored in the Excellent range. The mean SUS score was 90.0 ($SD = 6.67$), placing PreMatch in the Grade A band per Bangor et al. (2008) and 22 points above the widely accepted benchmark of 68 for above-average usability (Bangor et al., 2008; Lewis, 2018). The minimum individual score was 80.0 (P08). The standard deviation of 6.67 is tight, suggesting perceived usability was consistent across the sample rather than driven by isolated favourable responses. Individual scores are listed in Table 9 and visualised in Figure 7.

Table 9: SUS Scores per Participant with Descriptive Statistics

Participant	SUS Score (0–100)	Adjective Rating
P01	100.0	Excellent
P02	85.0	Excellent
P03	85.0	Excellent
P04	90.0	Excellent
P05	85.0	Excellent
P06	90.0	Excellent
P07	95.0	Excellent
P08	80.0	Excellent
P09	100.0	Excellent
P10	90.0	Excellent
Mean (SD)	90.0 (6.67)	Excellent (A)
Min / Max	80.0 / 100.0	–
Industry benchmark	≥ 68 (Good)	–

P08 scored lowest, with their session producing the most facilitator observations regarding navigation difficulties, specifically around the friend invitation flow and the chat input field, though the score still falls within the Excellent range. The fact that the most challenging session still yields 80.0 suggests the usability floor is relatively high across the sample, which is relevant for what follows, since negative pragmatic experiences tend to suppress hedonic ratings regardless of design intent (Hassenzahl, 2003), making it unlikely that usability issues have confounded the AttrakDiff results presented in Figure 7 below.

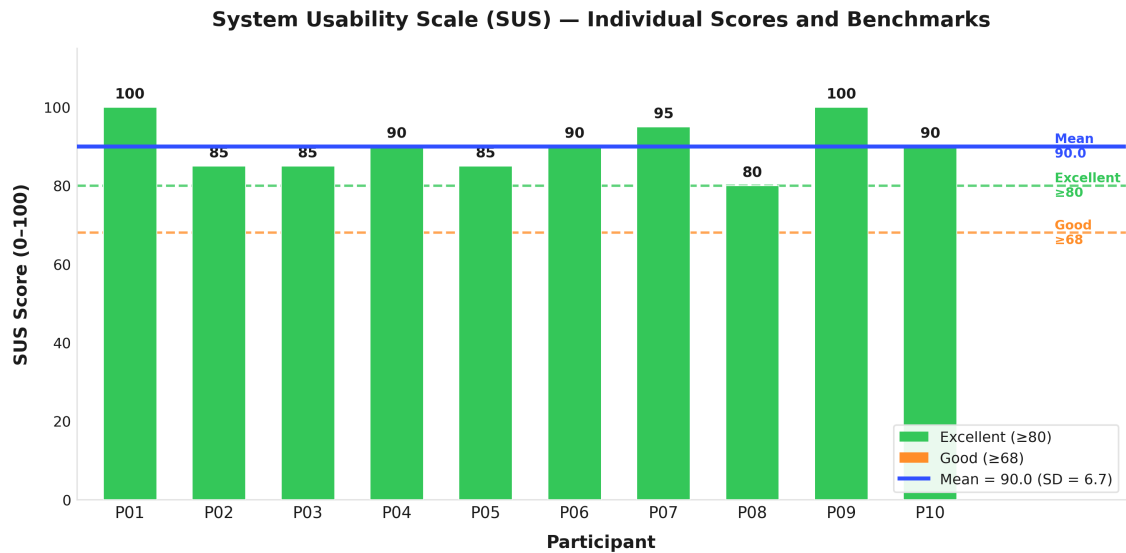


Figure 7: SUS scores per participant with benchmark thresholds (Good ≥ 68 , Excellent ≥ 80) and study mean of 90.0 ($SD = 6.67$). Benchmark thresholds per Bangor et al. (2008). All participants scored in the Excellent range.

AttrakDiff

AttrakDiff captures whether a product goes beyond usability to produce a genuinely engaging and identity-affirming experience, separating pragmatic from hedonic quality (Hassenzahl et al., 2003). For this thesis, the key question is whether the Positive Friction interventions elevated hedonic quality without pushing pragmatic quality into negative territory. Raw item scores (1–7 per bipolar pair) were averaged per dimension and converted to the -3 to $+3$ scale by subtracting 4 from the mean, per the standard AttrakDiff procedure (Hassenzahl et al., 2003). All four dimensions produced positive means, as shown in Table 10 and visualised per dimension in Figure 8.

Table 10: AttrakDiff Dimension Scores per Participant (Scale: -3 to $+3$)

PID	PQ	HQ-S	HQ-I	ATT	HQ combined
P01	1.14	1.14	3.00	1.29	2.07
P02	0.00	0.86	1.86	0.86	1.36
P03	-0.14	-1.00	1.43	0.43	0.21
P04	0.14	0.29	1.71	0.43	1.00
P05	0.86	1.43	1.86	0.71	1.64
P06	0.57	0.14	2.14	0.86	1.14
P07	0.29	0.29	1.29	0.43	0.79
P08	0.43	1.29	2.86	0.86	2.07
P09	0.29	1.29	2.86	1.00	2.07
P10	0.86	1.29	1.29	0.43	1.29
Mean	0.44	0.70	2.03	0.73	1.36
SD	0.41	0.77	0.66	0.30	–

PQ = Pragmatic Quality; HQ-S = Hedonic Quality Stimulation; HQ-I = Hedonic Quality Identity; ATT = Attractiveness; HQ combined = $(HQ-S + HQ-I)/2$.

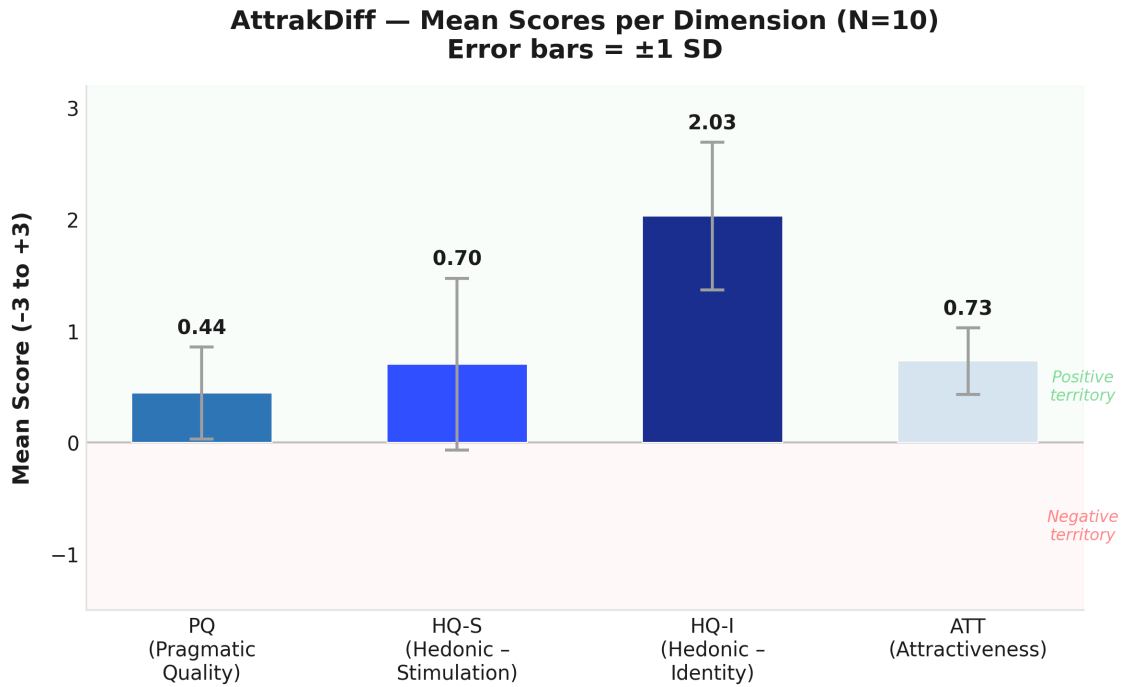


Figure 8: AttrakDiff mean scores per dimension (N=10). Error bars indicate ± 1 SD. HQ-I (2.03) substantially exceeds PQ (0.44), consistent with the hedonic quality hypothesis. All dimensions are in positive territory.

Pragmatic Quality. With a mean Pragmatic Quality (PQ) of 0.44 ($SD = 0.41$), participants generally experienced the prototype as functional and task-supportive. The only slightly negative score came from P03 (-0.14), whose session produced the most detailed facilitator notes on input field behaviour and onboarding issues. The modest magnitude of the mean is not unexpected, as the hold-to-confirm gesture introduces a small efficiency cost at the moment of prediction submission, though this does not appear to translate into a negative usability experience at the system level, as the SUS result already suggests.

Hedonic Quality – Stimulation. A mean HQ-S of 0.70 ($SD = 0.77$) came with the highest standard deviation across all AttrakDiff dimensions, which is consistent with stimulation being the measure most sensitive to prior expectations. Regular Kicktipp users such as P02 and P04 rated HQ-S lower than casual users, who brought fewer pre-formed expectations about what a prediction app should feel like. P03 was the only participant to produce a negative score at -1.00 , explicitly describing the app in their debrief as monotonous and lacking animation, a preference for richer visual interaction

that is internally consistent with a low stimulation rating.

Hedonic Quality – Identity. The strongest result across all AttrakDiff dimensions was HQ-I, with a mean of 2.03 ($SD = 0.66$). Every participant scored above 1.0 on this dimension, including P03, with P01, P08, and P09 producing particularly high scores of 3.00, 2.86, and 2.86 respectively. The relatively tight standard deviation of 0.66 suggests that participants broadly agreed that PreMatch reflects the kind of football fan they are and want to be seen as. Features such as the friend-scoped leaderboard, club logo integration, and prediction rationale prompt may all contribute to this reading by giving users a sense that their predictions are self-expressive rather than transactional (Ryan & Deci, 2000; Rigby & Ryan, 2011).

Attractiveness. Overall appeal was the most consistent result across the sample, with a mean ATT of 0.73 ($SD = 0.30$) and no participant scoring below 0.43, producing the lowest standard deviation of any AttrakDiff dimension regardless of individual differences on other measures.

Portfolio Zone. With mean PQ at 0.44 and combined HQ at 1.36, the product point falls in the Superfluous zone of the AttrakDiff portfolio, as shown in Figure 9 (Hassenzahl et al., 2003), meaning hedonic quality clearly dominates pragmatic quality. Rather than being an inadvertent outcome, this reflects a deliberate design position, since the platform was built to prioritise the quality of social and reflective engagement over transaction speed, consistent with what the SDT framework would suggest (Ryan & Deci, 2000; Hassenzahl, 2013). Both HQ-S (0.70) and HQ-I (2.03) exceed PQ (0.44), while PQ itself remains positive, a pattern broadly in line with the design intent, though the small sample size means these results should be treated as indicative rather than conclusive.

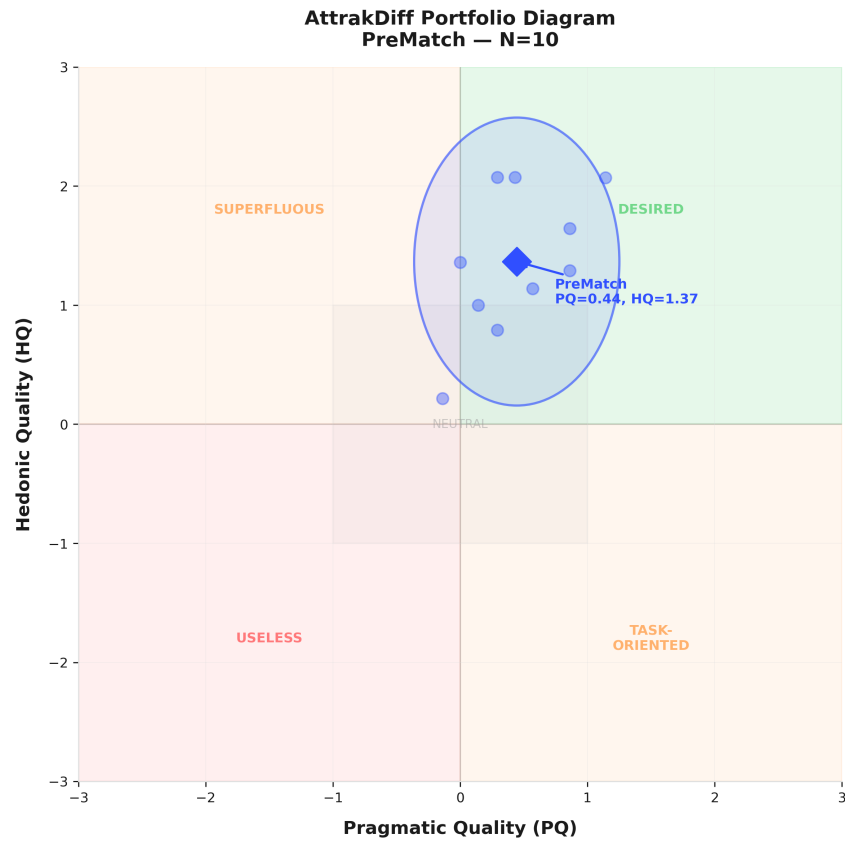


Figure 9: AttrakDiff portfolio diagram (N=10). The product point (PQ = 0.44, HQ = 1.36) places PreMatch in the Superfluous zone. Individual participant scores shown as semi-transparent dots; the ellipse represents the approximate 95% confidence region. Portfolio zone taxonomy per Hassenzahl et al. (2003).

INTUI

INTUI measures four facets of intuitive interaction, namely Gut Feeling, Magical Experience, Verbalisability, and Confidence (Ullrich & Diefenbach, 2010a), and the central question for this thesis is whether the Positive Friction interventions damage the overall sense of intuitive flow that mobile users expect. Raw items were reverse-coded where applicable and sub-scale means calculated following Ullrich and Diefenbach (2010a), with all scores on a one-to-seven scale and a midpoint of 4.0. All four sub-scales exceeded the midpoint, as shown in Table 11 and Figure 10.

Table 11: INTUI Sub-scale Scores per Participant (Scale: 1–7, Midpoint: 4.0)

PID	Gut Feeling	Magical Exp.	Verbalisability	Confidence
P01	5.00	5.80	3.25	5.00
P02	5.00	5.20	3.75	4.67
P03	5.20	4.80	5.75	4.33
P04	5.40	4.40	3.50	3.00
P05	5.00	4.40	4.25	4.67
P06	6.20	5.00	3.75	4.33
P07	5.20	3.60	5.00	4.67
P08	4.40	5.60	4.25	3.67
P09	6.00	5.20	3.50	4.33
P10	5.60	4.20	4.25	5.00
Mean	5.30	4.82	4.12	4.37
SD	0.53	0.68	0.77	0.62
vs midpoint	> 4.0 ✓	> 4.0 ✓	> 4.0 ✓	> 4.0 ✓

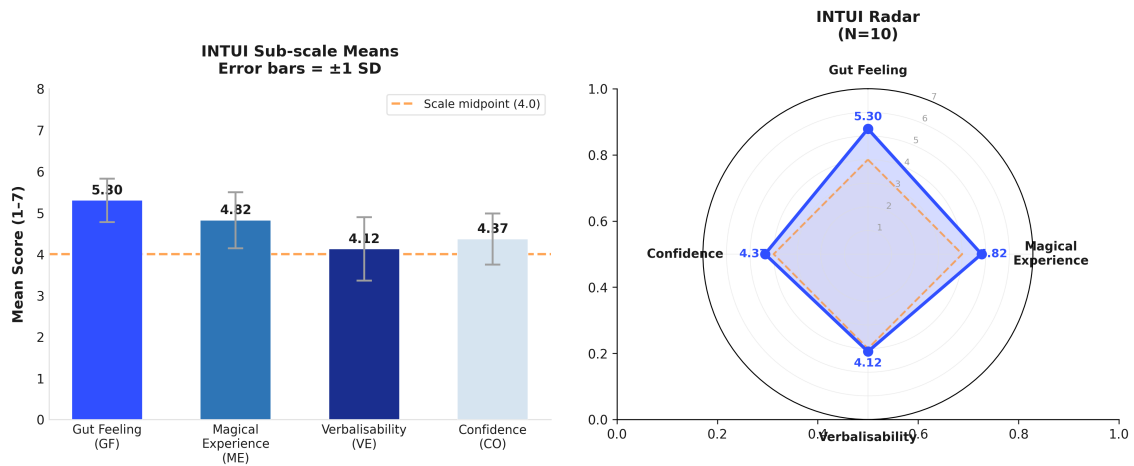


Figure 10: INTUI sub-scale means (N=10). Left: bar chart with ± 1 SD error bars and scale midpoint reference line (4.0). Right: radar chart. All four sub-scales exceed the midpoint, suggesting that positive friction does not substantially damage overall intuitive interaction quality (Ullrich & Diefenbach, 2010a).

Gut Feeling. With a mean of 5.30 ($SD = 0.53$), Gut Feeling produced the highest sub-scale result and the tightest standard deviation, with every participant scoring above 4.0. Given that this sub-scale measures effortless, automatic interaction quality, a mean of 5.30 in the presence of a deliberate friction mechanism is perhaps the most notable INTUI finding of the study, as the hold-to-confirm gesture interrupts the prediction submission flow locally but does not appear to alter the global intuitive character of the application, with participants generally experiencing the app as natural and instinctive overall.

Magical Experience. Above the midpoint at a mean of 4.82 ($SD = 0.68$), Magical Experience was nonetheless the sub-scale most likely to be affected by friction, since it captures the feeling that the interface recedes and the interaction transcends the device. Friction makes the interface briefly visible by design, so a perfect score was not expected here. Consistent with this, P07, the most experienced prediction app user in the sample, was the only participant below the midpoint at 3.60, suggesting that for a user whose mental model expects one-tap submission, the hold gesture interrupted an established routine, which is broadly what friction is intended to do.

Verbalisability. Sitting just above the midpoint at a mean of 4.12 ($SD = 0.77$), the Verbalisability sub-scale measures whether participants can articulate why an interaction felt the way it did. In frictionless products this tends to be low because System 1 processing renders decisions difficult to explain after the fact (Kahneman, 2011), making a score above the midpoint noteworthy and potentially suggesting that the rationale prompt is producing some of its intended effect. P03 scored the highest verbalisability result in the sample at 5.75, with their debrief explicitly stating that the prompt changed how they thought about their tipping behaviour, while P01 scored lowest at 3.25, a casual fan who navigated the entire session quickly without engaging deeply with the reflection step. The variance within this sub-scale reflects genuine individual differences in how participants engaged with the friction mechanism.

Confidence. At a mean of 4.37 ($SD = 0.62$), Confidence was above the midpoint across the sample with one exception. P04 was the only participant to score below the midpoint on any INTUI sub-scale at 3.00, with facilitator notes from that session recording back-navigation failures during onboarding and ambiguous step indicators that left the

participant briefly uncertain about their position in the setup flow. Removing P04 from the calculation raises the Confidence mean to 4.63, which points to the possibility that onboarding flow quality has a direct effect on downstream confidence perception, though this cannot be confirmed from a single case.

Net Promoter Score

NPS measures recommendation intent on a zero-to-ten scale, classifying respondents as Promoters (9–10), Passives (7–8), or Detractors (0–6), with the score calculated as the percentage of Promoters minus the percentage of Detractors (Reichheld, 2003). In this thesis it serves as a supplementary indicator of whether participants would actively advocate for an application of this kind within their social circle. Results are shown in Table 12 and Figure 11.

Table 12: Net Promoter Score Results (N=10, indicative)

Metric	Value
NPS Score	70
Mean rating	9.0 / 10 ($SD = 0.82$)
Promoters (9–10)	7 / 10 (70%)
Passives (7–8)	3 / 10 (30%)
Detractors (0–6)	0 / 10 (0%)
Classification	Directional result approaching world-class range (75%–80%+) per Reichheld (2003)

Note: With $N = 10$, the NPS is treated as a directional indicator rather than a statistically

generalisable metric (Reichheld, 2003).

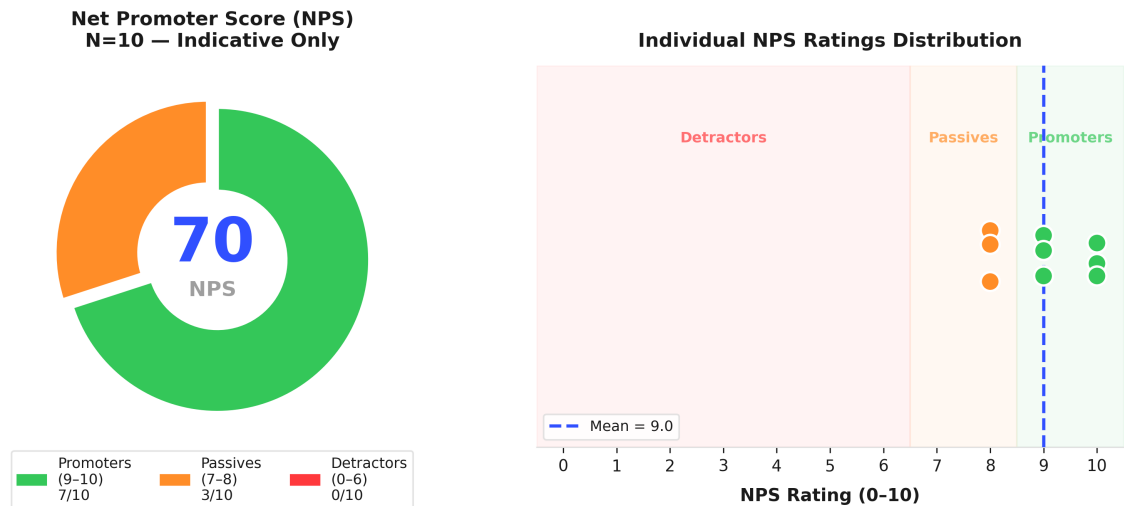


Figure 11: Net Promoter Score results (N=10). Left: donut chart showing the distribution of Promoters (7/10), Passives (3/10), and Detractors (0/10) with the NPS score of 70 at centre. Right: individual ratings distribution. No participant scored below 7, confirming zero Detractors across the full sample (Reichheld, 2003).

Seven participants scored 9 or 10 (Promoters), three scored 8 (Passives), and zero scored below 7, producing an NPS of 70 that approaches the world-class range identified by Reichheld (2003) and represents a directional result worth noting at this sample size. Most notable among the figures is the absence of any Detractors, meaning no participant indicated they would actively discourage others from using the platform.

The three Passive scores came from the most experienced prediction app users in the sample, namely P02, P07, and P10, all of whom were regular Kicktipp users. Their debrief responses are consistent with this pattern, reflecting feature requests rather than rejections. P02 wanted news integration and multi-matchday support, P07 noted that the friction broke their tipping rhythm, and P10 asked for richer social mechanics, none of which maps to would-not-recommend sentiment.

Cross-Instrument Summary

Table 13 consolidates all instrument results with benchmarks and directional assessments. The full aggregated results file is provided in Appendix A.

Table 13: Cross-Instrument Results Summary

Instrument	Dimension	Mean	SD	Assessment
SUS	Score (0–100)	90.0	6.67	Excellent (A)
AttrakDiff	PQ	0.44	0.41	Positive
	HQ-S	0.70	0.77	Positive
	HQ-I	2.03	0.66	Strong
	ATT	0.73	0.30	Positive, consistent
	Portfolio zone	Superfluous		Hedonic dominant, by design
INTUI	Gut Feeling	5.30	0.53	Above midpoint
	Magical Experience	4.82	0.68	Above midpoint
	Verbalisability	4.12	0.77	Above midpoint
	Confidence	4.37	0.62	Above midpoint
NPS	Score	70	–	Directional; approaching world-class
	Mean rating	9.0	0.82	Strong intent

Taken together, the four instruments paint a consistent picture. SUS at 90.0 places the prototype well above the accepted usability threshold, meaning usability is unlikely to be acting as a confound in the hedonic results. The AttrakDiff pattern shows substantially more hedonic value than pragmatic efficiency, most sharply on identity (HQ-I = 2.03), the dimension that maps most directly onto SDT’s autonomy and self-expression constructs (Ryan & Deci, 2000). All four INTUI sub-scales exceed the midpoint, with Gut Feeling (5.30) as the strongest result, suggesting the hold-to-confirm gesture does not substantially damage the intuitive character of the overall experience. NPS at 70 with zero Detractors points to strong recommendation intent across the sample. Taken together, these results are broadly consistent with the central hypothesis that Positive Friction can elevate hedonic quality without substantially damaging intuitiveness or usability, though the exploratory nature of the study and the sample size mean this should be treated as tentative rather than conclusive.

Qualitative Debrief: Thematic Summary

Following the psychometric instruments, each session concluded with the semi-structured debrief covering open questions O1 through O7, where the prefix O denotes open question rather than the numeral zero, as described in Section 4.3. Verbal responses were recorded across all ten sessions and transcribed in Appendix B.¹ Responses are summarised thematically below, triangulating the quantitative instrument scores with participant voice.

Positive Friction: The hold mechanic. Reactions to the hold-to-confirm gesture varied by prior app experience. Participants without established Kicktipp habits tended to accept or welcome it. P06 described it as “like a payment confirmation — that longer press makes you think for another second, or at least makes you consciously aware that you’re committing to something” (P06, l. 53.). P08 reached for a similar analogy spontaneously during the task phase: “this reminds me of Instagram’s ‘are you sure you want to post’ prompt — like a breath before committing” (P08, l. 22.). P10 reacted positively on first contact: “you have to hold to save. That’s cool” (P10, l. 19.). Among regular Kicktipp users, the response was more ambivalent. P07 accepted the mechanic in principle while registering the disruption: “I was briefly surprised by having to hold the button, but then I realised it’s actually not a bad idea — because you can quickly mis-tap or accidentally hit send” (P07, l. 34.). P05, who preferred quick picks, raised a scalability concern directly: “if I imagine tipping a World Cup with 46 teams and having to hold that button every single time, it would absolutely drive me crazy” (P05, l. 55.).

Positive Friction: The rationale prompt. The justification step produced more consistently positive responses than the hold mechanic. P02 reported that the prompt activated self-questioning: “It basically gave me the chance to question myself — whether I should spend a bit more thought on that particular match” (P02, l. 52.). The effect was not always a changed prediction: P02 noted that “it stayed the same. But it made me engage more with the data” (P02, l. 54.). P04 went further, describing a near-revision: “it made me reflect on why I’d just acted the way I did. And it actually made me reconsider the result” (P04, l. 63.). P05, despite disliking the hold mechanic, specifically approved the prompt:

¹All transcript citations follow the format (Pxx, l. n), referring to the participant ID and line number in the corresponding appendix section. P01 citations refer to written questionnaire responses, as no audio recording was available for that session.

“the justification prompts I actually thought were a cool idea — to make you think a bit more” (P05, l. 55.). P09 framed the prompt’s long-term value as retrospective data: “I’d find it really interesting to see that kind of retrospective analysis. Like: how often did you go with gut feeling and how often were you right?” (P09, l. 57.).

Autonomy and sense of control. Responses to O3 were consistently positive across the sample. P06 summarised the felt experience: “I always felt like I could go wherever I wanted. There was no dead end or loop. I was in charge throughout” (P06, l. 55.). P08 attributed this in part to the design’s visual restraint: “nothing pushed you toward one option more than another” (P08, l. 49.). The one exception was P04, whose Confidence score of 3.00 corresponded to onboarding difficulties that preceded the main task scenarios. P04 noted: “during the sign-up I did feel briefly frustrated when I had to start over — that annoyed me for a moment” (P04, l. 65.).

Competence through statistics. Responses to O4 suggested that the match data was generally experienced as competence-supporting rather than as noise. P02 described the consolidation benefit: “it’s just easier to have that directly in the app rather than opening seven different pages” (P02, l. 60.). P08 connected the data experience explicitly to competence: “it definitely made me feel more competent. And it reinforced my decision — I felt like I was better justifying my choice, and more confident that the decision was grounded in something” (P08, l. 51.). P10 positioned the app against other data platforms: “I normally use other football apps — like Fotmob or SofaScore — to look up stats. So here I don’t have to switch between apps” (P10, l. 55.). P01 noted in their written response that the statistics were “very good and informative — really helpful for someone who doesn’t know much about the sport” (P01, written response, Q4).

Relatedness and social features. Responses to O5 were the most varied of the debrief themes. P05 identified a specific relatedness function for the chat: “especially because you’re sometimes in prediction rounds with people you don’t otherwise stay in contact with, and having the option to write to them in that context is nice” (P05, l. 63.). P04 framed the ideal relatedness experience aspirationally: “when I have the app, I want to feel the way I’d feel sitting outside around a campfire with ten friends, able to talk about this app because we all use it” (P04, l. 73.). More sceptical assessments came from participants

with active WhatsApp group dynamics. P02 acknowledged the chat's quality while naming the competing channel: "the primary communication channel would remain WhatsApp when you're in a league like that" (P02, l. 64.). P10 contextualised the social features' limitations as prototype-specific: "it would feel different in a real league with real friends and actual profile photos" (P10, l. 55.).

Comparison with existing applications. Responses to O6 were notably consistent: across all ten participants, PreMatch was evaluated as preferable to Kicktipp. P10 put it most directly: "compared to Kicktipp — worlds apart. Kicktipp's design is really outdated. This feels modern and intuitive. With Kicktipp I feel like I'm using a website from 1998" (P10, l. 59.). P05 reached for similar language: "Kicktipp sometimes gives me retro vibes. This app is significantly more modern and more clearly structured" (P05, l. 65.). P03, despite being the most critical participant on hedonic stimulation, still concluded: "the app is significantly more modern. Kicktipp also has all that advertising, which is really annoying" (P03, l. 100.). P06 added the absence of advertising as a distinct differentiator: "much better. I find it much nicer than Kicktipp visually. The haptics are much more pleasant. And of course there were no ads" (P06, l. 61.). P08 offered a more concise verdict: "clearly better" (P08, l. 59.).

6 Discussion

6.1 Interpreting the Results Against the Research Questions

The three SRQs form a logical chain moving from market diagnosis through design response to empirical evaluation. Phase I answered SRQ 1, the design process of Phase II addressed SRQ 2 as a constructive claim, and Phase III tests whether that claim holds empirically. Rather than simply pointing in one direction, the results produce a specific pattern that is theoretically interpretable and worth examining in detail before drawing broader implications.

Turning to SRQ 3, which asked how the prototype performs across pragmatic and hedonic quality, intuitive interaction, and usability, the picture across all four measures is broadly positive. A SUS score of 90.0 places the prototype in the Excellent range (Bangor et al., 2008), all AttrakDiff dimensions are positive with HQ-I at 2.03 as the strongest result, all INTUI sub-scales exceed the midpoint, and NPS reaches 70 with zero Detractors. No instrument produced a finding that clearly contradicts the design intent. Within this overall positive direction, however, the shape of the results rather than their magnitude is where the more interesting theoretical observations lie, and these are taken up in the sections that follow.

Addressing SRQ 2, which asked how an MVP must be designed to balance gameful design principles with Positive Friction interventions, the design response was to treat friction not as a penalty imposed on usability but as a hedonic investment, accepting a small pragmatic cost at the moment of prediction submission in exchange for a richer sense of agency and self-expression. The AttrakDiff pattern, with PQ at 0.44 and HQ-I at 2.03, is broadly consistent with this logic, as the hedonic surplus is present and notable while the pragmatic cost appears marginal. Whether this balance was achieved by design or partly by chance cannot be determined from this study alone, but the directional result is encouraging.

Table 8 in Section 5.2 mapped each design decision to a specific theoretical construct and heuristic gap. The evaluation results now allow a provisional assessment of whether those mappings held. The hedonic surplus on HQ-I and the intact intuitiveness scores on INTUI suggest the SDT-grounded features translated into measurable experience outcomes, while the contained pragmatic cost at PQ is consistent with the Positive Friction logic the feature

set was designed around.

6.2 The Positive Friction Hypothesis: What the Data Suggests

The central empirical claim of this thesis was that deliberately slowing users down at the moment of prediction submission would elevate hedonic quality without pushing pragmatic quality below acceptable thresholds, and without substantially damaging the overall intuitive character of the interaction. Three separate instruments address this from different angles, and their results are broadly convergent.

The pragmatic cost is real but contained. PQ at 0.44 is the lowest AttrakDiff score in the study, which is not unexpected. The hold-to-confirm gesture requires a sustained press rather than a single tap, and the rationale prompt adds a step that would not exist in a frictionless design. Participants notice this. What the data tentatively suggests is that they notice it without it becoming a problem. SUS at 90.0 runs in parallel, a system-level usability score of that magnitude is difficult to reconcile with a substantially damaging pragmatic experience. The friction appears to introduce a local cost at one moment in the prediction flow without contaminating the global usability perception, though this interpretation should be treated cautiously given the sample size.

P06 articulated the hold mechanic's logic directly, comparing it to a payment confirmation: "that longer press makes you think for another second, or at least makes you consciously aware that you're committing to something" (P06, l. 53.). P08 reached for a similar analogy unprompted during the task phase: "like a breath before committing" (P08, l. 22.). Even P02, who would have preferred quick picks for casual use, acknowledged the mechanism's effect: "it basically gave me the chance to question myself, whether I should spend a bit more thought on that particular match" (P02, l. 52.).

Mecke et al. (2019) anticipated a pattern along these lines. Small motor-effort hurdles, a sustained press rather than a tap, are sufficient to interrupt automatic behaviour without being perceived as obstacles in usability testing. The PreMatch result is broadly consistent with this finding and extends it to a naturalistic prediction context with real football data, where the informational content of the rationale prompt may renew its value each matchday (Mecke et al., 2019).

Not all participants experienced the friction uniformly. P05, who explicitly preferred the speed of quick picks, raised a scalability concern: “if I imagine tipping a World Cup with 46 teams and having to hold that button every single time, it would absolutely drive me crazy” (P05, l. 55.). This concern is taken up in the implications for future design in Section 6.6.

The hedonic surplus is larger than expected. HQ-I at 2.03 out of a maximum of 3.0 is a notably strong result for a prototype evaluation. Nine of ten participants scored above 1.0 on this dimension, and the standard deviation of 0.66 indicates relatively high consistency, regardless of other differences in session experience, participants tended to feel that PreMatch reflected their sense of self as football fans. The identity dimension produced the strongest and most consistent result in the entire study.

P10, a regular Kicktipp user, pointed to what may have driven this result: “what I found really special was the design, compared to Kicktipp it’s genuinely, genuinely cool. Especially the use of club logos” (P10, l. 49.). P06 noted a similar contrast across several dimensions: “much better. I find it much nicer than Kicktipp visually. The haptics are much more pleasant. And of course there were no ads” (P06, l. 61.).

Ryan and Deci (2000) argue that autonomy satisfaction depends on the felt quality of volitional engagement, not just the presence of choice, but the sense that one’s actions are genuinely self-authored. The rationale prompt may operationalise this directly. Before a prediction is saved, users briefly articulate their reasoning, which could make the decision feel more like a statement than a tap. That the identity dimension scores substantially above pragmatic quality is consistent with this mechanism, though causal attribution is not available from this data.

HQ-S at 0.70 is positive but more variable ($SD = 0.77$). Stimulation is the dimension most sensitive to prior expectations, and the higher variance among regular Kicktipp users is plausible. They brought a higher novelty threshold and rated stimulation lower than casual users who had no established comparison point. P03 described the app as “a bit monotonous... it makes it feel a bit cold. Like it feels a bit lifeless” (P03, l. 73.), explicitly contrasting it with the micro-interactions they expected from a leisure product. P04 made a similar observation: “it’s monotone in its design. It feels more like a software application than a leisure app” (P04, l. 59.). Both were regular football prediction users

with established reference points, which is consistent with the higher novelty threshold interpretation.

Intuitive quality appears largely intact despite the friction. Gut Feeling at 5.30 is the INTUI result most relevant to the friction argument. Ullrich and Diefenbach (2010a) describe Gut Feeling as the effortless, automatic quality of interaction. A mean of 5.30 in the presence of a deliberate interruption to the automatic tipping gesture suggests the friction is experienced as local rather than global. It fires at one moment, and the rest of the interaction recovers. The hold-to-confirm does not appear to recharacterise the entire app as laborious, at least not within a single session.

Verbalisability at 4.12 adds some theoretical texture. In frictionless products, Verbalisability tends to be low because System 1 processing renders interaction decisions difficult to articulate after the fact (Kahneman, 2011). A score modestly above the midpoint may suggest the rationale prompt is producing some of its intended effect. Participants could explain their prediction decisions somewhat more readily than might be expected in a one-tap system.

P03, with the highest Verbalisability score in the sample (5.75), made the mechanism visible in their debrief: “it would’ve been cool to know in advance. Because in principle, it could be interesting to look back at my prediction reasoning over time” (P03, l. 78.). P09 extended this into a self-improvement framing: “I’d find it really interesting to see that kind of retrospective analysis. Like: how often did you go with gut feeling and how often were you right?” (P09, l. 57.). P01, a casual fan who navigated the session quickly without engaging deeply with the reflection step, scored lowest on Verbalisability (3.25). The contrast between these participant types maps loosely onto the System 1 and System 2 distinction (Kahneman, 2011). The prompt appears to activate deliberation in users who engage with the reflection step, while users who move through it quickly show less verbalisability benefit. Neither response is necessarily a failure; the design does not coerce reflection, it makes reflection available (Cox et al., 2016).

Magical Experience at 4.82 sits closest to the midpoint across all INTUI sub-scales, and this is worth accounting for. This dimension captures the sense that the interface disappears. Friction makes the interface briefly visible by design, so a lower score here relative to other sub-scales is not surprising. That it remains above 4.0 suggests the momentary

visibility of the submission mechanism does not persist substantially into the overall session experience.

6.3 The Superfluous Zone: A Deliberate Design Outcome

The AttrakDiff portfolio classification requires direct discussion. Superfluous is not, in the Hassenzahl taxonomy, an unambiguously positive label (Hassenzahl et al., 2003). It describes a product where hedonic value dominates pragmatic quality, where the experience goes beyond what the task alone demands. In consumer product evaluation this can point to over-engineering, more identity and stimulation than a product with this level of task support warrants.

For PreMatch, a result in the hedonic-dominant zone is broadly what the design was aiming for. The hold-to-confirm takes longer than a tap and the rationale prompt adds a step. These costs are not accidents; they are the mechanism. A platform designed around the argument that the quality of the prediction decision matters more than the speed of the transaction (Cox et al., 2016) would not be expected to land in the task-oriented zone. PQ at 0.44 is positive and above zero, which suggests the pragmatic function is present and acceptable. HQ-I at 2.03 and ATT at 0.73 sit substantially above it. Whether this profile is directly attributable to the friction intervention specifically, or to the visual identity, club logos, and social features, cannot be determined from this data, a point returned to in the limitations.

Hassenzahl (2013) distinguishes between pragmatic goals (achieving something) and hedonic goals (being someone). Prediction platforms built on the Kicktipp model address only pragmatic goals: enter a score, submit, move on. The PreMatch results suggest the platform at least partially addresses both, which is what an SDT-informed design would aim for (Ryan & Deci, 2000; Peters et al., 2018).

P05 captured the restraint behind the design: “it’s sufficient, it’s a football prediction app and nothing more. That’s all it needs to be” (P05, l. 55.). This response is consistent with the selective gameful design approach the MVP aimed for: hedonic depth through deliberate feature choices, not feature accumulation (Deterding et al., 2011; Tondello et al., 2016).

6.4 Social Features and Relatedness

Phase I identified Social and Community as the most severely underserved motivational category across the evaluated market, with a cross-application mean of 2.10 and SC2 (Cooperation) and SC3 (Social sharing) both at 1.25 across all four apps. The design response was to address this through friend-scoped leaderboards, group chat, shared achievement features, and prediction distribution display. The NPS and debrief results offer some qualitative evidence about how this response was received.

The zero Detractors from the instrument NPS result is the most direct indicator. No participant indicated they would actively discourage others from using the platform. Seven of ten were Promoters. The three Passive scores came from the most experienced Kicktipp users in the sample, and their debrief responses amounted to feature requests rather than rejections.

P05 identified a specific relatedness function for the chat that goes beyond casual social contact: “especially because you’re sometimes in prediction rounds with people you don’t otherwise stay in contact with, and having the option to write to them in that context is nice” (P05, l. 63.). P04 framed the aspiration more evocatively: “when I have the app, I want to feel the way I’d feel sitting outside around a campfire with ten friends, able to talk about this app because we all use it” (P04, l. 73.). Neither response suggests the kind of perfunctory engagement the benchmarked platforms’ Social and Community scores would predict.

Shapiro and Varian (1998) suggest that even a technically superior alternative may struggle to attract users away from an incumbent with strong network effects. The Teamtip benchmarking result, a platform that outperformed Kicktipp on most heuristics but holds marginal market share, is consistent with this dynamic. P02 articulated the tension directly: “the interface feels significantly better than Kicktipp . . . but I can still imagine that the primary communication channel would remain WhatsApp when you’re in a league like that” (P02, l. 64.). A participant spontaneously naming this trade-off is a plausible illustration of the lock-in mechanism at work. The social features generated appreciation in principle, but the practical cost of migration for users with established leagues elsewhere remains a real barrier.

6.5 Perspective of Experienced Users

The three Passive NPS scores (P02, P07, P10) are not noise to be explained away. All three participants were regular Kicktipp users. Their debrief responses shared a structural characteristic: they evaluated PreMatch not against an abstract standard of good design but against the established experience of their existing platform and their existing friend group workflow.

P07's response to the hold gesture was the clearest example. The mechanic's logic was accepted, but the disruption to an established routine was also registered: "I was briefly surprised by having to hold the button, but then I realised it's actually not a bad idea, because you can quickly mis-tap or accidentally hit send" (P07, l. 34.). P07's Magical Experience score of 3.60, the only INTUI sub-scale result below the midpoint in the entire sample, may reflect precisely that a user whose mental model expects one-tap submission encountering the hold as an interruption to an established routine, even when they can justify it intellectually.

Kahneman (2011) would frame this as resistance from a well-established System 1 pattern. For a habitual user, the hold gesture does not just interrupt a single prediction, it interrupts a routine reinforced across many matchdays. Whether repeated exposure to the friction mechanism would produce adaptation or sustained resistance cannot be determined from a single-session study (Cox et al., 2016). What the P07 result tentatively suggests is that the friction threshold may need to be calibrated differently for users arriving from established one-tap platforms, a point directly relevant to the adaptive friction direction outlined in the future research section.

The Kicktipp comparison across all three Passive scorers produced notably consistent language. P10 put it most directly: "with Kicktipp I feel like I'm using a website from 1998" (P10, l. 59.). P05 reached for similar terms: "Kicktipp sometimes gives me retro vibes. This app is significantly more modern and more clearly structured" (P05, l. 65.). P03, despite being the most critical participant on HQ-S, still concluded: "the app is significantly more modern. Kicktipp also has all that advertising, which is really annoying" (P03, l. 100.). Across very different responses to the friction intervention, the directional verdict on the market comparison was consistent across the sample.

6.6 Implications for Design and Research

The results point to several practical implications for both the design of social prediction platforms and the broader research agenda around Positive Friction in mobile applications.

The first concerns friction calibration for different user types. The Verbalisability variance ($SD = 0.77$) and the P07 Magical Experience score (3.60) together suggest the friction mechanism may produce different experiences depending on prior app familiarity. Casual users and new adopters appear to engage with the reflection step as a feature. Experienced prediction app users encounter it more as an interruption to a habitual pattern. P08 suggested the threshold could even be raised: “maybe it could even be pushed a little further, two seconds of holding doesn’t feel like it triggers that much in you, or maybe people have already become so desensitised to confirmation gestures from other apps” (P08, l. 47.). An adaptive friction system, one that modulates the hold duration based on observed tipping behaviour, could potentially address the range between P07 and P08 (Cox et al., 2016; Lavie & Meyer, 2010). The ethical implications of such a system are worth scrutinising. Adapting interface behaviour to individual patterns sits uncomfortably close to the personalised manipulation this thesis explicitly positions itself against (Newall, 2019). The line between responsive design and covert nudging is not always clean, and future work should treat this as a design problem rather than a given.

The second implication concerns onboarding. P04 produced the only Confidence score below the midpoint (3.00), and their session notes record back-navigation failures and ambiguous step indicators during setup. P04 described the experience directly: “during the sign-up I did feel briefly frustrated when I had to start over, that annoyed me for a moment” (P04, l. 65.). Confidence in the interaction appears to carry forward from early session experience. A participant who feels uncertain during onboarding may bring reduced trust into subsequent tasks. The current onboarding flow is listed as a high priority feature in the MVP feature map, and the P04 result is consistent with this priority being correctly assigned. A clearer step indicator, explicit back-navigation affordance, and a shorter onboarding sequence would likely raise the Confidence floor across the sample.

The third implication concerns the logo licensing constraint documented in the limitations. Club logos were identified in Phase I as a potential differentiator from Kicktipp, addressing IN2 (Thematic visual consistency) which scored the minimum possible value for the

market leader. HQ-I at 2.03 is at least partially attributable to the visual football identity the logos provide. A publicly distributed version of PreMatch could not use official club imagery without commercial licensing agreements. Whether the HQ-I result would hold if logos were replaced by colour-coded team identities or typographic treatments is an open empirical question. The identity surplus may be resilient, given that other identity-supporting features such as the rationale prompt, personalisation, and friend-scoped leaderboard are not logo-dependent. Future work should test this before any commercialisation pathway is pursued.

Taken together, these three implications do not undermine the central finding but rather refine it. The friction mechanism appears to work, though the degree to which it does depends on who is using it and in what context. What this study adds is a partial map of those conditions, making the design space around Positive Friction in mobile applications somewhat more navigable than it was before.

Note on Evaluation Scope

The results are interpreted descriptively throughout. With $N = 10$ and no comparative condition, causal attribution is not available. The limitations of the evaluation are documented in full in Section 7.2.

7 Conclusion

7.1 Summary of Contributions

This thesis set out to answer the central research question of how design principles based on SDT and Positive Friction can be embedded in a social football prediction application to produce a user experience that exceeds the current market standard. The three-phase DSR methodology produced three distinct contributions, each building on the last.

Phase I established an empirical baseline. The heuristic benchmarking of Kicktipp, Tackle, Teamtip, and Tipico using the Gameful Design Heuristics framework (Tondello et al., 2016) produced a quantifiable gap analysis across 28 motivational dimensions. The most consequential finding was not that Kicktipp performs poorly in isolation, it is that the entire market appears to fall short in the same areas. Social and Community (cross-app mean 2.10), Immersion and Narrative (2.25), and the reward structure of every social prediction platform collectively scored at or near the floor. This is not a design critique of one product but a structural description of what the current market has not built. The benchmarking data provided the feature prioritisation logic for Phase II by identifying precisely where the market has not gone and treating those gaps as design opportunities.

Phase II produced the artefact. PreMatch was designed and implemented as a functional Flutter application grounded in the three SDT constructs and the Positive Friction framework. The hold-to-confirm gesture and prediction rationale prompt operationalise the friction argument of Cox et al. (2016) in a concrete mobile interaction. The friend-scoped leaderboard, group chat, and prediction distribution display address the relatedness deficit the benchmarking identified. The visual identity built around club logos and semantic colour feedback addresses the immersion floor that Kicktipp exemplifies at a score of 1.0 on thematic visual consistency. Each design decision is traceable to a specific heuristic gap, and that traceability represents a methodological contribution in its own right, demonstrating that SDT-informed design does not require guesswork about what users need but can instead derive its requirements directly from empirical benchmarking data.

Phase III tested whether the artefact achieves its intended effect. SUS at 90.0 suggests a usability baseline that is unlikely to be acting as a confound in the hedonic perception results. AttrakDiff reveals a product that produces substantially more hedonic value relative

to pragmatic efficiency, most sharply on identity ($HQ-I = 2.03$). All four INTUI sub-scales exceed the scale midpoint, with Gut Feeling at 5.30 as the strongest result: the friction mechanism does not appear to substantially damage the intuitive character of the overall experience. NPS at 70 with zero Detractors points to broad recommendation intent across the sample. Taken together, the data is broadly consistent with the central hypothesis that Positive Friction can elevate hedonic quality without destroying intuitiveness or usability, though the exploratory nature of the study and the small sample size mean this should be treated as a tentative rather than a confirmed finding.

The practical contribution of this thesis is a replicable design template in the form of the combination of hold-to-confirm gesture, rationale prompt, and opt-in reflection layer, which together constitute an interaction pattern for ethical friction in mobile applications that is not limited to football prediction. Any domain where impulsive digital behaviour carries real costs, whether financial, social, or health-related, represents a potential candidate context for this pattern. The conceptual contribution is the empirical link between friction and identity. $HQ-I$ at 2.03 is consistent with the idea that slowing a user down at a decision point, and asking them to briefly own that decision, may not frustrate them but instead make the outcome feel more genuinely theirs. This is consistent with the autonomy principle of SDT expressed in measurement terms (Ryan & Deci, 2000), though the causal direction cannot be established from this data alone.

7.2 Limitations and Future Research

Limitations

Sample size and generalisability. The user study was conducted with $N = 10$ participants. This is consistent with the exploratory scope of a DSR-based Master's thesis and with established guidance on qualitative saturation in semi-structured interview research (Nielsen, 1994a; Guest et al., 2006). It does not support inferential statistical claims. The results from SUS, AttrakDiff, and INTUI are treated throughout as directional evidence rather than confirmed effects, and the NPS in particular is an indicator rather than a generalisable metric at this sample size. A larger and more diverse sample would be required to detect smaller effect sizes in the hedonic dimensions and to test whether the friction response varies systematically across demographic groups. The recruitment

method, personal and social media networks, also introduces a self-selection bias toward digitally active, football-interested adults. The sample is broadly representative of the target user group but not of the wider population of potential prediction app users.

Short-term evaluation and habituation. The study captures first-use impressions, and a single session of 25 to 55 minutes cannot address whether the friction mechanism retains its reflective value across repeated use. Cox et al. (2016) identify this as a fundamental open question in positive friction research, since a mechanism that produces System 2 engagement on first encounter may eventually be automated into a System 1 routine through practice (Kahneman, 2011). The P07 result, a Magical Experience score of 3.60 from the most experienced prediction app user in the sample, hints at what this might look like over time. Whether that response persists, adapts, or inverts across a full season of use cannot be determined from this data.

Absence of a control condition. The study evaluates the complete designed system. No version of the app without the friction features was tested against the friction version. This means the hedonic gains observed in AttrakDiff cannot be attributed to the friction mechanism specifically. Club logos, friend-scoped leaderboards, and the visual identity all plausibly contribute to HQ-I. Disentangling these contributions would require a controlled experimental design that falls outside the scope of a DSR evaluation study (Hevner et al., 2004). The causal claim available here is modest but honest: the complete artefact produces a hedonic and usability profile broadly consistent with its design intent. Which specific design decisions are doing the most work remains a question for future research.

Club logo licensing. The MVP uses official club crests sourced through API-Sports (API-Sports, 2024) for research purposes. A publicly distributed version could not do so without commercial licensing from the Deutsche Fußball Liga, the Premier League, UEFA, and other rights holders. This is a real barrier for any commercialisation pathway. The licensing constraint matters for the evaluation results because club logos were a deliberate design response to IN2 (Thematic visual consistency), which Kicktipp scored at the minimum possible value. Whether the HQ-I result of 2.03 would hold if logos were replaced by colour-coded team identities or typographic treatments is an open empirical question. The identity surplus may be resilient, given that the rationale prompt, personalisation features,

and friend-scoped leaderboard are not logo-dependent. Future work should test this before any commercialisation step is taken.

Network effects and switching costs. The thesis assumes that a better product can attract users away from Kicktipp, though the Teamtip benchmarking result complicates this assumption. Teamtip outperformed Kicktipp on most heuristics and yet holds only marginal market share, suggesting the reason is structural rather than design-related, as switching prediction platforms means persuading an entire friend group to move together (Shapiro & Varian, 1998). A user who leaves Kicktipp does not just change apps, they leave behind years of shared league history, established scoring rules, and the social routines built around matchday tipping. The NPS result suggests participants would recommend PreMatch. Whether they would actually migrate, and whether they could bring their group with them, is not something this study measures. It is probably the most important practical question the platform faces.

Future Research

Longitudinal field study across a full season. The friction habituation question can only be answered with time, and a field study in which a cohort of participants uses PreMatch across an entire Bundesliga season of 34 matchdays would generate the temporal data this study cannot produce. The key outcome variable would be the trajectory of the Verbalisability score, specifically whether the rationale prompt continues to surface genuine deliberation after matchday ten, twenty, or thirty-four, or whether it becomes a rote step that participants complete without engagement. A repeated-measures design administering SUS, AttrakDiff, and INTUI at three time points across the season would allow these trajectories to be tracked and compared against the first-use baseline this study provides.

Adaptive friction. A fixed hold duration and a fixed rationale prompt treat all users identically, yet the data suggests they are not, with P07 and P03 representing opposite ends of the friction response spectrum, one disrupted and one enriched by the same mechanism. A system that calibrates the friction level to observed tipping behaviour, applying a longer hold for impulsive patterns and a lighter prompt for more deliberate ones, could potentially address this range (Cox et al., 2016; Lavie & Meyer, 2010), though the ethical stakes of

such an approach are not trivial. Adapting the interface to individual behavioural patterns sits uncomfortably close to the personalised engagement manipulation this thesis explicitly opposes (Newall, 2019), making the tension between responsive design and covert nudging a research problem worth investigating in its own right.

Logo-free visual identity. A licensing-compliant version of PreMatch would need an alternative to club crests. Candidate approaches include colour-coded team systems derived from official club palettes, typographic treatments using abbreviated team identifiers, and abstract icon systems that suggest team identity without reproducing protected imagery. A comparative evaluation of these alternatives against the logo-present condition used in this study would help establish whether the hedonic identity result is driven by football visual culture specifically or by the broader sense of personalisation the logos facilitate. The answer has direct design implications for any commercialisation effort.

Personality moderators and user type differentiation. The sample is treated as broadly homogeneous in its response to friction. The data suggests it is not. A user with high football domain expertise may find the rationale prompt valuable precisely because it externalises a reasoning process they would have run anyway. A casual fan tipping primarily for the social dimension of a friend group may experience the same prompt as an unwelcome imposition on a low-stakes activity. Future work should incorporate validated user-type measures, such as the Hexad scale (Tondello et al., 2019) or established impulsivity instruments, to identify which profiles benefit most from the friction intervention and which require a lighter layer. This would contribute a more precise account of when Positive Friction helps, when it is neutral, and when it hinders.

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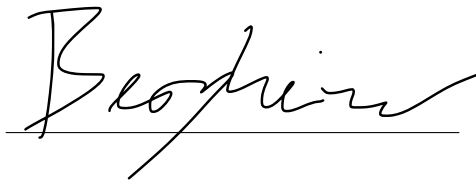
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Affidavit

I hereby declare, under oath, that this Master's Thesis has been my independent work and has not been aided with any prohibited means. I declare, to the best of my knowledge and belief, that all passages taken from published and unpublished sources or documents have been reproduced whether as original, slightly changed or in thought, have been mentioned as such at the corresponding places of the thesis, by citation, where the extent of the original quotes is indicated.

The paper has not been submitted for evaluation to another examination authority or has been published in this form or another.

Innsbruck, June 27, 2026

A handwritten signature in black ink, appearing to read 'Bergheimer', written over a horizontal line.

Erik Bergheimer

AI Disclaimer

Documentation of the use of AI-supported tools

The following AI-supported tools were used for the following purposes.

AI-supported tool	Type of support	Contribution
Claude	Translation	Translating the transcripts
Antigravity, Claude Code	Code Generation & Development	Writing the Flutter code
Claude	Paraphrasing & Editing	Rewriting sentences out of sources
Gemini, Claude	Research & Analysis	Deep research
Claude	Content Summarization	Summarizing content out of sources

I hereby declare that when AI has been used not only as a tool but also as a source, APA citation guidelines have been applied. All content generated using AI tools and included in the thesis, whether as direct or indirect citations, is clearly identified in accordance with APA citation style.



Appendix

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A Study Data and Materials

The following links provide access to the raw data collected during the user study. All files are stored under pseudonymised participant IDs (P01–P10) in line with the data protection framework described in Section 4.3.

Individual Participant Response Sheets

- P01 Session Report
- P02 Session Report
- P03 Session Report
- P04 Session Report
- P05 Session Report
- P06 Session Report
- P07 Session Report
- P08 Session Report
- P09 Session Report
- P10 Session Report

Aggregated Results

- Google Forms Raw Results
- Full Results File (all instruments, calculated scores, cross-instrument summary)

Study Materials

- Participant Information Sheet and Consent Form
- Study Session Protocol and Task Scenarios

Transcripts for all ten sessions are provided in Section B.

Design and Code

- Figma Design File (PreMatch Prototype)
- Application Source Code
- Setup and Deployment Instructions (README)

B Session Transcripts

The following transcripts document the verbal data collected during the ten user study sessions across the think-aloud task phase and the semi-structured debrief (O1–O7). Facilitator prompts are marked with **F:**, participant responses with **P:**. Observer notes and contextual remarks are shown in *italics*. Transcripts have been lightly edited for readability: filler words and false starts removed, all substantive content preserved verbatim.

All ten sessions were conducted in German. Audio was recorded using the Apple Voice Memos application and subsequently transcribed and translated into English using Claude (Anthropic), with the translated output reviewed and verified by the primary researcher to ensure semantic accuracy and preservation of conversational tone. All inline citations in the thesis text reference the translated and researcher-verified versions.

Note on P01: No audio recording was available for the P01 session due to a technical failure. The P01 transcript has been reconstructed entirely from the researcher’s observer notes entered in Google Forms and P01’s written questionnaire responses. Verbal exchanges are not documented; task behaviour is recorded as observer notes only. Citations to P01 refer to written questionnaire responses rather than spoken utterances and are marked accordingly as (*P01, written response, Qn*).

Transcript citations follow the format (*Pxx, l.~n*), referring to the participant ID and line number within the corresponding appendix section. For example, (P03, l.~47) refers to line 47 of the P03 transcript in Appendix B.3.

B.1 P01 – Session Transcript

In-Person Session | 16.05.2026 | Language: German | Note: No audio recording available. Transcript reconstructed from researcher observer notes and written questionnaire

responses only.

PreMatch User Study | In-Person Session | 16.05.2026, 05:45

Note: No audio recording is available for this session. This transcript has been reconstructed from the researcher's observer notes entered in Google Forms and P01's written questionnaire responses. Verbal exchanges are not documented; task behaviour is recorded as observer notes only.

BRIEFING & CONSENT

[Observer note: Standard briefing delivered. P01 gave consent to all five consent items (Yes / Ja).]

[Observer note: Session language: German. Mode: In-person. Session start: 05:45.]

[Observer note: Screen recording started.]

TASK 1 – Account Creation & Joining a League

[Observer note: Observer note: Email sign-up found and used without difficulty. Password icon noticed and used. Language set to German. Age entry worked well. P01 selected quick picks over deliberate tipping. Proceeded through onboarding quickly overall.]

TASK 2 – Exploring the App & Placing a Prediction

[Observer note: Observer note: P01 navigated to match details. Found win probability useful. Described the spider/radar diagram as 'nice and helpful.' Squad information also noted positively. Found the prediction input element directly and intuitively. Held the button to confirm and selected statistics as justification. Task completed successfully.]

TASK 3 – Checking the Leaderboard

[Observer note: Observer note: Navigated immediately to My League. Found position 1 with 0 points. Reacted with amusement ('lol').]

TASK 4 – Finding & Changing a Setting

[Observer note: Observer note: Initially confused when looking for the prediction setting. Eventually found and changed it.]

TASK 5 – Viewing Other Players' Predictions

[Observer note: Observer note: Tapped on a player's name in the leaderboard —

nothing happened. Became confused and clicked around without finding the Tips section. Did not find it independently.]

TASK 6 – Sending a Chat Message

[Observer note: Observer note: Found the chat quickly and sent a message without difficulty.]

TASK 7 – Inviting a Friend

[Observer note: Observer note: First tapped the button in the top centre of the screen, then navigated to Settings. Eventually found the correct button (top right in My League) after some searching.]

QUESTIONNAIRE RESPONSES

AttrakDiff — Pragmatic Quality (PQ)

PQ1 (obstructive–supportive): 5 (1 = obstructive, 7 = supportive)

PQ2 (complicated–simple): 2 (1 = complicated, 7 = simple)

PQ3 (inefficient–efficient): 6 (1 = inefficient, 7 = efficient)

PQ4 (confusing–clear): 7 (1 = confusing, 7 = clear)

PQ5 (boring–exciting): 2 (1 = boring, 7 = exciting)

PQ6 (unruly–manageable): 7 (1 = unruly, 7 = manageable)

PQ7 (impractical–practical): 7 (1 = impractical, 7 = practical)

AttrakDiff — Hedonic Quality: Stimulation (HQ-S)

HQ-S1 (conventional–inventive): 2 (1 = conventional, 7 = inventive)

HQ-S2 (unimaginative–creative): 7 (1 = unimaginative, 7 = creative)

HQ-S3 (cautious–bold): 5 (1 = cautious, 7 = bold)

HQ-S4 (conservative–innovative): 4 (1 = conservative, 7 = innovative)

HQ-S5 (ordinary–novel): 7 (1 = ordinary, 7 = novel)

HQ-S6 (undemanding–challenging): 4 (1 = undemanding, 7 = challenging)

HQ-S7 (dull–captivating): 7 (1 = dull, 7 = captivating)

AttrakDiff — Hedonic Quality: Identity (HQ-I)

HQ-I1 (isolating–connective): 7 (1 = isolating, 7 = connective)

HQ-I2 (unprofessional–professional): 7 (1 = unprofessional, 7 = professional)

HQ-I3 (tacky–stylish): 7 (1 = tacky, 7 = stylish)

HQ-I4 (cheap–premium): 7 (1 = cheap, 7 = premium)

HQ-I5 (alienating–integrating): 7 (1 = alienating, 7 = integrating)

HQ-I6 (separates me–brings me closer): 7 (1 = separates me, 7 = brings me closer)

HQ-I7 (unpresentable–presentable): 7 (1 = unpresentable, 7 = presentable)

AttrakDiff — Attractiveness (ATT)

ATT1 (unpleasant–pleasant): 1 (1 = unpleasant, 7 = pleasant)

ATT2 (ugly–beautiful): 7 (1 = ugly, 7 = beautiful)

ATT3 (disagreeable–likeable): 7 (1 = disagreeable, 7 = likeable)

ATT4 (rejecting–inviting): 7 (1 = rejecting, 7 = inviting)

ATT5 (bad–good): 1 (1 = bad, 7 = good)

ATT6 (repelling–attractive): 7 (1 = repelling, 7 = attractive)

ATT7 (discouraging–motivating): 7 (1 = discouraging, 7 = motivating)

INTUI

INTUI1: 4 (1 = 1, 7 = 7)

INTUI2: 6 (1 = 1, 7 = 7)

INTUI3: 1 (1 = 1, 7 = 7)

INTUI4: 4 (1 = 1, 7 = 7)

INTUI5: 7 (1 = 1, 7 = 7)

INTUI6: 2 (1 = 1, 7 = 7)

INTUI7: 6 (1 = 1, 7 = 7)

INTUI8: 1 (1 = 1, 7 = 7)

INTUI9: 1 (1 = 1, 7 = 7)

INTUI10: 7 (1 = 1, 7 = 7)

INTUI11: 1 (1 = 1, 7 = 7)

INTUI12: 7 (1 = 1, 7 = 7)

INTUI13: 1 (1 = 1, 7 = 7)

INTUI14: 1 (1 = 1, 7 = 7)

INTUI15: 7 (1 = 1, 7 = 7)

INTUI16: 1 (1 = 1, 7 = 7)

INTUI17: 7 (1 = 1, 7 = 7)

SUS (System Usability Scale, 1–5)

I think that I would like to use this system frequently: 5 (1 = strongly disagree, 5 = strongly agree)

I found the system unnecessarily complex: 1 (1 = strongly disagree, 5 = strongly agree)

I thought the system was easy to use: 5 (1 = strongly disagree, 5 = strongly agree)

I think I would need support from a technical person to use this system: 1 (1 = strongly disagree, 5 = strongly agree)

I found the various functions were well integrated: 5 (1 = strongly disagree, 5 = strongly agree)

I thought there was too much inconsistency in this system: 1 (1 = strongly disagree, 5 = strongly agree)

I would imagine most people would learn to use this very quickly: 5 (1 = strongly disagree, 5 = strongly agree)

I found the system very cumbersome to use: 1 (1 = strongly disagree, 5 = strongly agree)

I felt very confident using the system: 5 (1 = strongly disagree, 5 = strongly agree)

I needed to learn a lot before I could get going with this system: 1 (1 = strongly disagree, 5 = strongly agree)

NPS

Score: 10 / 10

Reason: "Because you can easily invite people with a link, you can play with people who don't know football well, and then the statistics help them feel more competent."

DEMOGRAPHIC INFORMATION

Age: 22

Gender: Woman / Frau

Football interest: Casual fan / Gelegenheitsfan

Prior prediction apps: None / Keine

Frequency of use: Never used one / Noch nie genutzt

Smartphone OS: iOS

Technical competence (self-rated): 2 / 5

Has used a betting app: No

OPEN QUESTIONNAIRE RESPONSES (written)

Q1 — Overall impression:

"The app is very clean and well-coordinated — everything is very consistent. Very intuitive. Icons are well chosen. Inviting people wasn't found immediately — maybe make that element bigger? Radar chart — how are goals conceded / goals scored displayed? Positive or negative?"

Q2 — Moments of deliberate friction (justify/hold to save):

"Good to justify the prediction — gamification — am I actually sure about this prediction? Take a moment to think about why you tipped that."

Q3 — Sense of control:

"No, there were no such moments."

Q4 — Match data & statistics:

"Very good and informative — really helpful for someone who doesn't know much about the sport."

Q5 — Social features (leaderboard / chat):

"Didn't feel strongly connected — but the prediction username in particular was a nice touch."

Q6 — Comparison to Kicktipp / other tools:

"—" (no prior experience with prediction apps)

Q7 — Anything else to share:

"I really like the app! Great for people who don't know much about the Champions League or similar tournaments. Information is well prepared."

— End of transcript —

This transcript was reconstructed entirely from researcher observer notes (Google Forms) and written questionnaire responses. No audio recording was available for this session.

B.2 P02 – Session Transcript

In-Person Session | 18:21 | Language: German

PreMatch User Study | In-Person Session | 18:21

BRIEFING & CONSENT

1. **F:** So what this is about is a prototype — basically a walkthrough of a new social football prediction app. I’m investigating how principles of self-determination theory and positive friction can be integrated into interaction design in order to create a better, more engaging, and ethically responsible user experience. The idea is that you’ll receive a series of realistic tasks within this app prototype. The important thing is that while you interact with it, you think out loud — describe what you see and what you’d expect. After that, you’ll get a questionnaire with some questions to answer — mostly scales between 1 and 7 or 1 and 5. And then a short, open conversation about your experiences and a bit more detail.
2. **F:** The time estimate is very rough — it definitely won’t take the full 45 minutes. The important thing to remember is that the app is being tested, not you. That means there are no right or wrong answers. If you think something is confusing, that’s okay, and you’re allowed to say so. That’s not a criticism of me personally — the criticism is directed at the product. If something doesn’t work the way you expect it to, give that feedback. It helps improve the design, and the more honest the reaction, the more valuable it is.
3. **F:** These are the data protection details. If you’d like, you can read through them. It covers the fact that I pseudonymize — I use participant IDs. I also record the session. None of it will be published. Quotes will be anonymized as well, and data collection follows GDPR. You can stop the session at any time, no reason required, and if you have questions afterwards or want to know the results, you can contact me via email. If you’d like to read through all of this and agree to it, you can just click through and accept.

[Observer note: Pause — P02 reads and accepts consent form]
4. **F:** Good, wonderful. We’ll start now. [Notes: Language: German, In Person, Time: 18:21] So, in this part of the study you’ll be asked to complete a series of tasks in the app. Please think out loud, describe what you see as I mentioned, and try to do what

you'd expect. There are no right or wrong answers. Take the time you need. I'll type a few things into the answer fields — you don't need to do that. Feel free to read the text again. And now I'll hand you the app.

TASK 1 – Account Creation & Joining a League

5. **F:** This is the app. Your first task: you open the app for the first time. Please create an account with your name, a fictional prediction username, the email address eb2000@gmx.de, and then join the existing league with the invitation code YW5UH. It's also important that when the settings come up later, you accept them as they're currently set.
6. **P:** So I'm opening the app, and there's a home screen with 'Get Started'. There's also some subtext you can read through, before you get three options regarding sign-up — and if you already have an account, a login. So here I'd go with 'Sign up with your Email'. And there I'd enter the email address eb2000@gmx.de. I can create a password — I'll redo that — and then 'Continue'. That takes a little moment.
7. **P:** Now I get to select my language — German or English — I'll go with German. My name, my full name — I'll type the full name first, but I have a short nickname for myself. Then it asks for my age. My national team that I'd like to support — Germany as well. And I can pick a favourite league from six options — the top five and Austria. I'd assume the Austrian one is only included because we're in Austria — otherwise nobody would need it. [laughs] And my favourite club — I don't want to share that publicly.
8. **P:** Now I can join a league or create one. I'd like to join a league with a code: YW5UH. And I can join.
9. **F:** How do you prefer to make predictions — quick picks or deliberate picks?
10. **P:** I actually much prefer quick picks, because I like to do it casually on the side.
11. **F:** [Notes selection] — Right. You're now in Standard Mode, where the exact score system works. Okay, let's go.
12. **P:** Okay, that works for now.
13. **F:** Wonderful. You can set that down. Thank you very much. That was the first scenario. I need to adjust one small thing — to make a later scenario make sense. [Takes device briefly, changes a setting] Exactly.

TASK 2 – Exploring the App & Placing a Prediction

14. **F:** The next task: you're now in the Champions League prediction league. Just take a look around the app — upcoming matches, past matches, whatever you like. Get an overview of the Champions League fixtures and all the information the app provides, and then find the Champions League Final between Paris Saint-Germain and Arsenal London, and place a prediction for that match. Just follow all the steps the app shows you.
15. **P:** I'm on the home screen. At the top we have the league selector — let me take a look. Champions League or a new league — I'll stay in the Champions League. Matchday points, prediction accuracy 0% — great. Prediction accuracy — one out of one correct. Points per matchday. Next match: Paris vs Arsenal. My predictions. Okay, here I can see every single matchday — that's cool — with the results. I can also go into match details — very cool. Club position — is that the position based on the matchday, or...? After eight matches. Line-up, head-to-head — okay. Are those the last five times they've ever played each other? [looks] Top scorer and Player of the Tournament — do you probably have to predict those beforehand? League, Final. My league — I'm in first place! [laughs] As it should be. Erik in last — last matchday, top 3 predictions — round of...
16. **P:** Right, I actually wanted to look at the app in more detail first. There's a chat. [navigates] And my predictions. [tries to submit a prediction] You really have to hold it for a long time.
17. **F:** 'Why this prediction?'
18. **P:** Gut feeling, I'd say. [looks at match statistics] The next match — some metrics stronger... So here it's like: further out means Paris has more goals conceded — so a larger weakness in that aspect. If Arsenal went further out and were larger, they'd have a greater strength in that aspect.
19. **F:** Okay, wonderful. You can set that down. Next scenario: find out what position you're currently in within the league, and how many points separate you from the next person.

TASK 3 – Checking the Leaderboard

20. **P:** I go to my league and can see right away that I'm in first place — and three points

separate me from the next person. [looks] In the last match I scored zero points — what a joke. [laughs]

TASK 4 – Finding & Changing a Setting

21. **F:** You remember that the app asked you during onboarding — where you clicked 'Quick Predictions' — to justify your prediction. I've now changed that setting before the prediction, in order to test the justification feature. Your task: find this setting and change it to the opposite of the current selection.
22. **P:** [navigates to settings] 'Prediction Settings' — okay, done.
23. **F:** I'd also like to put the app in dark mode — it has a few issues, but you're welcome to try it.
24. **P:** No, leave it.

TASK 5 – Viewing Other Players' Predictions

25. **F:** Second-to-last scenario: find out what the other people in your league predicted in the last matchdays.
26. **P:** I go to my league. Last matchday. And then I'll go to 'Matchday Predictions' and then probably scroll down to 'Predictions'. [looks]
27. **F:** You're already in the right place — there are simply no predictions for the last matchdays. The other players didn't make picks in the recent rounds. You're in the right spot, there's just no data. Feel free to navigate to the semi-final — there should be predictions there.
28. **P:** [navigates] Right, everyone made predictions there — I can see.

TASK 6 – Sending a Group Chat Message

29. **F:** Then that's done. Second-to-last task: write a message in the Champions League group chat about something that's on your mind right now.
30. **P:** I need to find the chat first. [navigates] League group chat — cool. [types message]

TASK 7 – Inviting a Friend

31. **F:** Last task: you'd like to invite a friend to your league. Find the invite function and share the invitation code.
32. **P:** My league — top right. [brief pause] Cool.
33. **F:** Done. Those were your tasks. Now you'll receive some questionnaires with scales

to rate the product experience.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

34. **P:** (human / technical) More on the human side — also because of the chat, relatively low on the technical side. (isolated / connected) Connected, because I joined a league — the community. (impractical / practical) Practical, because I like predicting football. (unstructured / structured) More on the structured side. (unpredictable / predictable) Predictable — what needs to happen is very easy to follow. (conventional / novel) Novel, because it combines things. (timid / bold) Bold, but... could be bolder. (uncreative / creative) Also creative — the spider web design is something you rarely see. (repelling / attracting) Attracting is good. (demotivating / motivating) Motivating, as long as I'm in the lead — if I were at the bottom, it'd be demotivating.

INTUI (Sentence Completion)

35. **P:** (deliberate / spontaneous) Deliberate — I have to think about who plays how and against whom. (ease of use) Ease — well, both, but... (magical experience) Okay...

SUS (System Usability Scale, 1–5)

[Observer note: P02 completes SUS questionnaire]

NPS

36. **F:** Last question: How likely is it that you would recommend this app to a friend or colleague who is interested in football predictions — and why?

[Observer note: P02 responds — exact wording not clearly recorded]

DEMOGRAPHIC INFORMATION

37. **F:** Do you also want to add the line-up feature?
38. **P:** Yes, that's planned.
39. **F:** Your age?
40. **P:** 25.
41. **F:** I could've guessed — the youth factor. [laughs] You're a passionate football fan — I know you. Which prediction apps have you used before?
42. **P:** [names apps, e.g. Kicktipp] ... and KickBase.
43. **F:** Frequency of use. Smartphone OS. How would you rate your technical compe-

tence?

44. **P:** A four out of five.

45. **F:** Have you used betting apps before?

46. **P:** Yes. [names betting app]

OPEN INTERVIEW

47. **F:** Okay. Now we'll go through the open questions — and after that you're done. Looking back at the entire session: what's your overall impression of the app? What stands out the most to you — positive or negative — just your honest, open opinion?

48. **P:** The positive is the ease of use. It was quite easy to find the league and place predictions. I liked the prediction flow for individual matchdays. But I don't know how it would've felt with multiple matchdays in a row — I found the overview briefly a bit tricky at one point... not tricky, but...

49. **F:** Which overview do you mean?

50. **P:** When I had to find the past predictions — there was only one result entered. Other than that, I found the interface nice. Clean and well-structured. I'm not a big fan of blue — but cool.

51. **F:** There were moments where the app deliberately slowed you down and prompted you — for example, to justify your prediction or to hold the button to confirm it. Did those moments feel special in any way, or did you actually find them annoying and unnecessary — or did they change something about how you interacted?

52. **P:** I didn't find it annoying. But it made me think again about whether that should really be my prediction. It basically gave me the chance to question myself — whether I should spend a bit more thought on that particular match.

53. **F:** Did that make you change your prediction, or did it stay the same?

54. **P:** It stayed the same. But it made me engage more with the data.

55. **F:** So the prediction stayed the same, but you just then spent more time with the data?

56. **P:** I think for other matches I'd probably go deeper into the analysis.

57. **F:** Did you feel in control of your user experience the whole time — or were there moments where you felt pushed in a direction you didn't choose?

58. **P:** In control.

59. **F:** Did the match data and statistics make you feel more competent and better informed

- when predicting — or was it more of a 'nice to have' or just extra noise?
60. **P:** It made me feel more competent. But I also like sites like Insider and everything else where you can get expert views and information. So for me it's just easier to have that directly in the app rather than opening seven different pages.
61. **F:** Do you think it could also be helpful for people who aren't as deeply into football and statistics?
62. **P:** The statistics can be very helpful, yes. Even for someone who's not that interested in football.
63. **F:** Did the social features — like the leaderboard or the chat — feel particularly special, or did they make you feel more connected to the other players — or was it more of a 'nice to have'?
64. **P:** I'd say more 'nice to have', simply because it would essentially be yet another social media channel. The chat feature is very well integrated — I can imagine it would be used. But I can still imagine that the primary communication channel would remain WhatsApp when you're in a league like that.
65. **F:** How does the app compare to apps like Kicktipp or other prediction tools you know? Does anything feel better? Worse?
66. **P:** The interface feels significantly better than Kicktipp, for example. And with Kicktipp you can really only predict — that's it. It's nice to have something like a daily information platform. What would also be cool would be something like news — news about your favourite team that you've set, or about all the clubs that are relevant in the league you're predicting. So: if I'm predicting the Champions League, I want to get news about the teams still in the Champions League.
67. **F:** As an example — you saw the squad section in the app. That was a sample implementation that hasn't been fully tested yet, but is planned: when 48 hours before a match the clubs are required to publish their injury report, that would also be pushed to users via push notification. Is that the kind of thing you mean — or more transfers?
68. **P:** Not transfers — there are too many other platforms competing for that on social media. More match-relevant stuff: injuries, manager changes, that kind of thing.
69. **F:** Okay. Last question: is there anything else you'd like to share — about the app, the experience, anything we haven't talked about yet or that just came to mind?

70. **P:** Very nice. Cool thing, man. [laughs]

71. **F:** Thank you very much.

72. **P:** You're welcome.

— *End of transcript* —

Note: Some passages (e.g. during SUS completion, specific app/betting names mentioned) were unclear in the original recording and have been summarised or marked accordingly.

B.3 P03 – Session Transcript

In-Person Session | Language: German

PreMatch User Study | In-Person Session

BRIEFING & CONSENT

1. **F:** Welcome to the user study. You've been invited to take part in my study for my Master's thesis at MCI, which involves evaluating a prototype I developed — a football prediction app. I'm investigating how self-determination theory, also called SDT, and positive friction can be implemented in interaction design. You'll be given a series of tasks in the prototype. The important thing is that while you interact with it, you think out loud and describe what you see. After the tasks, you'll fill out a questionnaire. Then we'll briefly talk about your experience and potential improvements.
2. **F:** The important thing is that the app is being tested, not you. There are no right or wrong answers. If something is confusing or doesn't work the way you'd expect, that's exactly the feedback I need to improve the app and the design.
3. **F:** Regarding data protection: all data is pseudonymised under a participant ID — for example P01 — and no names are recorded. Session recordings are used exclusively for research analysis and will not be published. Anonymised quotes will be used in the Master's thesis. Data processing follows GDPR, and you can request deletion of your data at any time by emailing me. Participation is completely voluntary. You can end the session at any time, without giving a reason and without any consequences. You can also withdraw your data after the session by contacting me. Results will be shared as a Master's thesis at MCI Innsbruck, expected this autumn, probably 2026. If you'd like a summary of the findings, please let me know. If you have any questions before or after, feel free to reach out.
4. **F:** Now I'll give you a few data protection consent questions, which you can answer with 'yes' if you understood everything I just said.
[Observer note: P03 answers consent questions]
5. **F:** Okay. In this part of the study you'll be asked to complete a series of tasks in the app. Please remember to think out loud, describe what you see and what you're trying to do and expect. There are no right or wrong answers. Take the time you need. I'll

start the screen recording now. Three, two, one. Let's go.

TASK 1 – Account Creation & Joining a League

6. **F:** The first task: open the app — you already have it open — and create an account with your name, a fictional prediction username, and the email address eb2000@gmx.de. Then join the existing league with the invitation code YW5UH.
7. **P:** Okay. So I need to create an account or log in.
8. **F:** You need to create a new account.
9. **P:** Okay, then — because I'm a lazy dog — I'd press Google. [laughs]
10. **F:** You need to sign up with the email address given here.
11. **P:** Okay, I can do that. [types] Oops, typo. Another one.
[Observer note: P03 types the email address; makes a few typos]
12. **P:** So I've gone with the password 123456. [laughs] And I'm in. First piece of feedback right away — that password obviously isn't secure. It's only six characters, no special character, and it's a sequential string from 1 to 6. You could think about a decent password strength check.
13. **P:** Language — German, because German is the best. What's my name? Marc. Prediction username: Mr. Tipp. I'm in.
14. **P:** Oh, let me quickly turn the sound on. There's no sound. For example, right here — this would be a good opportunity to use Apple's Haptic Engine natively. Also, I notice you're not using a gradient overlay — the number snaps hard to the new opacity. You could do a smooth overlay instead. And in my opinion, these blue bars in the middle are too much. I actually think the bars would look even cooler if they were removed entirely.
15. **P:** Okay, enough feedback — let me enter my age. Very cool though, so far. Nice. My national team: Germany, obviously. Favourite league: Bundesliga. Favourite club: St. Pauli.
16. **P:** Now I want to join a league. I have an invitation code from my mate — I'll enter that now. [types code] There's no league for this code. Wait, I entered the code wrong. [tries again] Damn. Also, this is a bit annoying — I can't tap directly into the input field.

[Observer note: P03 finds the Champions League and joins with the correct code]

17. **P:** Found it — Champions League, I'd guess. And the validation is happening in real time too.
18. **F:** You're a user here, not a developer — so no security comments for now. [laughs]
19. **P:** Right, right. Okay.
20. **F:** How do you prefer to make predictions — quick picks or deliberate picks?
21. **P:** I'd say here it would be cool if I could tap some kind of info icon that opens a pop-up — maybe a scrollable view with a few UX examples or screenshots showing what the difference is, so I can better picture it. But honestly, I'd go with quick picks.
22. **F:** You can select 'deliberate' for now — but it's good to know you'd naturally choose quick picks.
23. **P:** Okay, I'm in. So: Standard Mode. New result in advance. Points for correct result or correct tendency. Perfect for a quick round. Tips, table, league settings. Let's go.
24. **F:** Scenario 1 done. Great.

TASK 2 – Exploring the App & Placing a Prediction

25. **F:** Scenarios 2 and 3: explore the matchday and predict the Champions League Final. Have a look at the upcoming matches, get an overview of the fixtures — including ones that may already have been played — and the information the app provides. Then find the Champions League Final between Paris Saint-Germain and Arsenal London and place a prediction. Just follow the steps the app shows you.
26. **P:** Okay, first I need to find the matches. I can already see upcoming matches directly on the home page. But I'm not sure if I'd place that above the matchday section — right at the very top. I think I'd put it at the top, because depending on how many matches there are — if it's just one, fair enough, but maybe if you cap it at three and make it a horizontal scroller, that could be quite nice. Because it's still interesting to see these points right there too. Points per matchday. And actually, what I'd also find cooler here — because the question is: is all of this already tied to the league? — would be to group it inside a league object.
27. **F:** What do you mean by that?
28. **P:** Well, everything here seems to relate to a specific league, right? But what happens if I'm in multiple leagues at the same time?
29. **F:** We'll get to that later — I can show you.

30. **P:** Ah, up here — okay, I see it. Got it.
31. **P:** Okay, I'll go to my predictions because I want to place a tip. Arsenal won. Oh, that hasn't been played yet — it's fake data. Okay. Win probability. Match details. But I actually want to predict. Where do I predict for this matchday? [navigates] I need to go into the match. It's already entered. Oh wait — I need to enter it at the top... that was already pre-filled. Was that intentional?
32. **F:** Yes.
33. **P:** Okay, fine. But I think even if it shows 0-0, it's not immediately obvious. Also — there's no way to close this once I've opened it; if I tap outside, it doesn't close. And scrolling doesn't close it either. The other thing is: if I tap on the score, I'd expect to jump directly into the input. Or you could make it look more obviously like an input field — just a white box — so it looks tappable without jumping straight into it. Fair enough that it's intentional. But I still don't immediately read it as interactive.
34. **P:** Okay, let me place a prediction. I say Paris will absolutely destroy them. 3-1 for Paris. Actually no — make it 3-0. They're getting blown out. [holds button to save] Hold to save — so it's locked in? Haptic feedback here again would be nice. 'Why this prediction?' — okay, being asked why. I'll say: strong form. Save. Next match? No — just save. Saved. Good.

TASK 3 – Checking the Leaderboard

35. **F:** Next task: find out what position you're currently in within this league, and tell me your position and how many points separate you from the next person.
36. **P:** I'm in there. I'm obviously first. But actually everyone is first. And here's some feedback: if everyone is tied first, they should all be shown as first. And instead of highlighting the rank, I'd highlight my name in colour — so I can instantly see where I am. That would've been much faster. That would've been my expected design — a colour highlight on my name so I know where I stand. And for positions: I'd show first, second, and third in bronze, silver, and gold.
37. **F:** Cool idea.
38. **P:** Yeah. Or even like medals. A gold medal, bronze medal, silver. You could do that too. That would bring a bit of colour in here too, without it looking gaudy. Medals — if you want to keep it minimal, you could just use a filled circle. But I actually think

actual medal icons would be cool.

TASK 4 – Finding & Changing a Setting

39. **F:** You remember that right at the start, in the app settings during onboarding, you were asked to set whether you want quick picks or deliberate picks. Find that setting and change it to the opposite of the current selection.
40. **P:** Yeah, the deliberate mode was already getting on my nerves with the justification prompt — it felt too cumbersome. [navigates] Profile, scroll down, Prediction Settings — quick pick is selected. Found it immediately. Okay, done.
41. **P:** Also here — you can tap it to change it, but some visual feedback that I've pressed the button would help. Like a click state. I get it with the auto-select style, but it's a matter of taste. If you still want to keep it minimal, you could add very thin lines between the options and then have a full-width hover state. But fine.

TASK 5 – Viewing Another Player's Predictions

42. **F:** Now find out what Macher2000 predicted for the last matchday's matches.
43. **P:** [navigates to league] Oh wait, sorry — you can't view that yet, because the Champions League Final hasn't happened. Okay, show me what Macher2000 predicted in the last matchday they played.
44. **P:** Macher2000... Macher2000... [navigates] That doesn't work. I would've tapped on Macher2000, right? [looks around] I found last matchday here. But I can't see Macher2000's picks.
45. **F:** You're in the right area. The actual route to get there is up here. [points]
46. **P:** Ah, okay — via 'Tips'. But I can already tell you, every user who's done this study so far hasn't found it — which is itself good feedback.
47. **P:** Has anyone had as good a username as mine? [laughs]
48. **F:** Not yet — Mr. Tipp is the best so far.

TASK 6 – Sending a Private Message

49. **F:** Next task: use the chat and send a private message to someone in the league.
50. **P:** Okay. [navigates to chat] I don't know what the study agreed on. [looks at chat] So I'll write to the first person. They have a profile picture but I can't see it clearly. [types] 'Hello. Great tip — cheers, Mr. Tipp. Ask me for a tip next time.'
51. **P:** Wait, I have feedback before I send. First of all, I can't scroll in here. And the

UI isn't great — personally I'd find it nicer if the input field expanded when I write a longer message. You can take inspiration from WhatsApp here. Also — [notices something] I think your storage is full?

52. **F:** I actually have unlimited storage.

53. **P:** Oh right. Okay. [sends message] Done.

TASK 7 – Inviting a Friend

54. **F:** Last task: you want to add another friend to your league. Find the invite function.

55. **P:** [navigates] League — plus icon — copy — share. Oh, it works! Nice, the share sheet.

56. **F:** Very good. And that's it — we're done with the tasks.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P03 completes AttrakDiff scales]

INTUI (Sentence Completion)

[Observer note: P03 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P03 completes SUS]

NPS

57. **F:** Last one: how likely is it that you would recommend this app to a friend or colleague who's interested in football prediction apps — and why?

[Observer note: P03 responds — exact wording not clearly recorded]

DEMOGRAPHIC INFORMATION

58. **F:** How old are you?

59. **P:** 23.

60. **F:** How would you identify gender-wise?

61. **P:** Male.

62. **F:** How would you describe your level of football interest? Are you a casual fan who follows the World Cup and Euros, a regular fan who follows the Bundesliga, or someone who really has a team they always support?

63. **P:** Casual fan. But lately I've been watching a bit more football — it's become a bit

more interesting to me.

64. **F:** Have you used prediction apps before?

65. **P:** Not really — well, maybe Kicktipp once or twice. But really only briefly, with a group, a long time ago.

66. **F:** How would you describe your usage frequency in that case — very rarely, not multiple times a week?

67. **P:** Yes, really rarely — exactly.

68. **F:** Your smartphone OS?

69. **P:** iOS.

70. **F:** How would you rate your technical competence out of 5?

71. **P:** Very high. I'd put myself at a 5.

OPEN INTERVIEW

72. **F:** Okay, now we're in the open debrief. Looking back at the entire session — what's your overall impression of the app? What stands out the most? What was positive, what was negative? Just whatever comes to mind first. Try to keep it relatively brief rather than going into too much detail.

73. **P:** I think the app is a bit monotonous.

74. **F:** In what way?

75. **P:** I feel like there are always very few components and large components on the screen. It makes it feel a bit cold. Like it feels a bit lifeless. I'm missing a bit of life in there. I don't know how to describe it — I also wouldn't know exactly how to fix it. But I'm just missing something. Maybe something like... I'm thinking of Duolingo — very different app, obviously, but there's a bit more animation in there. Like, when you do something, something happens. Maybe when you place a prediction, a ball gets kicked somewhere. You know what I mean? Some kind of little moment.

76. **P:** What I liked: it was all very clean. I found my way around instantly. You immediately knew what to do where. And I think even if you're in multiple leagues, it's very convenient that you don't have to switch to a different interface — you can just jump right to what you need and place your tip. I liked that. And overall — for what the app is meant to do, I think it does it very well. There's not a lot of unnecessary clutter around it. And I think that's actually the point. If you want a

prediction app, you don't want a lot of extra fluff. Kicktipp feels significantly more cluttered and a bit too nested by comparison.

77. **F:** Thank you — really great and detailed feedback. There were moments where the app deliberately slowed you down — for example, when you had to justify your prediction. Did that moment feel special in any way, or was it just annoying and slowing you down — or did you actually find it useful?
78. **P:** I found myself wondering: what's the point of this? It would've been cool to know in advance. Because in principle, it could be interesting to look back at my prediction reasoning over time. Maybe. But overall it was a bit surprising — like suddenly something extra pops up. Also because I pressed it very lightly and suddenly there's this power-down effect. Maybe you could've had a ball being kicked there. [laughs] No, but seriously — yeah.
79. **F:** Did you feel in control of your experience the whole time, or were there moments where you felt pushed in a direction you hadn't chosen and couldn't get out of?
80. **P:** I felt safe. I knew what I was doing at all times. I could basically see the source code directly. [laughs]
81. **F:** Did the match data and statistics you saw while predicting make you feel more competent — or did you mostly gloss over them?
82. **P:** Honestly — no, because I predicted mostly on gut feeling. But I think for someone who bases their picks on that kind of data, the question becomes: are you actually more competent than your fellow predictors if they have the same information? Anyway. But yes — objectively, you do get a lot of information. Especially with that spider-web chart showing playing styles and so on — that's actually pretty cool. You can look at it and think, okay...
83. **P:** But it can also introduce a bit of bias, right? Because if you follow the statistics, and football isn't really a statistics-driven sport...
84. **F:** Where does the win probability come from, by the way?
85. **P:** From the data platform you're using?
86. **F:** Yes — it uses all the seasonal data from both teams and calculates the win probability.
87. **P:** I'd actually consider removing that metric.

88. **F:** Why?
89. **P:** Because I feel like that belongs more in a betting app than a prediction app. The sport-specific, match-relevant metrics are cool — you can see how teams play. But showing the raw win probability directly... I'm not sure that's the right fit here. Because then people think: 'oh, they're just picking based on win rates.' You know what I mean? That can make it feel a bit less fun. But would it change anything if the win probability were explained in more detail?
90. **F:** Would it?
91. **P:** Probably not. I think it would have to be a group agreement — like, 'we're not looking at odds.' So would it be cool to have an option to turn it off in the league settings?
92. **F:** Yes, that could actually be a cool feature.
93. **P:** And then when you join the league, you agree not to look at odds. You can't enforce it in the app, but the group is playing with friends anyway. It's a trust thing — whether they actually look or not. But that's a really cool idea.
94. **F:** Did the social features — like the leaderboard or the chat — feel particularly meaningful? Did you feel more connected to the others, or was it more of an extra layer on top of what you wanted to do with the app?
95. **P:** That's a question of whether your prediction group already has a WhatsApp group. If yes, I can imagine most activity stays there. But you can see — for example — if you write in the group chat: 'haha, you predicted so badly, what was that?' More than that probably won't happen. Or some football player might get dissed.
96. **F:** That's a very good point about WhatsApp groups — I can tell you it's already come up multiple times in this study. You could think about whether the chat is really a useful feature, since most prediction rounds already have a WhatsApp group and it just ends up being duplicated.
97. **P:** I think what could be cool — and kind of funny — is if you integrated the prediction history more into the social layer. So that you can react to predictions directly. Like reactions on tips. Or — when you're looking at someone's predictions — you hold down on a tip and get a message input right there, and then you send it not as a private message to that person but to the group. Because honestly, I wouldn't know why I'd

send you a private message through this app specifically. More of a group thing.

98. **P:** Actually — I'd remove the chat tab at the bottom entirely. And instead, maybe on the leaderboard when you scroll down, the other players' predictions from the last match appear there. Maybe the top 3 and the bottom 3 — and then you can react to them. Either with an emoji or with a long-press that opens a message. So you could write something like 'what was that?' on someone's pick. And then it's visualised somehow in that view. I think that would actually drive more interaction — reacting to other people's picks rather than having a separate chat. And in the predictions overview with all the others, you'd see those reactions inline. I think that could also increase the sense of connection — because right now it's quite data-heavy, it's just numbers. But if there's a heart, a rocket, or a thumbs down because someone predicted badly — or a middle finger — [laughs] — then you'd feel more connected. Maybe you could even animate the players with reactions. Actually no, that's probably too much.
99. **F:** How does the app compare to Kicktipp, which you mentioned using a long time ago — if you can still remember?
100. **P:** The app is significantly more modern. It's not overloaded. Kicktipp also has all that advertising, which is really annoying.
101. **F:** Is there anything else you'd like to share — about the app, the experience, or anything from the session we haven't covered?
102. **P:** I want to buy shares. [laughs] No — no further comments.
103. **F:** Thank you very much.

— End of transcript —

Note: Some passages (e.g. during questionnaire completion, exact NPS wording, specific app names) were unclear in the original recording and have been summarised or marked accordingly.

B.4 P04 – Session Transcript

In-Person Session | 18:12 | Language: German

PreMatch User Study | In-Person Session | 18:12

BRIEFING & CONSENT

1. **F:** Welcome to my user study. You've been invited to participate in the user study for my Master's thesis at MCI Innsbruck. This study evaluates a high-fidelity prototype of a new social football prediction app. My research investigates how principles of self-determination theory — also called SDT — and positive friction can be applied in interaction design to create an even better user experience. You'll receive a series of tasks to complete in the prototype. Depending on how long you take and how much you share, it'll be around 20 to 30 minutes. The important thing is that while you interact with it, you think out loud and describe what you see and expect. After the tasks, you'll fill out a questionnaire. And then we'll have a couple of open questions in a short debrief where we talk about your experience and go into a few things in more detail.
2. **F:** The important thing is that it's the app being tested, not you — there are no right or wrong answers. If something is confusing or doesn't work as expected, that's exactly what I want to know — it's the app's fault, not yours. I need honest reactions, because those are the most valuable data for my study.
3. **F:** Regarding data protection: all data is pseudonymised and stored under a participant ID. Your name will not be recorded at any point. Recordings are used exclusively for analysis and will not be shared publicly. Quotes and the transcript will be used in the Master's thesis. You can request deletion of your data at any time by emailing me — I'll give you that address afterwards. Participation is completely voluntary. You can end the session at any time, without a reason, and without any negative consequences. You can also withdraw your data after the session by contacting me. Results will be published as part of the Master's thesis at MCI Innsbruck, likely sometime in autumn. If you'd like a summary of the results beforehand, just let me know. If you have any questions before or after, feel free to reach out. If you've read all of this and agree, please tick all the boxes to confirm, and then we can get started.

[Observer note: Researcher fills in session metadata: Language: German, In Person,

18:12]

[Observer note: Researcher starts screen recording]

TASK 1 – Account Creation & Joining a League

4. **F:** As part of this study, you're asked to complete a series of tasks in the app. Think out loud, describe what you see, what you're trying to do, and what you expect. There are no right or wrong answers. Take the time you need. I'll take a few notes alongside to capture the key points for the transcript. The first task: open the app for the first time and create an account with your name, a fictional prediction username, and the email address eb2000@gmx.de. Then join an existing league with the invitation code YW5UH.
5. **P:** Okay. So we're now here on what you could call the start screen. What I notice right away is that the email option isn't at the same visual level as the Google or Apple buttons. I'd find it really nice if everything were aligned. But I'll just go with email. And immediately — very smooth — I'm redirected and I can see the steps at the top showing how many I need to complete.
6. **P:** Now I'll enter eb2000@gmx.de. Correct. If I tap here — okay — I've entered that. Now I need to choose my own password. [creates password] Everything feels very intuitive so far and looks like a standard sign-up process I've seen in other apps.
7. **P:** One thing I'd note: bottom right there was a checkmark that confirmed the input for that field. It might be even smoother if that checkmark jumped directly to the next input field — so I go straight to the next step. Because then I'd just continue without thinking about it.
8. **P:** Now I'm up here — let me do this quickly. Oh — if I go one step back because I forgot something, I get kicked all the way back to the beginning. That shouldn't happen — it shouldn't jump all the way out.

[Observer note: P04 navigates back, gets redirected to start; uses a slightly different email to re-register]
9. **P:** Now I'll select German. What I notice is that I'm already on step one of nine — and now it still says step one of nine. But this is actually the second step, right? So for me it would make sense if it showed two of nine here. Now my name is shown. I'll enter a normal name. And now I can enter my prediction username. [types username]

Continue.

10. **P:** Now I enter my age. I like the age slider — the visual bar makes it actually fun to set your age. The switch animation is a bit fast when switching through, but you could improve that slightly for an even smoother experience. I like the 3D effect. I'll select Germany as my national team — and I like that it says 'You can change this later' right there, because these are just the initial settings. It would be a cool feature to support multiple nationalities — I'm thinking of people like German-Turks who might want to support both teams, or German-Italians.
11. **P:** My favourite league is the Bundesliga. I like how there's always a blue outline around the selected item — so you can see what you've picked without it selecting immediately, and you confirm by pressing Continue. Club. [selects club]
12. **P:** Now the step to join a league. What I find good right away is how simple this is — either you join a league or you create one. There are only these two options. [enters code YW5UH] What I see next: quick picks or deliberate picks.
13. **P:** I don't fully understand this at first glance, because I don't really know what a 'prompt' is supposed to be here. You'd probably want to inform the user beforehand about what this app brings with it — what features it has. A short explanation that doesn't make the feature itself seem bad, but just walks the user through it so they understand. In that case, I'd maybe choose 'deliberate' — and this long description here reminds me a bit of AI. You could cut that down.
14. **F:** Go ahead and select 'deliberate' — we'll come back to that later. Good to know you'd naturally choose quick picks.
15. **P:** What I like is how simple and easy this sign-up process is. As a potential user of the app, it's really important to me to get through the onboarding quickly. I don't want to fiddle around with it for long. [completes onboarding] I'm in. Now I don't quite understand — what's 'CL'? And what's 'Standard Mode'? It's not explained, but that's something you'd probably want to clarify more.
16. **F:** CL is the name of the prediction league. Standard Mode is the currently active mode — and the rules for it are explained below. In the future there'll be different modes, including something called 'Mastermind Mode', which would also be explained in onboarding. But for now this is the starting point.
17. **P:** Okay, so for me it would be really useful to understand what Standard Mode means

in context. As a user, it's very important to me that I understand things simply and immediately right from the start. Too many features at once actually hinders me. My acceptance of an app is very high if it's simple. Simple features — that's it.

18. **F:** That was scenario 1. Wonderful. You can set it down for a moment.

TASK 2 – Exploring the App & Placing a Prediction

19. **F:** You're now in the prediction league 'CL' — as you noted, that name was chosen by another user. Have a look at the upcoming matches in the app. Feel free to explore, get an overview of the fixtures and the information the app provides. Then find the Champions League Final between Paris Saint-Germain and Arsenal, and follow all the steps the app shows you.
20. **P:** What I notice first is this 'CL' at the top — that's my prediction round. And in the future there might be several, so I find this field quite small. I'd find it more appealing if it were larger up there, so if I'm in multiple prediction rounds I can navigate easily. Also, the right edge of the prediction accuracy bar seems slightly cut off — oh, I see, I can swipe through interactively.
21. **P:** What I like about the home screen: there are no left-right swipe areas. A phone is already narrow, and having that kind of swipe navigation where you swipe between sections isn't intuitive to me. Having buttons at the top to navigate to different screens, and a bottom navigation bar — that's the logic I prefer as a user. Side panels are a no-go for me.
22. **P:** Okay, now I'm looking for the Champions League Final. There it is — it came up right at the bottom. I found it very intuitively. Now let me try to place a prediction.
23. **P:** I'm going to go for 2-0. What bothered me slightly just now was that when I entered a number, the selection button disappeared immediately. But in the moment where I was still entering, it didn't disappear yet. For me it would be good if I enter a number and it's just done — confirmed immediately. Or I can just quickly confirm it again, in case I mis-typed. Very intuitive. And now I can see all the statistics as well. That was also immediately intuitive to me — after placing my prediction, the stats came up at the bottom. I actually wanted to look at the other features first to validate my prediction a bit. I really liked the radar chart overview. Now: 'Why this prediction?' — yes, that's what comes to mind first, just gut feeling. I'd like to leave it blank — can I? No, in this mode it's required, because we selected deliberate

- picks earlier. In quick picks mode it wouldn't show at all. [submits justification]
- And now I got a success message. Final feedback: very intuitive flow overall. What did bother me was the score input field at the top — when I enter a number and then want to confirm it, something like a checkmark like before in the email step would be nice. Because maybe I'm still thinking: one — no, actually two — no, zero. The hold-to-save mechanic is good. I just find it slightly too long. If I think about predicting nine Bundesliga matches and having to hold every single time, that starts to feel tedious. I want my user experience to be better when I can give predictions quickly, because often you're on the go and thinking: oh, I need to quickly tip a few results. Speed matters. People often don't have time and just want to do it quickly.
24. **F:** Noted. That was the next scenario. Wonderful.

TASK 3 – Checking the Leaderboard

25. **F:** Now it's about checking your position in the league — your current standing and how many points separate you from the next person. As I mentioned, this is still a prototype, so not everything works 100% yet. But the goal is to see whether the basic setup works and whether you can find the features.
26. **P:** Intuitively, from the home screen, I'd go directly to 'My League' because that's what it points me towards. Now I can see right away — in brackets — that that's me. What I'd like even more is if my name stood out a bit more. I want to see who I am in the list — not just have it written there, but have it visually differentiate itself somehow. If my name were slightly elevated or highlighted, I'd intuitively feel a bit better. What's my current position? Position 1 — I recognised that immediately. And I immediately understood why, because I can see the average score and the points on the right. The reasoning is intuitive.
27. **F:** Good — you've found it. You remember the setting from onboarding where you chose deliberate picks. Find that setting and change it to the opposite.

TASK 4 – Finding & Changing a Setting

28. **P:** Right. I'd intuitively go to Profile. Then I'd look through the individual setting categories. I'd have two options: FAQs — I might think about that — or actually Prediction Settings. I'm someone who wants to do things quickly, so I'd go straight to Prediction Settings. If I were someone who wanted to be thorough, I'd probably check

FAQs first. [navigates] Prediction Settings — and I can see 'Quick Picks' right there. Found it very quickly intuitively — because the settings are organised by category and not nested. I find it very hard to find things when they're buried in subcategories.

TASK 5 – Viewing Other Players' Predictions

29. **F:** We have three quick scenarios left. Find out what the other predictors have already tipped in the previous matchdays — the exact predictions of other users in the previous Champions League round.
30. **P:** Because I've seen the other players before, I'd go directly to My League. Then I'd look at 'Last Matchday'. [navigates] I'd look at 'My Tips' — but it's not there. I'd go back to My League because I think I might find it there. Now I tap on a person — and I see nothing. So I can't tap on the other players to see their predictions. That would've been my intuition. I can see my own stats, but I need to see the other players' stats.
31. **F:** You're already in the right place — I'll help you out. It's the 'Tips' button up here on the right. It's definitely good feedback that it's hard to find — everyone so far has missed it, which means it needs to be more prominently surfaced.
32. **P:** Ah — so my feedback would be that this needs to be re-framed or relabelled. Because what I understood by 'Tips' was suggestions for me. That's why I didn't intuitively go there. The label should make it clearer what you actually see there. Something like 'What did the others predict?' as a starting point — just to signal that it shows other people's picks, not just tips for me. Also: when exactly can I see what others predicted? Only after the match kicks off?
33. **F:** Yes — exactly.
34. **P:** That's really important to me, because otherwise I could use it to influence my own picks. That wouldn't be fair.

TASK 6 – Sending a Chat Message

35. **F:** Second-to-last scenario: you want to use the chat and send a short message to another member of the CL league.
36. **P:** I'd intuitively tap 'Chat' at the bottom, then 'New Chat'. I want to write to another player, so I'd go for a one-on-one chat. I'll write to Lukas. [types] 'Good luck.' Now I have two options — a checkmark and another icon. That confuses me a bit, because

I feel like one option is enough. I'm very used to WhatsApp, and in WhatsApp there's a space bar at the bottom and the send button. So I'd remove one of those and just have the send button — bottom right, like an Enter key in WhatsApp.

37. **P:** What I also notice here — and in the app generally — is that the different colours are very similar in tone. That means there's not much contrast. It looks modern, but a bit flat. Here, for example, I can barely see my own messages. I think I already have full brightness. With low brightness or in bright sunlight I'd see even less. So my feedback would be to differentiate the colours a bit more from each other — while still staying within the same design style.

TASK 7 – Inviting a Friend

38. **F:** Last scenario: you want to invite a friend to the league. Try to find that function.
39. **P:** [navigates to Profile] Now I'm thinking. I could go to FAQs but I want to do it intuitively. I don't see anything on my profile that points me towards inviting someone. So I'll go to the home screen. Not there either. Back to Profile. I want to invite a friend to this league — right? Now: My League. And up there on the right I see a plus icon. Found it. But honestly, it took me three attempts — because intuitively I went to my profile first.
40. **P:** If I think about it, that probably comes from how I share things in other apps — like WhatsApp, where I find my QR code or share link via my own profile. So my intuitive thought pattern followed the logic of other apps I already use. I think the app would be more user-friendly if the navigation logic were aligned with apps people already use.
41. **F:** That's a great point. There is also the possibility of placing it in multiple locations. The plus icon in My League is there, and we could also add it to the settings. I actually think it's a really good idea to have it in the profile too — like a QR code icon top right that you can tap, so you can show someone and they scan it to join directly. That's also a cool second way to join a league. I think this feature is incredibly important and needs to be prominent — but not intrusive. Because the app lives and dies by how quickly you can get a group set up. At a party, you'd say: 'Let's do a prediction round' — and it needs to happen instantly. For me that's QR code or a direct WhatsApp link share.
42. **P:** Yes — and that link share would've worked if you'd tapped 'Share', you could've

sent it via WhatsApp. But because the function was placed somewhere else first, that's definitely valid and important feedback. The 'add friend' plus icon — I did see it, but it didn't intuitively click for me because it felt too small.

43. **F:** Great. We can stop the screen recording now — we're through the tasks.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P04 completes AttrakDiff scales]

INTUI (Sentence Completion)

[Observer note: P04 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P04 completes SUS]

NPS

[Observer note: P04 completes NPS — exact wording not clearly recorded]

DEMOGRAPHIC INFORMATION

44. **F:** Your age?
45. **P:** 20.
46. **F:** How would you identify gender-wise?
47. **P:** Male.
48. **F:** How would you describe your level of football interest — casual fan, regular fan, or passionate supporter?
49. **P:** Passionate fan.
50. **F:** Have you used prediction apps before, like Kicktipp or TeamTipp?
51. **P:** I've used Kicktipp.
52. **F:** How often did you use it?
53. **P:** Multiple times per week.
54. **F:** Your smartphone OS?
55. **P:** iOS.
56. **F:** How would you rate your technical competence out of 5?
- [Observer note: P04 responds — exact number not clearly recorded]*
57. **F:** Have you used a betting app before, like Tipico or Bet365?

[Observer note: P04 responds — answer not clearly recorded]

OPEN INTERVIEW

58. **F:** Now we have a couple of final open questions where you can give me honest feedback. Don't go too deep into detail — just share your first thoughts, and I'll follow up if needed. First question: looking back at the entire session — what's your overall impression? What stands out the most, both positively and negatively?
59. **P:** Let me start with the negative. What strikes me is that the app isn't exciting. I'd say — maybe a bit exaggerated — it's monotone in its design. It feels more like a software application than a leisure app. And this app is supposed to generate a sense of joy and anticipation — I'm looking forward to something through the predictions. But intuitively, because of the colour palette being so uniform, it feels very professional. And that professionalism I associated with a work tool rather than a fun one.
60. **P:** What I find very positive is the simplicity — the ease of use. Because what do I actually want from this app? I want to not spend a lot of time on it, but generate that positive feeling. That feeling of being part of a community, of discussing with my friends. So really I don't want to be on the app for long — I want to drop my predictions, check my rank, and be done. For that kind of use case, extreme simplicity is the key to adoption. And my intuition was mostly right — I found my way around quickly. That was positive. The sign-up process was also very smooth and understandable. I liked being shown statistics that helped me with my predictions. I found it positive that I then had to decide why I'd tipped what I did — though what I found negative was not being clearly told why I was being asked to do that. I guessed it was related to tracking my reasoning for statistical purposes, but then I thought: what is being done with these data? Personally I don't mind, but in Germany especially, some users might have concerns about their data being analysed.
61. **P:** Also, the settings were very well structured — clearly organised by category.
62. **F:** Very valuable feedback, thank you. Second question: there were moments where the app deliberately slowed you down and prompted you — for example, to justify your prediction. Did that moment feel special in any way, or was it annoying — and did it change anything about your interaction?
63. **P:** It made me think. In that moment I questioned whether the prediction I'd entered intuitively was actually right — and it almost made me change my pick. So what did

- it do to me? It made me reflect on why I'd just acted the way I did. And it actually made me reconsider the result.
64. **F:** Did you feel in control the whole time, or were there moments where the app navigated you somewhere you couldn't get out of?
65. **P:** In terms of the app and its features — no, not constrained. The one thing that did feel constraining was the score input field jumping away when I wanted it. That's the kind of thing where I feel restricted as a user. And also the keyboard not dismissing when I didn't want it. Other than that I didn't feel constrained — partly because there were no left or right side panels, everything was reachable from the bottom navigation. Side panels would be a dealbreaker for me. During the sign-up I did feel briefly frustrated when I had to start over — that annoyed me for a moment.
66. **F:** Did the match data and statistics make you feel more competent and better informed — or was it more of a nice extra you didn't really need?
67. **P:** It made me feel confirmed in what I had predicted. I felt very well informed.
68. **F:** Did the social features feel particularly meaningful, or were they more of a functional extra for you?
69. **P:** That's an area where I'd say it still felt a bit too functional for what's supposed to be a shared social experience among friends. I would've wanted more incentives that encourage interaction between users. Maybe insights — like this player has been on a massive winning streak recently. Or maybe a head-to-head challenge, so I can see how I've performed compared to a specific other user. I know the kind of users this targets — they love that competitive element between each other, which is why they play in a league, why there's a leaderboard. The challenge is integrating that in a way that stays simple and doesn't feel like a separate feature — but still amplifies that competitive aspect and therefore the communication in the app. Because that's what keeps people engaged and coming back.
70. **F:** You mentioned using Kicktipp before. How does this app compare?
71. **P:** I'd describe it as a smoother prediction experience overall — the individual buttons, the visual design — it feels more modern, more intuitive, more future-oriented.
72. **F:** Is there anything else you'd like to share that we haven't covered?
73. **P:** I've been hinting at it throughout. The core question is: what is the 'job to be done'

with this app? And that needs to come through somehow. What do I want to achieve? I want that when I open the app, I feel something positive and think of my friends — that feeling of fun with the people I care about, that comes through the app. And I was missing that a bit. Maybe through slogans that pop up here and there, maybe quotes that someone shared in the chat at some point, or something like that — to improve interaction a bit more. Like, imagine something floating in at the top: 'See how the others are doing' or 'Lucas is on a winning streak' — something like that. So I get a little reminder: hey, I'm in a community here. Basically: when I have the app, I want to feel the way I'd feel sitting outside around a campfire with ten friends, able to talk about this app because we all use it and I enjoy it when I think about it. More of that community feeling needs to be integrated.

74. **F:** Very, very cool. Thank you so much. Have a great day.

75. **P:** Thank you.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, exact NPS wording, technical competence rating, betting app usage) were unclear in the original recording and have been summarised or marked accordingly.

B.5 P05 – Session Transcript

In-Person Session | 24.05.2026, 21:30 | Language: German

PreMatch User Study | In-Person Session | 24.05., 21:30

BRIEFING & CONSENT

1. **F:** Welcome to the user study for my Master's thesis — 'From Comparative Benchmarking to Realisation: Designing and Prototyping a Next-Generation Football Prediction Experience.' First, the purpose of the study: you've been invited to take part in a user study for my Master's thesis at MCI Innsbruck. This study evaluates a high-fidelity prototype of a new social football prediction app. The research investigates how principles of self-determination theory and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
2. **F:** What's going to happen: you'll receive a series of realistic tasks in the app prototype. While completing them, you're asked to think out loud, describe what you see and what you expect. Afterwards you'll fill in a short questionnaire, and then the interview ends with a brief open debrief about your experience where we'll go into more detail on certain topics.
3. **F:** What is being tested: the app, not you. That's very important. There are no right or wrong answers. If something is confusing or doesn't work the way you'd expect, that's exactly what I want to know — it's the feedback I need to improve the app and design. Your honest reactions are incredibly important, so please be completely honest, not accommodating.
4. **F:** Regarding data protection: your data is pseudonymised and stored under a participant ID, for example P01, and your name is not recorded anywhere. Session recordings are used exclusively for research analysis and are not shared publicly. Quotes will be used anonymously in the Master's thesis. Data processing follows GDPR and you can request deletion of your data at any time. Participation is completely voluntary — you can end the session at any time without giving a reason and without any negative consequences, and you can also withdraw your data after the session. Results will be published as part of my Master's thesis at MCI, likely sometime in autumn. If you'd like a summary of the results earlier, please let me

know — contact Erik Bergheimer at the email address listed here. If you have any questions before, during, or after, feel free to reach out. Please tick all the boxes to give your consent for the data protection topics.

[Observer note: P05 gives consent]

5. **F:** Great. Today is 24.05., in-person session, participant 5, 21:30. I'll start the screen recording now. As I mentioned, you'll be asked to complete a series of tasks in the app. Please think out loud, describe what you see and what you're trying to do and expect. There are no right or wrong answers — that's very important. Just take the time you need. If you have no further questions, we can start with the first scenario.

TASK 1 – Account Creation & Joining a League

6. **F:** The scenario is as follows: you open this social football prediction app for the first time. Please create an account with your name, a fictional prediction username, and an email address — note that signing in with Google and Apple doesn't work yet in this prototype. Just enter whichever email you like. Then join the existing league with the invitation code YW5UH.
7. **P:** Okay, so first of all it's clear that I get to the email option here. The button is clearly recognisable in first position. I'll enter my email address where it says 'Email' — very easy to find. The keyboard, if that's also relevant, is the one I'm used to from Apple. I'll enter my password. I can also check whether I've typed it correctly. It's a very simple password and I'm not getting any feedback on whether it's secure or not — but for an app like this I don't really care. I'll tap Continue and immediately select German as my language. It's nice that there's a flag emoji next to each language option. I notice that at the top it says 'Choose your language' in English — so if I were German and couldn't read English, I wouldn't be able to read the sentence, but I can recognise the flag and the language name and figure out the rest. I tap Continue, and from now on Continue is also shown as 'Weiter' in German.

[Observer note: Brief technical issue — screen navigates back unexpectedly]

8. **P:** Okay, so now everything is in German because I selected the language. I'll enter my name — I had to think briefly about what exactly was meant. My full name. And now: prediction username, optional. I'm wondering — if I don't enter one, will I just appear as Stefan in the prediction game? But of course I have a legendary name: Two Win. [types] Age — required for legal requirements and age-restricted features. I'm

curious what age-restricted features that might be. I'll enter my age. Date of birth — I'm not sure that's necessary, because my birthday might be coming up and if it's about legal age that could become relevant. National team — select your favourite national team for international predictions. You can change this later. Very cool, looks very fresh. I'm guessing it doesn't yet have all 46 teams due to the prototype status. I like that Germany appears first — that's my pick. I wonder: would England be first if I'd selected English? Interesting. The Netherlands is there too.

9. **P:** Favourite league — nice logos, clean overview, all the important leagues are there except the second and third divisions. Probably also due to the prototype. Favourite club — I immediately tapped FC Bayern. Nice that they appear first. Interesting that BVB isn't second — I'm wondering how the clubs are sorted here. If it's by popularity, then Wolfsburg in third place is definitely off. And great that Paderborn somehow ended up in there — that's clearly an error, Erik.
10. **F:** You're right — it's a data issue. Because of relegation being included in the dataset, Paderborn accidentally made it in. Known issue.
11. **P:** One last step: join a league with a code or create your own for friends and family. Right, I have the invitation code. I'll tap 'Join League'. [enters code YW5UH] Oh, I'd find it very cool if the letters were all the same size — and nice, it found it right away, shown in green below. Easy. I'm not sure if it's the keyboard on this phone, but I can also see numbers on there. When you have codes with numbers in them, I'd find it quite nice to have a dedicated code input keyboard so I don't have to switch. But that's a bit old school.
12. **P:** Oh — quick picks or deliberate picks? I don't fully understand this. I can sort of figure it out: quick means I just enter the result, deliberate means I can also justify why I think it'll go that way. I think that's a feature — I'd like an info icon here to tap that explains it in detail, because I don't know: do I have to justify? Is there a prompt I have to respond to? In that case, I'd say quick picks — because deliberate sounds like it'll be too much effort. I already can't be bothered to always tip carefully, especially for teams I don't care about. I don't want to also justify why Paderborn will beat Hamburg by a certain score.
13. **F:** Let's leave deliberate ticked for now — we'll come back to that. Good to know you'd choose quick picks naturally.

14. **P:** That was step 9 of 9. I'm in. 'How it works' — points for correct result or correct tendency. I notice that 'predict the exact result in advance' and then 'tendency' seems slightly contradictory. Otherwise, where you find everything: place predictions, follow the table, league settings. Icons are appropriate, looks clear. Font size is fine for my eyes. I'm not sure if the greyed-out text under 'place predictions' might be hard to read for some people. I tap 'Let's go' and that seems fine for now.
15. **F:** Great, that was the first scenario. Thank you.

TASK 2 – Exploring the App & Placing a Prediction

16. **F:** Second scenario: exploring the matchday and predicting the Champions League Final. You're now in the prediction round 'CL' — Champions League. Your task is to look at the upcoming matches, explore freely, get an overview of the fixtures including ones that may have already been played, and then predict the Champions League Final between Paris Saint-Germain and Arsenal. Just follow all the steps the app shows you.
17. **P:** Okay, the UI at the top — nice. My league, CL Champions League, ten members, I'm a member, currently apparently in 4th place. 'Selected' — that's in English, even though I chose German. That must be a bug. 'Hey Stefan' — my real name is showing. Oh, I see — my name is how I'm addressed, and probably I appear to others only under my prediction username 'Two Win', but the app itself addresses me as Stefan, which is how I want to be called. If I could give feedback: at the start, when I was unsure what name to enter, it might help to add something in brackets like 'how you want to be addressed by the app', or just explain the name field better.
18. **P:** Matchday points — Matchday 32. I don't quite understand that in a Champions League context. Probably Bundesliga logic. Prediction accuracy 0% — probably because I haven't predicted anything yet. I'd find it really cool to know what that means exactly — tendency accuracy or full result accuracy? Points per matchday — seems abbreviated, probably because it's a test, and there's a gap between rounds 16 and 32. Next match countdown — 140 hours until I miss this game. 18:00 is the wrong time, but let's go in. The Champions League Final is in Istanbul but at 18:00? Oh well. Bayern might still have had a chance. [laughs]

[Observer note: Brief technical issue with screen recording — resolved]

19. **P:** Okay, we can see both teams. Date, time, statistics, win probability — how on

earth does Arsenal get 45%? But the colours are well divided. There's an info icon at the top too — very good — 'based on historical data, form and strength.' Form — also explained with an info icon, nice. I see W, L, D here for the form — but in the display it shows W for Win in English. And T somewhere — my English isn't good enough. [laughs] The table — okay, that's the Champions League past table. That's not so important right now — I'd find it more relevant to see the last head-to-head meetings in the Champions League specifically for the final. Very cool graphic for the strength comparison. I'm missing a passing percentage or shot accuracy metric there. Shall I comment on everything?

20. **F:** No, no — just place a prediction as you normally would.
21. **P:** I tap the two grey zeros at the top and — 3-2, I'll go with that even though I'm not confident. The numbers turn blue. I'm not sure why blue specifically, given PSG is a blue team. Maybe black would be better, but you probably have reasons — it's probably better for readability. 'Hold to save your prediction' — I find that quite interesting. I've seen this before in one of your sessions. The explanation is that it makes you consciously aware you're committing to a kind of bet — I think that's nicely done, especially because it's explained. 'Why this prediction?' — oh, now I understand what was meant earlier. I can tap reasons: home advantage — no; strong form — yes; goal-heavy match — yes, because I tipped 3-2; statistics — no. I'm missing a few more options, or the ability to add something custom. I'll leave it at that. And now my prediction is confirmed at the bottom — nice that it stays visible. If I scroll down and tap it again — nothing happens. Good.
22. **F:** Thank you very much. That's done. Next scenario.

TASK 3 – Checking the Leaderboard

23. **F:** Find out what position you're in within the current prediction league, and tell me your current position and how many points separate you from the next person.
24. **P:** Without even tapping — 4th place — but I'll do it properly. I go to match details, then back via the arrow. Now: I could tap the small Champions League indicator at the top — I can see my ranking there. Currently 4th. But that's very small. I'm guessing they designed something specific for this, which is 'My League' in the bottom centre — with the trophy icon. And here — I'm actually not 4th, I'm 1st. [laughs] Are all of these previous test participants?

25. **F:** The data in the table isn't accurate since you haven't predicted anything yet. But can you tell me your position?
26. **P:** Yes, 1st. And how many points separate me from the next person?
27. **F:** Zero.
28. **P:** Everyone's on three — everyone's in joint first place. [laughs] Then the question is whether I'm allowed to be critical: in that case we're all in joint first place.

TASK 4 – Finding & Changing a Setting

29. **F:** You remember that right at the start, the app asked whether you wanted quick or deliberate tipping. Please find that setting and change it to the opposite.
30. **P:** For me as Gen Z — no problem. I go to the profile icon bottom right. I see several options. Scrolling down, I don't see a separate 'Settings' button — just League Settings and Prediction Settings. Otherwise I'd have gone to 'My Account'. I'm unsure. But because I want to be right, I'll try Prediction Settings first. There it is. You'd find it in three taps: profile first, then the distinction between Prediction Settings and the other options. Can I change it now?
31. **F:** Yes, go ahead.
32. **P:** [taps to change setting] Done. And I assume it saves automatically without needing an explicit save button?
33. **F:** Correct.

TASK 5 – Viewing Other Players' Predictions

34. **F:** Now find out how the others in the league predicted the Champions League semi-final — try to find their predictions for past matches.
35. **P:** [navigates to My League] Then I go to 'Last Matchday' — no, that's about points. I try 'Matchday' — no, those are my stats. 'Tips'? No, those are all my own predictions. [tries various routes] I'll try 'My Tips'. I go to the match I just predicted. I scroll until I see what the others predicted. [confused] Nothing.
36. **F:** You've hit the same issue as everyone else so far. But — wait, do you want to try once more?
37. **P:** I want to be the one who finds it. [navigates again] I'm still not finding it on first attempt.
38. **F:** You were actually already in the right place — in 'Tips', in the matchday tips.

The issue is that the prototype data starts with Qualifying Rounds and the test data is sparse, so it wasn't visible. But the path was correct.

39. **P:** Also — it wasn't in German. I think that's why it didn't immediately click for me. And the teams shown weren't Champions League teams, just qualifying rounds. That threw me off.

TASK 6 – Sending a Chat Message

40. **F:** Next scenario: write a message to another person from the league in the chat.
41. **P:** I go to Chat. Then New Chat. [curious] I tap the group option first — ah, same thing. Okay. So either via the icon top right, or — I'll go for a 1-on-1 chat. It says 'with a specific team member' below. I want to know who 'Teufeltipper' is. I tap on them. Now I have a classic chat input at the bottom and I write: 'Who are you?' [types] Let me test whether there are emojis. [tries] No built-in emoji keyboard integration? Okay, I'll use this one. [sends] Sent.

TASK 7 – Inviting a Friend

42. **F:** Last task: you want to invite a friend to your league. Find the invite function and share the code or link.
- [Observer note: P05 briefly gets distracted navigating to their own profile]
43. **P:** Right — invite someone. Should I create a new league, or add someone to the existing one?
44. **F:** Add someone to the existing league.
45. **P:** I go to My League. Top right there's a plus button — it's a bit small, but I see it. There it is — I can just copy the code and paste it into a chat, or I can use the share sheet to select different apps directly. [uses share] Brilliant. Done.
46. **F:** Perfect. That's all the scenarios. We can stop the screen recording now.
47. **P:** Just so you know — because I tapped on my contacts during that, there might be data protection implications.
48. **F:** Noted, thank you.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P05 completes AttrakDiff — notes that one pair appears twice ('conventional'), identifies this as an intentional technique to catch non-attentive

responses]

INTUI (Sentence Completion)

[Observer note: P05 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P05 completes SUS — comments that Kicktipp will probably be replaced]

NPS

49. **F:** Last one: on a scale of 0 to 10, how likely is it that you would recommend this app to a friend or colleague who's interested in football predictions — and why?

[Observer note: P05 completes NPS and written justification]

DEMOGRAPHIC INFORMATION

50. **F:** How would you rate your football interest?

51. **P:** Not extremely high — somewhere in the middle.

[Observer note: Remaining demographics completed — exact responses not clearly recorded (age, gender, prior app usage, OS, technical competence, betting app usage)]

OPEN INTERVIEW

52. **F:** We have one or two final open questions where you can respond with your first intuition. Looking back at the entire session — what's your overall impression of the app? What stands out most to you, both positively and negatively?

53. **P:** I find it minimalist and at the same time multifaceted. There are more options — or rather, the options are presented in a clear visual way. The statistics for example are displayed very nicely — I've never seen anything like that in Kicktipp. I like that you can customise how you engage — whether you want to predict more deliberately or just quickly. I find the chat function and generally the user interface very appealing and modern. And it's still minimal in a good way. It's sufficient — it's a football prediction app and nothing more. That's all it needs to be.

54. **F:** There were moments where the app deliberately slowed you down and prompted you — for example, to justify your prediction or to hold the button to save your pick. Did that feel special in any way? Did you find it annoying? Did it change your interaction?

55. **P:** If I imagine tipping a World Cup with 46 teams and having to hold that button

- every single time, it would absolutely drive me crazy — especially because I use an app like this precisely because I can't be bothered with real betting apps. If I tip too quickly in a pub quiz and get it wrong, that's on me. That's the same logic here. So I found the hold mechanic more annoying. The justification prompts I actually thought were a cool idea — to make you think a bit more. I did wonder: is it just nice context for others to see why they predicted what they did? Or does it earn extra points? And I'd want more options beyond what was given — or the ability to enter something custom.
56. **F:** The plan is eventually to make the reasoning visible as part of the social layer — and to provide statistics like 'how often did you tip based on gut feeling' and so on. The idea behind it is to get the user to reflect on why they made a decision. And it's already happened several times that people adjusted their pick after having to justify it.
57. **P:** That's interesting.
58. **F:** Did you feel in control the whole time, or did the app ever push you in a direction you couldn't navigate out of?
59. **P:** Yes — it was fine. I felt in control throughout.
60. **F:** Did the match data and statistics make you feel more competent and better informed while predicting, or was it more of a nice-to-have extra?
61. **P:** It's definitely nice to have. Personally I wouldn't need it to be that detailed, but I think it's good for people who might not be as into football as I am — it gives them a better and more direct chance to participate. And ultimately it's also an incentive: if the win probability shows 45% for Arsenal, then for me — from a competitive standpoint against others in the table — the temptation to tip PSG is actually higher.
62. **F:** Did the social features — like the leaderboard or the chat — feel particularly meaningful, or more like nice extras?
63. **P:** The leaderboard is typical and expected. The chat I actually think is a genuinely good feature — especially because you're sometimes in prediction rounds with people you don't otherwise stay in contact with, and having the option to write to them in that context is nice.
64. **F:** How does this app compare to Kicktipp or other prediction tools you know?

65. **P:** Kicktipp sometimes gives me retro vibes. This app is significantly more modern and more clearly structured.
66. **F:** Is there anything else you'd like to share that we haven't covered?
67. **P:** Get this thing on the market.
68. **F:** Very cool. Thank you so much for your time — have a great evening.
69. **P:** Thank you, likewise.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, exact NPS score, demographic details) were unclear in the original recording and have been summarised or marked accordingly.

B.6 P06 – Session Transcript

In-Person Session | Language: German

PreMatch User Study | In-Person Session

BRIEFING & CONSENT

1. **F:** You've been invited to take part in my user study for the Master's thesis at MCI Innsbruck. The study evaluates a high-fidelity prototype of a new social football prediction app. The research investigates how principles of self-determination theory — SDT — and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
2. **F:** What happens now: you'll be given a series of tasks to complete in the prototype. The important thing is that you think out loud while interacting and describe what you see and what you expect. Afterwards you fill out a questionnaire — that takes around 10 to 15 minutes depending on how long you take. Then we'll have a short open conversation about your experience. The 45 to 60 minutes I estimated is way too long — most people finish in 25 to 30 minutes.
3. **F:** The most important thing to remember: the app is being tested, not you. There are no right or wrong answers. If something is confusing or doesn't work the way you expect, that is exactly the feedback I need to improve the design. Be 100% honest — don't hold back out of fear of seeming unintelligent or of hurting my feelings. The app is the problem, not you.
4. **F:** Data protection: all data is pseudonymised and stored under a participant ID, e.g. P01. Your name is not recorded in any file. Session recordings — audio and screen — are used exclusively for research analysis and not shared publicly. Anonymised quotes from your responses may be used in the thesis. Data processing follows GDPR. You can request deletion of your data at any time by contacting me — email or phone, you have my number. Participation is completely voluntary. You can end the session at any time, no reason needed, no consequences. You can also withdraw your data after the session. Results will be published as part of my Master's thesis at MCI Innsbruck, likely in autumn. Let me know if you'd like a summary of the results, and feel free to reach out with any questions before, during, or after. Please tick all the boxes to confirm you've been informed about the above.

[Observer note: P06 gives consent]

[Observer note: Researcher notes session metadata and starts screen recording]

5. **F:** As I mentioned, you'll be asked to complete a series of tasks. Please think out loud, describe what you see, what you're trying to do, and what you expect. There are no right or wrong answers. Take the time you need. Your first task: open the app for the first time — a social football prediction app. Create your own account with your name, a fictional prediction username, and your email address. Note that Apple and Google sign-in don't work yet in the prototype. Then join an existing league with the invitation code YW5UH.

TASK 1 – Account Creation & Joining a League

6. **P:** I tap the button with the email option. Then I take my own email. I'll choose a password — I'm aware it's not very secure. The app tells me the password needs at least six characters. It doesn't tell me whether the password I've chosen is strong or not, but I know that myself. Select language — I choose German. What should I be called? My name first — Timotheus. That comes out. 'Toys 42' — obviously, that's my prediction username. My age, my national team is Germany. I'll select that. Favourite league: Bundesliga — very nicely done with all the club crests. My favourite club is Borussia Dortmund — it shows up right away and I can see it's also changeable later. So I'll go with Borussia Dortmund, that's what I'm most interested in.
7. **P:** Now I should join a league. I have the code, or I could create my own. [enters code YW5UH] Join league.
8. **F:** Before you tap — what's your first intuition for the next screen? What would you tap?
9. **P:** Quick picks or deliberate picks? I think I'd intuitively go with quick picks, because I don't really know what to expect from deliberate. I mean, I can kind of figure it out — quick means just enter the result, deliberate means I can also justify why I think it'll go that way. That's a feature. But I'd want an info icon here to tap that explains it fully, because I don't know: what exactly am I being asked to justify? Is there a prompt I have to answer? I'd go with quick picks — because deliberate sounds like too much effort. I already can't always be bothered to think carefully about every single match.

10. **F:** Very cool — let's select deliberate for now, we'll come back to it later.
11. **P:** I'm in. 'CL' — what does that stand for? Champions League?
12. **F:** CL is the name of the prediction league — you'd know what it means if you already knew who you were playing with.
13. **P:** 'How it works' — predicts the exact result in advance, points for correct result or tendency, quick round — cool. Where to find everything: place predictions, follow the table, see the leaderboard in My League tab, league settings. Open My League. Got it. Continue? Yes, done.
14. **F:** Great, you went through that really fast. Well done.

TASK 2 – Exploring the App & Placing a Prediction

15. **F:** You're now in the prediction round CL. As you may know from following football, there's only one match left in the Champions League — the final between Paris Saint-Germain and Arsenal. Find that match and place a prediction for it. Just follow all the steps the app shows you. You can also use all the information the app provides.
16. **P:** Okay, I'm being greeted. Next matches — there's the match — match details, statistics, table from probably the group stage, all the information. I know where the match is being played. My predictions — I can tap on that. And I can just tap the number and it's already there. I'll say 3-1 for PSG. Hold to save your prediction — cool, very simple. It's made very clear. 'Why this prediction?' — okay, now this comes with the deliberate mode. 'Optional — helps the league see your reasoning. Reflection prompts. These prompts help you think through your picks and let the league see your reasoning.' — ah okay, so it's basically that others in my league can see why I predicted that — the social thinking behind it. Makes sense. I'll say no to home advantage. Goal-heavy match — yes, because I've got a lot of goals. I go with gut feeling because I think Paris just scores more goals. Win probability says Arsenal wins — okay, interesting.

[Observer note: P06 navigates to squad section to check injury data]

17. **P:** Where does all this information come from? Is this AI? Five players injured or suspended? Only two. That would then — wow, they've never played each other before? Interesting.
18. **F:** You're actually the first person to look at the squad section. The injury data has a

- note because official injury reports are only released 48 hours before the match. So right now you can't fully rely on that information. The note is there, but apparently not prominent enough — otherwise you'd have noticed it and thought 'I can't fully work with this yet.'
19. **P:** Yeah, it should maybe be a bit more visible. But it is interesting that I went there.
20. **P:** One thing that bothers me: the green 'prediction successfully saved' banner doesn't go away. I think it would be nicer if it disappeared after two or three seconds, or if you could dismiss it. Or — what if the prediction entry itself turned green once saved, and then the banner disappeared?
21. **F:** What if the score field itself changed colour — like turning green — when saved?
22. **P:** Yes, I like that too. Or black. Some kind of colour change to show that something has been updated. Or just being able to dismiss it. I think those are both solid options.
23. **F:** Great — that was already the next scenario. You moved through it really quickly. Well done.

TASK 3 – Checking the Leaderboard

24. **F:** Find out what position you're in within the league, and tell me your current position and how many points separate you from the next person.
25. **P:** I go back from match details and then to My League. I'm currently in first place — Toys 42. But we're all tied on points — Teufeltipper, Larissa Winsz, Lukas Brühl, they all have three points and 0.6 average. I'm not sure why I'm first. Probably because I joined last. But maybe if we're tied on points it would make more sense to show the same rank number for everyone.
26. **F:** That's good feedback — it's come up before. Of course in a working product you'd start with zero points, but since we're testing mid-season the data situation is a bit messy.

TASK 4 – Finding & Changing a Setting

27. **F:** You remember the app asked you about quick versus deliberate tipping at the start — you intuitively would have chosen quick picks. Find that setting and change it to the opposite.
28. **P:** Right — I saw it somewhere just now. I'm on the home screen. My Predictions... actually, let me think. It would have to be in My League. My Statistics — no. League

Settings — league-wide settings. Prediction Settings — there it is. Quick picks. I've changed it.

29. **F:** Perfect.

TASK 5 – Viewing Other Players' Predictions

30. **F:** Find out what the other predictors in the league tipped for the recent Champions League matches.

31. **P:** My League — and then I'd click on 'Last Matchday' — could look at how the league looked then. But probably I'd just tap... [navigates] not that... Tips at the top. There are the tips from the last matches. These are Third Qualifying Round — oh, that's from the very beginning. Interesting. But there it is — the matchday tips. No one tipped anything, weirdly. In the semi-final, Arsenal vs Atlético, there were tips and people got points. But what's interesting is that clicking on a player to see their individual tips is something I also tried — and I think about four out of six people so far have done the same. Very revealing.

32. **F:** Exactly — that's really useful data.

TASK 6 – Sending a Chat Message

33. **F:** Use the chat and write something to someone from your prediction league.

34. **P:** I go back. There's Chat. I'll send Erik Bergheimer a message. I tap on his name — he even has a profile picture. I write: 'Hello, shall we test the app?' Sent — 19:56.

TASK 7 – Inviting a Friend

35. **F:** Last task: you want to invite another friend to the league. Find the invite function and share the link or code.

36. **P:** Back to My League — top right, there's a share icon. There's the code and I tap Share. And then I can share it to whoever I want on various platforms. Fantastic.

37. **F:** Great — that was the prototype. Now on to the questionnaire.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P06 completes AttrakDiff — notes unfamiliar German vocabulary (e.g. 'schwerfällig', 'ungehorsam') and requests translations]

INTUI (Sentence Completion)

[Observer note: P06 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P06 completes SUS]

NPS

[Observer note: P06 completes NPS — researcher reads out the question and notes the response]

DEMOGRAPHIC INFORMATION

38. **F:** How old are you?

[Observer note: P06 responds — exact age not clearly recorded]

39. **F:** Gender identity?

40. **P:** Male.

41. **F:** How would you describe your football interest — casual fan, regular fan, passionate supporter?

42. **P:** Passionate fan, unfortunately.

43. **F:** Which prediction apps have you used before?

44. **P:** I've used Tipico at some point — does that count? And Kicktipp.

45. **F:** How often did you use Kicktipp?

46. **P:** Weekly.

47. **F:** Smartphone OS?

[Observer note: P06 responds — iOS or Android not clearly recorded]

48. **F:** Technical competence self-rating?

[Observer note: P06 responds — exact number not clearly recorded]

49. **F:** Have you used a betting app before, like Tipico or Bet365?

[Observer note: P06 confirms yes — Tipico mentioned earlier]

OPEN INTERVIEW

50. **F:** Looking back at the entire session — what's your overall impression of the app? What stands out, both positively and negatively?

51. **P:** I found it nice — simply and nicely designed. That's what stuck with me the most. It was easy to use. I'm not sure if 'haptics' is the right word, but I had the feeling that when you tap something, something responds. That worked really well. In terms of

- negatives: on first impression it's a bit — not cluttered, but there's a lot of information and a lot going on at once. But I think that's normal for new apps. Once you know where everything is, you know where everything is.
52. **F:** There were moments where the app deliberately slowed you down — for example, asking you to justify your prediction or to hold to save. Did those moments feel special, or were they annoying? Did they change your interaction or your prediction behaviour?
53. **P:** Definitely. The justification prompts and the multiple-choice reasons made me think more carefully. I'd already placed my prediction and then looked at the stats — the justification step probably made me reconsider or think harder about my pick, rather than just tapping something and moving on. And the hold-to-save mechanic — that felt very special. I liked it. It's like a payment confirmation — that longer press makes you think for another second, or at least makes you consciously aware that you're committing to something.
54. **F:** Did you feel in control of your experience the whole time, or did the app ever push you somewhere you didn't choose or couldn't get out of?
55. **P:** No — I always felt like I could go wherever I wanted. There was no dead end or loop. I was in charge throughout.
56. **F:** Did the match data and statistics make you feel more competent and better informed while predicting, or was it more like extra noise you didn't really pay attention to?
57. **P:** Better informed, definitely. Not necessarily more competent, but definitely better informed.
58. **F:** Did the social features — the leaderboard, the chat — feel particularly meaningful? Did you feel connected to the other players, or were they more like optional extras?
59. **P:** The chat function I think is an extra. Something I personally wouldn't necessarily need. But I can imagine — depending on the group and context — it could be very useful. Especially the group chat. Then again, I wonder why you'd write there beyond making fun of people who predicted badly. [laughs] My first instinct is that I wouldn't need it — but I can see how it might get used.
60. **F:** How does this app compare to apps like Kicktipp that you've used?
61. **P:** Much better. I find it much nicer than Kicktipp visually. Kickbase I also remember

- as a nice, modern app — but haven't used it in a long time. Compared to Kicktipp specifically: significantly better looking. The haptics are much more pleasant. And of course there were no ads. That definitely counts too. Overall — nicer, more enjoyable.
62. **F:** Is there anything else you'd like to share that we haven't covered?
63. **P:** Maybe — I found it a bit plain once you were on the actual main screens. The onboarding with all the club crests, flags, and colours was nice and visually engaging. But once you were actually in the league, it felt quite flat. I could imagine, for example, a strip at the top showing your favourite club and your country — something personalised. Because the top felt a bit empty to me.
64. **F:** Very cool — thank you so much. You'll hear from me. Fantastic. Get the app finished by next year.
65. **P:** We'll see.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, exact age, OS, technical competence rating) were unclear in the original recording and have been summarised or marked accordingly.

B.7 P07 – Session Transcript

In-Person Session | 21:43 | Language: German

PreMatch User Study | In-Person Session | 21:43

BRIEFING & CONSENT

1. **F:** Welcome to my user study. You've been invited to take part in my user study for my Master's thesis at MCI Innsbruck. This study evaluates a high-fidelity prototype of a new social football prediction app. The research investigates how principles of self-determination theory and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
2. **F:** What will happen: you'll receive a series of realistic tasks in the prototype. While completing them, it's important that you think out loud and describe what you see and expect. After that you fill out a short questionnaire, and then we have a short open conversation about your experience. The total duration shown in this document is 45 to 60 minutes — it will be significantly shorter, around 25 minutes on average.
3. **F:** What is being tested: the app, not you. There are no right or wrong answers. If something is confusing or doesn't work the way you think it should, that's exactly the feedback that helps improve the design — and your honest reactions are extremely important. Please really say what you think and what you'd expect. Your data will be pseudonymised and stored under a participant ID — something like P0 followed by a number. Your name will not be recorded. Session recordings are used exclusively for my analysis and are not shared publicly. Anonymised quotes from your answers may be used in my Master's thesis. Data processing follows GDPR and you can request deletion of your data at any time. Participation is completely voluntary — you can end the session at any time without giving a reason and without any negative consequences. You can also withdraw your data after the session by contacting me via email. Results will be published as part of my Master's thesis at MCI Innsbruck, likely sometime in autumn. If you'd like a summary of the results, please let me know. If you have any questions before, during, or after, feel free to reach out — you have my email and number, Julia. If you've read and understood all of that, please tick all the boxes and then we can get started.

[Observer note: P07 gives consent]

[*Observer note: Researcher notes: Language: German, In-Person, 21:43. Starts screen recording.*]

4. **F:** As mentioned, you'll be asked to complete a series of tasks. Please think out loud, describe what you see, what you're trying to do, and what you expect. There are no right or wrong answers — take the time you need. Let's start with the first scenario.

TASK 1 – Account Creation & Joining a League

5. **F:** You're opening this social football prediction app for the first time. You want to predict in a league with your friends. Create an account with your name and a fictional prediction username, and your own email address. Then join an existing league with the invitation code YW5UH. Note that Google and Apple sign-in don't work yet — you can only sign in via email.
6. **P:** Okay — sign in with email here at the top. [taps] Password. I can see the verification too. I'll tap Continue. Language: German. Username — what do I enter as two usernames? Luca 187 — my first name and my birthday. The only thing that might be slightly misleading is that it says 'Name' here, not 'Username'. That could make someone think you're asking for very personal information. I could read it as the username, but it could also be misleading.
7. **F:** Good observation.
8. **P:** Prediction username — that's the one shown when active in a league, right? Then I'll just not use the same as my name. Age: 21. National team: Germany. Favourite league: Bundesliga.

[*Observer note: Technical issue — app navigates back to start unexpectedly; P07 re-registers with a slightly modified email address*]

9. **P:** Okay, doing it again. Username and display name. Done. Age: 21. National team: Germany. Are there more leagues to choose from, or just these for now?
10. **F:** Just these for now unfortunately.
11. **P:** What would I personally want? I'd pick the third division — my home club plays there. I'd also say first division is the most exciting to follow. But once you select a league, the next step automatically shows the clubs from that league — and there's no search field to find a club that isn't in that league. But I'll just continue. I'll go with Bayern Munich.

12. **P:** Now join a league with the invitation code. [enters YW5UH] How do I prefer to tip — deliberate. [selects] Okay, a quick section on how everything works. Where to find things. I think I'd choose 'following league settings'. Let's go. End of first scenario.

13. **F:** Great. You can set it down for a moment.

TASK 2 – Exploring the App & Placing a Prediction

14. **F:** You're now in the prediction round. Have a look around, get an overview of the fixtures, and then find the Champions League Final between Paris Saint-Germain and Arsenal and place a prediction for it. Follow all the steps the app shows you.

15. **P:** Okay — so the match against Paris and Arsenal, I should look at it and place a prediction. Down here at the bottom it's already suggested under 'Matches' with a countdown running — it's highlighted, which is nice. I tap on it. I need to place a prediction. [scrolls] Squad section — probably not relevant here. [taps into the score fields] I'll go with 3-1 for Paris. 'Hold to save your prediction' — I press the button. Something's happening — good. 'Why this prediction?' — six options are shown. I tap the question mark out of curiosity. 'To help...' Okay, that's the explanation for the prompts. Cancel. I haven't actually looked at any statistics, so I'll go with gut feeling and strong form — both work. And then I tap save. Done.

16. **F:** Perfect. That was the scenario already.

TASK 3 – Checking the Leaderboard

17. **F:** Find out what position you're currently in within the prediction league CL, and tell me your current position and how many points separate you from the next person.

18. **P:** I go back to the home screen and then to My League. I'm shown as first place — 'You' next to my name. I have three points and an average of 0.6.

TASK 4 – Finding & Changing a Setting

19. **F:** You remember that the app asked you to choose between quick and deliberate tipping in the settings at the start. Find that setting and change it to the opposite.

20. **P:** Home screen — nothing there. My Predictions — I try 'League Settings'. That's only league-wide settings. I go up to my profile icon. My Account — I don't think I'd find it there. Prediction Settings — yes, quick picks. I'll change that. Done. Going back one step. Done.

TASK 5 – Viewing Other Players’ Predictions

21. **F:** Find out what predictions the other players in your league submitted for the semi-final, and how many points they received for them.
22. **P:** I go back to My League. I tap on the player below me — Teufeltipper. Nothing happens. Last Matchday — also nothing when I tap it. [navigates] Semi-final — that’s what I predicted earlier, right? No, that was the final. Ah, here — Tips. That’s the Third Qualifying Round — this is the final. [uses arrow navigation] Here I’ve found the semi-final — Atlético Madrid vs Arsenal. And those are the only ones with predictions for that match. Arsenal vs Atlético — Larissa predicted 2-1. Lukas predicted 2-1 as well.
23. **F:** Yes, perfect.

TASK 6 – Sending a Chat Message

24. **F:** Write a message to someone from your prediction league.
25. **P:** Chat. Group chat for the whole league, or one specific person?
26. **F:** One specific person.
27. **P:** I go to the 1-on-1 chat and pick Teufeltipper. A chat opens and I can write to them. I type: ‘Sucker.’ [sends]

TASK 7 – Inviting a Friend

28. **F:** Last task: you want to invite another friend to your prediction league. Find the invite function and share the invite link or code.
29. **P:** I go back to My League and I can see a symbol top right that looks like adding something. [taps] That’s right. I tap Share. [share sheet appears] And now I can choose who to send it to on various platforms. Done.
30. **F:** Great — that’s the prototype done. Let’s end the screen recording. Now on to the questionnaires.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P07 completes AttrakDiff independently]

INTUI (Sentence Completion)

[Observer note: P07 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P07 completes SUS]

NPS

[Observer note: P07 completes NPS including written justification]

DEMOGRAPHIC INFORMATION

[Observer note: P07 completes all demographic fields in-app — age, gender, football interest, prior app usage, OS, technical competence, betting app usage]

[Observer note: Researcher notes: P07 mentioned Tipico (betting app) when asked about prediction apps; also used Kicktipp]

OPEN INTERVIEW

31. **F:** Looking back at the whole session — what was your overall impression? What stood out positively and negatively?
32. **P:** Positively: the UI elements — everything was pretty self-explanatory. Finding things was fine, the settings took a tiny bit of searching but it also might have been that I didn't look carefully enough — it wasn't actually hard to find. The statistics feature when predicting — I found that cool too. The only thing I'd flag is the visual design — it's not dull, but in terms of colour it's maybe a bit... unremarkable.
33. **F:** There were moments where the app deliberately slowed you down — asking you to justify your prediction or hold to save. Did those moments feel significant, or more like an annoyance? Did they change your interaction?
34. **P:** I was briefly surprised by having to hold the button, but then I realised it's actually not a bad idea — because you can quickly mis-tap or accidentally hit send. It wasn't too long either. That was fine. As for the justification — you mean where I had to tap options?
35. **F:** Exactly — you chose gut feeling and team strength.
36. **P:** Why did I do that? I mean it didn't really change anything in my interaction, but I noticed the app wanted an extra piece of information from me, so I selected it. I quickly scanned the options and picked what felt most fitting for my prediction. It didn't make me feel more uncertain or anything. I didn't feel like I'd chosen something wrong.
37. **F:** Did the match data and statistics make you feel more competent and better informed

- while predicting, or was it more like extra noise?
38. **P:** I didn't look at them specifically this time, but I found it cool that they were there. I'm not a particularly experienced predictor and I haven't followed football super closely recently — but I can imagine that for people who do this regularly or seriously, it's a really cool feature.
39. **F:** Did the social features — the leaderboard and chat — feel particularly meaningful? Did they make you feel more connected to the other players, or were they more like optional extras?
40. **P:** More like additional extras — but cool extras.
41. **F:** How does this app compare to other apps like Kicktipp that you've used?
42. **P:** As far as I can remember — it's been over five years. I feel like this app felt a bit more dynamic. Everything ran more smoothly. More dynamic, more fluid — I think it looks quite good.
43. **F:** Is there anything else you'd like to share that we haven't covered?
44. **P:** Maybe: during the tasks I kept trying to tap on players in the leaderboard — in my prediction group. It might be quite cool if you could get to their profile from there, or at least see what they've predicted in recent matches within the group.
45. **F:** Very cool. Thank you so much — that's everything.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, exact demographic responses) were unclear in the original recording and have been summarised or marked accordingly.

B.8 P08 – Session Transcript

In-Person Session | Language: German

PreMatch User Study | In-Person Session

BRIEFING & CONSENT

1. **F:** Welcome to my user study for my Master's thesis at MCI Innsbruck. My study evaluates a high-fidelity prototype of a new social football prediction app, and my research investigates how principles of self-determination theory — SDT — and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
2. **F:** What will you do? You'll receive a series of realistic tasks in the prototype. While doing them, it's very important that you think out loud, describe what you see and what you expect. After the tasks you fill out a questionnaire, and then there's a short open conversation about your experiences. The total duration won't be the 45–60 minutes listed here — the average has been around 25 to 30 minutes. Most importantly: the app is being tested, not you. There are no right or wrong answers. If something is confusing or doesn't work as expected, that's exactly the feedback I need to improve the design — and your honest reactions are the most valuable data for my study.
3. **F:** Data protection: your data is pseudonymised and stored under a participant ID. Your name is not recorded in any file. Session recordings are used exclusively for research analysis and are not shared publicly. Anonymised quotes from your answers may be used in the thesis. Data processing follows GDPR and you can request deletion of your data at any time. Participation is completely voluntary — you can end the session at any time without giving a reason and without consequences, and you can also withdraw your data after the session. Results will be published as part of my Master's thesis at MCI Innsbruck. If you'd like a summary of the results, let me know. Please tick all the boxes to confirm and then we can get started.

[Observer note: Side conversation: P08 notes the researcher could've just had them read and sign a consent form]

[Observer note: P08 gives consent. Researcher starts screen recording.]

4. **F:** You'll now be asked to complete a series of tasks in the app. Please think out loud,

describe what you see, what you're trying to do, and what you expect. There are no right or wrong answers. Take the time you need. This is a football prediction app — you predict match results, not bet on them. You're now on the start screen as if opening the app for the first time.

TASK 1 – Account Creation & Joining a League

5. **F:** Open the app for the first time and create an account with your name and a fictional prediction username. You can use your own email address. Note that Google and Apple sign-in don't work yet. Afterwards, join an existing prediction league with the invitation code YW5UH. You can read the code from the screen later.
6. **P:** Signing in — I'm not thinking too much about this part, it's very clearly set up. You enter your email and password. I like that it's bilingual — but I want my native language. At 'Name' I'll enter 'Tipp Profi' — because I'm very confident in my prediction abilities. At the prediction username, I wonder what the difference is from the name field. I'll just enter the same thing. Age: 24. National team — we're not in Germany right now, but cool that you can just select it. Why only a limited selection of countries? I'll take Germany because of my superior knowledge there. Leagues — why are there two Bundesligas? Oh, one is the Austrian Bundesliga. I'll take the German one. Favourite club: VfB Stuttgart — because they're fantastic. Join the league — yes, that's what you said earlier. [enters code YW5UH]
7. **F:** Before you tap 'let's go' — what would you choose here, quick or deliberate tipping?
8. **P:** Probably deliberate, because it's already selected and you're already primed towards it.
9. **F:** Go ahead and continue.
10. **P:** Done — that was quick already.
11. **F:** Great, thank you. Second scenario now.

TASK 2 – Exploring the App & Placing a Prediction

12. **F:** You're now in a prediction league called CL — that stands for Champions League. Someone else created it and you've joined. The Champions League Final hasn't been played yet. You want to predict the final between PSG and Arsenal. Take your time to get an overview of the matches already played and the upcoming final, look at the

information the app gives you, and then place your final prediction.

13. **P:** Okay, let me see what I can find here. [navigates] I've already noticed the final, but I was figuring out how to actually place a prediction. I need to get an overview first since I missed the last few football matches. What information is the app giving me right now?
14. **P:** The win probability — it's showing Arsenal quite high. And there's an info button right there — I was just wondering what the win probability is and how it's calculated, and then there was the info button directly. That was very intuitive.
15. **P:** And this radar chart is nice — it shows the general strengths and weaknesses of each team and their direct comparison, so you can think about what matters for this specific match. The red highlighting — does that mean whichever metric is higher than the other team's? Okay. I'd assume the abbreviations are clear to football people. Defense, obviously. Maybe it could be cool to also score or rate each area, or show which player is currently on form — like a standout performer this season, rated 1–10 or something.
16. **F:** Good idea — showing current in-form players, rated across the season?
17. **P:** Yes, or just: this player has done really well this season. And then try placing a prediction with the data you've just looked at.
18. **P:** Okay — the win probability shows Arsenal higher. But what confuses me is that looking down here, the statistics actually seem better for Paris Saint-Germain. Do those two things not connect?
19. **F:** They come from the same data provider. The win probability also factors in team form, defensive strength — Arsenal's defensive stats in the spider chart are very strong, while Paris excels in attack. So it all feeds from the same source.
20. **P:** Okay — I think it'll be close. Both very strong teams. But I'm going for Arsenal: 2-1. [places prediction] Now this banner bothers me — it's not going away. [tries to dismiss it]
21. **F:** You can tap Continue.
22. **P:** Let me re-enter the 2 for Arsenal — there we go. And now this reminds me of Instagram's 'are you sure you want to post' prompt — like a breath before committing. I went with statistics of course, so the others will see 'Lisa predicted 2-1 based on

statistics.’ [saves prediction] Done.

23. **F:** Perfect, that was the scenario.

TASK 3 – Checking the Leaderboard

24. **F:** Find out what position you’re currently in within the league with the other predictors. Tell me your current position and how many points separate you from the next person.
25. **P:** Back to the main screen, then My League. I can see Teufeltipper is leading, and I’m in 13th place. But actually everyone has roughly the same four points. It might be cool to have little arrows showing whether you’re moving up or down — like in Kahoot, you know, the trend indicator. Can I see the others’ predictions now?
26. **F:** Not yet — but good that you asked. Let’s continue.

TASK 4 – Finding & Changing a Setting

27. **F:** You remember that at the start I briefly asked what you’d choose — quick or deliberate tipping. Find that setting and change it to the opposite.
28. **P:** I go to Profile — settings are usually there. Why does the accessibility icon look like a bikini? [laughs] I briefly considered tapping ‘My Account’ but then ‘Prediction Settings’ caught my eye and it was obvious. I tap on it and select quick picks. Saved — I also like that you don’t need to confirm again separately. Done.

TASK 5 – Viewing Other Players’ Predictions

29. **F:** Find out how the others in the league predicted the Champions League semi-final.
30. **P:** I’ll look in My League. I tap on Tips. These matches — I can’t do much with them because [navigates through rounds] I go to Matchday Tips — I’d have expected something to happen here. It would be nice if there were a bit more shown there, or some explanation of why I can’t see it yet. I’m assuming Lukas’s tip would appear here if I could see it.
31. **F:** Right — you can’t see the predictions yet because the match hasn’t been played. But you’re right that it’s not explained. You haven’t found the semi-final yet though — want to try again?
32. **P:** Last Matchday maybe. [tries] Okay, never mind.
33. **F:** It would’ve been in the Tips section, scrolling through the rounds at the top. That navigation is apparently not obvious enough — you’re not the first to miss it. A

dropdown or some kind of navigation pattern there might help.

34. **P:** A dropdown, or something at the edge. I didn't really notice you could scroll/tap through there. Maybe make it more consistent with the navigation used elsewhere in the app — you already use a similar pattern somewhere else, so applying it there too would help.

TASK 6 – Sending a Chat Message

35. **F:** Write a message to another user in your league.
36. **P:** Chat — that's logical. 1-on-1, because I want to be mean. Teufeltipper — they're at the top and doing well, so I'll write to them. I can attach something — let me check — a photo, that's cool. Stickers — why Padini stickers? [laughs] I write: 'Watch out.' [sends] Can you have read receipts like WhatsApp?
37. **F:** Not yet — do you think that would be a nice feature?
38. **P:** Yes. And a sent confirmation too. It's just something you're used to from every other app these days.

TASK 7 – Inviting a Friend

39. **F:** Last task: you want to invite a friend to join the league. Find the invite function and share the link.
40. **P:** I go to my Profile — that's where you usually find that kind of thing. My Account — no, that's just for profile picture. Home screen — checking there. Back to Profile. [doesn't find it] I wouldn't end up inviting my friend.
41. **F:** You'd find it in My League, top right — the invite function. From there you can share the link. That's a known discoverability issue.
42. **P:** Got it.
43. **F:** That's the prototype done — on to the questionnaire now.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P08 completes AttrakDiff independently]

INTUI (Sentence Completion)

[Observer note: P08 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P08 completes SUS]

NPS

[Observer note: P08 completes NPS — written response: noted the app is very well designed and intuitive, good even for users less familiar with statistics]

DEMOGRAPHIC INFORMATION

[Observer note: P08 completes demographic fields in-app — prior to open debrief. Mentioned Kicktipp as prior experience.]

OPEN INTERVIEW

44. **F:** Looking back at the session — what's your overall impression? What stood out positively and negatively?
45. **P:** Positively: the UI elements — everything was basically self-explanatory. Finding things was easy; the settings took a tiny moment to locate, but it might have been that I didn't look carefully enough — it wasn't actually hard. The statistics when predicting — I found that cool. The only thing I'd flag is the colour design — it's not dull, but maybe a little unremarkable visually.
46. **F:** There were moments where the app deliberately slowed you down — asking you to hold to save, and then to justify your prediction. Did those moments feel meaningful, or more annoying? Did they change your interaction?
47. **P:** It made me think again about whether what I was predicting was actually right — a kind of reflection, so you're not just deciding purely on gut feeling. I think that's also somewhat unique to a prediction app like this, because it's built around emotions and engagement. Maybe it could even be pushed a little further — two seconds of holding doesn't feel like it triggers that much in you, or maybe people have already become so desensitised to confirmation gestures from other apps. Maybe the barrier should be even higher — like a pop-up: 'Are you sure? Here are the consequences' — or something like that.
48. **F:** Did you feel in control the whole time, or were there moments where the app pushed you in a direction you hadn't chosen?
49. **P:** No — I didn't find any element that was overly dominant or pushy. I think it helps that the design isn't visually loud. Too many different colours can be distracting, and here everything was relatively balanced in size and weight — nothing pushed you

toward one option more than another.

50. **F:** You looked at the match data and statistics quite carefully. Did that make you feel more competent and better informed, or was it more like extra information on top?
51. **P:** It definitely made me feel more competent. And it reinforced my decision — I felt like I was better justifying my choice, and more confident that the decision was grounded in something. Maybe you could even extend that into a kind of advisory layer: 'In matches where you based your pick on X, your results were better/worse.' Like a retrospective insight. So if the goal is to nudge users toward statistics-based picks rather than pure gut feeling, you could show them: when you used statistics, your accuracy was Y%. And maybe tie it back to the justification step — like 'last time you were in this situation and went with statistics, here's how it turned out.'
52. **F:** Could it also be interesting to make it time-based? If the app detects that you've entered a prediction within 10 seconds, it triggers a gentle check-in?
53. **P:** Yes, though that's kind of related to what we already discussed. I did wonder about explicitly prompting what to look at, but then everyone would look at the same things — and then there's less point in predicting individually.
54. **F:** Did the social features feel particularly meaningful, or more like optional extras?
55. **P:** The chat is cool — and the photo sharing is actually fun. The fact that it's not just text and also supports voice messages is nice. You also jokingly mentioned Panini stickers earlier — do you have an actual idea there?
56. **F:** Yes — something like that could be cool. And you also mentioned the idea of team challenges within a league — like Jannik and I competing as a team against you and Marc. That could add to the community feeling.
57. **P:** Yes, exactly — a 2- or 3-person team format. That's a good idea.
58. **F:** You mentioned Kicktipp as prior experience — how does this app compare?
59. **P:** Clearly better. [laughs]
60. **F:** Is there anything else you'd like to share that we haven't covered?
61. **P:** I'm actually quite surprised — and I find it very user-friendly, even for people who don't do this often or don't know their way around statistics. I don't know how many apps integrate statistics this strongly. And seeing how you yourself have performed — I didn't look at that very carefully. It would be interesting to reflect on: how did

I do in the last few matches? I think there's something there that could be pulled out better — personal prediction statistics. Like your points total and accuracy trend across recent matchdays.

62. **F:** Very cool — thank you so much. That's everything.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, exact demographic responses) were unclear in the original recording and have been summarised or marked accordingly.

B.9 P09 – Session Transcript

In-Person Session | Female, 26 | Language: German

PreMatch User Study | In-Person Session | Female, 26

BRIEFING & CONSENT

1. **P:** Hello, I'm *Name of the Participant*, 26 years old, and I'm ready.
2. **F:** Welcome to my user study for my Master's thesis at MCI Innsbruck. My study evaluates a high-fidelity prototype of a new social football prediction app, and my research investigates how principles of self-determination theory — also called SDT — and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
3. **F:** What will you do? You'll receive a series of realistic tasks. While completing them, your job is to think out loud and describe what you see and expect. After that you fill out a questionnaire, and then there's an open conversation about your experience. The listed duration of 45–60 minutes is way too high — most people have taken around 25–30 minutes. What's being tested: the app, not you. There are no right or wrong answers. If something is confusing or doesn't work as expected, that's exactly the feedback I need to improve the design. Your honest reactions are the most valuable data in my study.
4. **F:** Data protection: all data is pseudonymised and stored under a participant ID. Your name won't appear in any file, and I'll remove your name from the transcript. I'll record the session via audio and screen recording, used exclusively for research analysis and not shared publicly. Anonymised quotes from your answers may be used in the Master's thesis. Data processing follows GDPR and you can request deletion of your data at any time by email or phone. Participation is completely voluntary — you can end at any time without giving a reason and without consequences. You can also withdraw your data after the session by contacting me. Results will be published as part of the Master's thesis at MCI Innsbruck, likely in autumn. If you'd like a summary of the results in advance, feel free to reach out. Please tick all the boxes to confirm.

[Observer note: P09 gives consent. Researcher fills in session details, starts screen recording.]

5. **F:** In this part of the study you'll be asked to complete a series of tasks in the app. Please think out loud, describe what you see, what you're trying to do, and what you expect. There are no right or wrong answers — take the time you need. The context is that you're opening a new football prediction app for the first time, similar to Kicktipp, which you've probably used before.

TASK 1 – Account Creation & Joining a League

6. **F:** You're on the start screen, opening the app for the first time. Create an account with your name, a fictional prediction username, and your email address. Note that Gmail and Apple sign-in don't work yet — just use a regular email. Then join an existing prediction league with the invitation code YW5UH, which you can read from my screen if needed.
7. **P:** I'm entering my email now — hopefully without typos — and I'm thinking of a password. [tries to log in] What's the app called? PreMatch. My email wasn't accepted. [tries again] The workspace version worked. I'd like to use the app in German. My name is Emma. Good name choice, if I do say so myself. I'm 26. My national team is Germany — though I have to say, Italy is also a favourite, but they're such blues, so that's not happening. I'll say Germany. My favourite league is the Bundesliga. My favourite club is FC Augsburg. Now joining — the code was YW5UH. That's usually how you get it when a friend shares the league link.
8. **F:** Before you tap Continue: how do you prefer to tip — quick or deliberate?
9. **P:** Quick. My head works that way — I'm a very intuitive person. I'd definitely just go with quick.
10. **F:** Interesting. In this case please tap deliberate — we'll come back to that later. And then continue.
11. **P:** Okay, I'm in. CL — Champions League, I guess, that's what we're predicting today. 'How it works' — points for correct result or tendency. Where to find things: place predictions, follow the table, league settings. Okay, done.
12. **F:** That was the first scenario. Well done.

TASK 2 – Exploring the App & Placing a Prediction

13. **F:** You're now in the prediction league CL — the Champions League final between Paris and Arsenal hasn't been played yet. Get an overview of all the matches and past

results, and then find the Champions League Final between Paris Saint-Germain and Arsenal and place your prediction. Follow all the steps the app shows you and use any information the app provides.

14. **P:** Okay, I'll get an overview first and then find the match. [navigates] This is actually pretty clear. If I'd already tipped here I'd have seen it. I keep coming back to this matchday view. Match details — interesting. I can see my tip. I can chat with people — even all members at once, or one-on-one. I can change the language. Accessibility with the glasses icon — nice. Sorry, should I be doing all of this?
15. **F:** Not necessarily, but it's all good. The dark mode isn't 100% finished yet, but feel free to try it. The accessibility features — you're actually the first person to look at those, and they took a lot of work.
16. **P:** I'll turn on bold text — I like that. Okay, now the match between Paris and Arsenal. [navigates to the final] I'll just say — Paris should only win at 10% probability, but I'm always rooting for the underdog, so I'm going with Paris winning 4-0. Hold to save — I have to hold this? [holds] 'Why this prediction?' — gut feeling. And also statistics, because I'm going against the statistics. Does that make sense?
17. **F:** Yes, you can select both.
18. **P:** Save. Done.
19. **F:** You chose gut feeling and statistics — we'll come back to why that matters later.

TASK 3 – Checking the Leaderboard

20. **F:** Find out what position you're in within the league.
21. **P:** I'm in 13th place. Why? Why do I only have two points — did I not get the exact result? [looks at data] I don't even know how many matches have been played. The match hasn't happened yet, but how do I have points at all?
22. **F:** That's because it's a prototype — the data is pre-loaded.
23. **P:** Oh right, makes sense. Anyway — I'm 13th. I'll end up in 1st. Teufeltipper had better luck than me.

TASK 4 – Finding & Changing a Setting

24. **F:** You remember the app asked whether you wanted quick or deliberate tipping — you wanted quick, but I asked you to select deliberate so we could test a feature. Now change the setting to the opposite.

25. **P:** [navigates directly to Prediction Settings] Found it. Done.
26. **F:** Great — that was quick.

TASK 5 – Viewing Other Players’ Predictions

27. **F:** Find out what the other players predicted for the Champions League semi-final.
28. **P:** Easy — I’ll go to My League and tap Tips. Then I’ll go to the Final. And there are question marks everywhere, because that match hasn’t been played yet. I can only see tips after the match — otherwise I could just copy them. I’d never do that in a prediction round at the World Cup. [laughs]
29. **F:** That’s actually the first time anyone has understood that on their own — that the tips aren’t visible because the match hasn’t happened. Everyone else assumed no one had tipped yet.
30. **P:** Maybe the others didn’t read the rules carefully enough — or there’s something in the design that could make it clearer.

TASK 6 – Sending a Chat Message

31. **F:** You’ve already found the chat — but please write a message to one specific person from the league.
32. **P:** [navigates to 1-on-1 chat, selects Teufeltipper] Oh, the text colour is still white — I’ll write ‘Good luck, 1st place.’ [sends]

TASK 7 – Inviting a Friend

33. **F:** Last task: invite another friend to the league. Find the invite function.
34. **P:** [navigates to My League, taps the share icon top right] I could even invite myself — that’s fun. [shares link] Done.
35. **F:** That’s everything — you were the fastest participant so far, which is great. Now on to the questionnaires.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P09 completes AttrakDiff, thinking out loud throughout — selected responses include: human, practical, structured, predictable, novel, bold, creative, fascinating, connecting, motivating, sympathetic, inviting]

INTUI (Sentence Completion)

[Observer note: P09 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P09 completes SUS — remarks include: 'Kicktipp will be discontinued', 'magical experience']

NPS

[Observer note: P09 completes NPS — written justification provided]

DEMOGRAPHIC INFORMATION

- 36. **F:** Age?
- 37. **P:** 26.
- 38. **F:** Gender?
- 39. **P:** Woman.
- 40. **F:** Football interest — casual fan, regular fan, passionate supporter?
- 41. **P:** Regular fan — passionate is a bit of a stretch.
- 42. **F:** Which prediction apps have you used before?
- 43. **P:** Kicktipp. And there was another one — blue and white design, I used to tip with my dad. TeamTipp, I think?
- 44. **F:** Yes, TeamTipp — blue and white.
- 45. **P:** Yes, exactly, that one.
- 46. **F:** How often did you use them?
- 47. **P:** During the World Cup and similar tournaments — multiple times a week then. Not daily in everyday life.
- 48. **F:** Smartphone OS?
- 49. **P:** iOS.
- 50. **F:** Technical competence self-rating out of 5?
- 51. **P:** Very high — 5.
- 52. **F:** Have you used a betting app before?
- 53. **P:** No, never. I'm against betting.

OPEN INTERVIEW

- 54. **F:** Looking back at the whole session — what's your overall impression? What stood out positively and negatively?

55. **P:** Positively: everything was intuitive. I didn't have to think long about anything. The app invites you to explore — to tap through the icons and see what's where. Settings were clear and organised. Negatively: nothing obvious came to mind. Maybe that I couldn't create a profile picture right away. That's always a thing for me. Oh — and my actual wish would be: not just a profile photo, but also being able to create a custom avatar. That would be really cool. Like your own little mascot with a jersey — similar to Apple Memoji. Your face, your style, and then you can switch the jersey depending on the competition. Like wearing the Germany shirt during the World Cup, then your club's shirt in the regular season.
56. **F:** There were moments where the app deliberately slowed you down — asking you to hold to save, or to justify your prediction. Did those feel meaningful, annoying, or did they change your interaction?
57. **P:** At first I found it a bit odd, but then I thought — actually it's quite cool to look back on. Because I'm a numbers person. I love data — I have a Garmin watch and I find it really satisfying when after two months I can see: when I tipped on gut feeling vs. statistics, what worked better? I'd find it really interesting to see that kind of retrospective analysis. Like: how often did you go with gut feeling and how often were you right? That I'd find genuinely cool. And I think that's also the plan, right?
58. **F:** Yes, exactly — that would be the intended direction. You'd eventually get a breakdown of when you tipped what and how your accuracy compared across different reasoning types. And one suggestion that came up a couple of times from other participants was also adding a free-text field to the justification prompt — for when your reasoning doesn't fit the given categories.
59. **P:** Oh yes, absolutely.
60. **F:** Did you feel in control throughout, or were there moments where the app pushed you somewhere you didn't choose?
61. **P:** No — I felt in control the whole time.
62. **F:** Did the match data and statistics make you feel more competent and better informed, or was it more of a nice extra?
63. **P:** The squad section and everything around it — I found that great. Really cool.
64. **F:** Did the social features — the leaderboard and chat — feel particularly meaningful,

or more like optional extras?

65. **P:** I like them. And actually — something like on Instagram, where instead of sending a Reel you send someone's prediction — like 'mate, were you drunk?' — would be really cool. So you could tap on an individual prediction and share it directly in the chat. Like: at the Arsenal match someone tipped 4-0 for Arsenal — I see that, tap on it, and send it into the chat with a message like 'Luca, what was that?'
66. **F:** Very cool. Would you prefer that to happen in the app's chat, or shared out to WhatsApp?
67. **P:** In the app. It has more charm that way. You open the app, you click send — but it stays in the app, not redirected to WhatsApp. That's the same annoying thing as when someone sends you an Instagram Reel via WhatsApp — it just feels like an unnecessarily convoluted path.
68. **F:** How does this app compare to Kicktipp or TeamTipp?
69. **P:** Solid. I think it fits right in — and honestly improves on them.
70. **F:** Is there anything else you'd like to share?
71. **P:** Chapeau to you — I found it really well facilitated. Respect. [laughs]
72. **F:** Thank you so much — that's everything!
73. **P:** Cool! Ciao ciao.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, partial verbal responses) were unclear in the original recording and have been summarised or marked accordingly.

B.10 P10 – Session Transcript

In-Person Session | 17:42 | Language: German

PreMatch User Study | In-Person Session | 17:42

BRIEFING & CONSENT

1. **F:** Are you ready? Good. You've been invited to take part in my user study for my Master's thesis at MCI Innsbruck. This study evaluates a high-fidelity prototype of a new social football prediction app. The research investigates how principles of self-determination theory — SDT — and positive friction can be applied in interaction design to create a more meaningful and ethically responsible user experience.
2. **F:** What will you do? You'll complete a series of realistic tasks in the prototype, thinking out loud while interacting and describing what you see and expect. Afterwards you'll fill out a short questionnaire, and then we'll have a brief debrief about your experience. The total duration is not 45–60 minutes — it'll be closer to 25–30. What's being tested: the app, not you. There are no right or wrong answers. If something is confusing or doesn't work as expected, that's exactly the feedback I want. It helps improve the design, and your honest reactions are the most valuable data.
3. **F:** All data is pseudonymised and stored under a participant ID. Your name won't appear in any file. Session recordings are used exclusively for research analysis and aren't shared publicly. Anonymised quotes may be used in the thesis. Data processing follows GDPR, and you can request deletion of your data at any time. Participation is completely voluntary — you can end at any time without reason or consequences, and can withdraw your data afterwards by contacting me. Results will be published as part of the Master's thesis at MCI Innsbruck, likely in autumn. If you'd like a summary of results, just let me know. Please tick all the boxes to confirm.

[Observer note: P10 gives consent]

[Observer note: Researcher fills in session metadata: Language: German, In-Person, 17:42. Starts screen recording.]

[Observer note: Brief technical interruption with screen recording — resolved]

4. **F:** In this part of the study you'll be asked to complete a series of tasks in the app. Think out loud, describe what you see, what you're trying to do, and what you expect. No right or wrong answers — take the time you need.

TASK 1 – Account Creation & Joining a League

5. **F:** You're opening the football prediction app for the first time. Create an account with your name, a fictional prediction username, and your email address. Note that Google and Apple sign-in don't work yet. Then join an existing league with the invitation code YW5UH.
6. **P:** Okay, I'll tap the email button. I'll enter my email now. [enters email] What I don't like: the password strength requirements aren't shown. Whether you need numbers, special characters, how many characters — that should be displayed. [attempts to continue] Oh right — it needs at least six characters. That should've been shown upfront. [creates password] Continue. I want German. [selects] Name — I'll use my own prediction username. Continue. Age: 27. My national team — I'm a Germany fan. But I do wonder why Italy is in there, since they didn't qualify for the World Cup. Same with Poland. That would be confusing in a real-world context.
7. **F:** Noted — that's a prototype data issue.
8. **P:** My favourite league: Bundesliga — I like the logos, you can immediately recognise them. Favourite club: Eintracht Frankfurt. [continues] Join league — you mentioned a code earlier.
9. **F:** Yes — YW5UH.
10. **P:** [enters code] Found it — 'join league'. Cool.
11. **F:** Before you tap the next button — quick or deliberate tipping?
12. **P:** I'd honestly go with quick picks.
13. **F:** Why?
14. **P:** Because I just click through and tip quickly. Deliberate sounds too slow to me.
15. **F:** For now, select deliberate — you'll find out why later.
16. **P:** [selects deliberate] Okay. Rules — okay. Let's go. Done — that was quick.

TASK 2 – Exploring the App & Placing a Prediction

17. **F:** Have a look at the upcoming matches, get an overview of the fixtures and all the information the app provides. Then find the Champions League Final between Paris Saint-Germain and Arsenal — we'll assume it hasn't been played yet — and place your prediction. Follow all the steps the app shows you.
18. **P:** Okay, let me have a look. Oh cool — the match is right there at the bottom. Win

probability only 10% for Paris. I like that it's explained — the pop-ups. Position — that's probably in the Champions League table. And I find it cool that you can look at the squad.

19. **P:** Okay, I'll go with 1-3 — you have to hold to save. That's cool. But yeah, with many matches on a Bundesliga matchday it could get a bit tedious. 'Why this prediction?' — that's cool. And it's good that you can change it directly here. I'll go with statistics and gut feeling. Save. Done.
20. **F:** Perfect, that was the scenario.

TASK 3 – Checking the Leaderboard

21. **F:** Find out what position you're currently in within the league CL, and how many points separate you from the next person.
22. **P:** I'd go back and then to My League. There are quite a few people — I'm in 13th place. But I only have two points, which feels a bit odd. I have one point more than Erik Bergheimer. I also think it's cool that people with the same number of points share the same position. And what I'd also like to see: green and red arrows to show if people are moving up or down — like a trend indicator.
23. **F:** Good idea. And the points are just prototype data — you haven't actually predicted anything yet.
24. **P:** Yes, makes sense.

TASK 4 – Finding & Changing a Setting

25. **F:** You remember the setting that controls how the app prompts you when tipping. Find it and change it to the opposite of your current selection.
26. **P:** I'd go to my profile. [navigates] League Settings — Tips — no. Prediction Settings — yes, there it is. Quick picks. [taps to change] Back. Done.

TASK 5 – Viewing Other Players' Predictions

27. **F:** Find out how the others in the league predicted the Champions League semi-final.
28. **P:** I'd go to My League. [taps on another player's name] Nothing. Then Last Matchday — that's just the table view. Maybe Tips? [taps] Yes — here I can see how others predicted. In My Tips as well.
29. **F:** You're actually one of the few people who's tried tapping on another player to find it — that's very useful feedback. You did find the right path though.

TASK 6 – Sending a Chat Message

30. **F:** Write a message to another person from the league.
31. **P:** New chat — 1-on-1. I'll write to Erik Bergheimer. [types] 'Hello, this is a test.'
[sends] Cool.

TASK 7 – Inviting a Friend

32. **F:** Last task: invite another friend to the league. Find the invite function and share the link.
33. **P:** I think I saw this earlier. Back to My League — top right, that's the invite icon.
[taps] And then Share. Done.
34. **F:** Perfect — that's all the scenarios. Now the questionnaire.

QUESTIONNAIRE RESPONSES

AttrakDiff (Semantic Differential)

[Observer note: P10 completes AttrakDiff independently]

INTUI (Sentence Completion)

[Observer note: P10 completes INTUI]

SUS (System Usability Scale, 1–5)

[Observer note: P10 completes SUS]

NPS

[Observer note: P10 completes NPS with written justification]

DEMOGRAPHIC INFORMATION

35. **F:** Age?
36. **P:** 27.
37. **F:** Gender identity?
38. **P:** Male — passionate fan.
39. **F:** Which prediction apps have you used?
40. **P:** Kicktipp.
41. **F:** How often?
42. **P:** Multiple times a week, or weekly — probably multiple times a week.
43. **F:** Smartphone OS?
44. **P:** iOS.

45. **F:** Technical competence self-rating?
46. **P:** Very high.
47. **F:** Have you used betting apps like Tipico or Bet365?

[Observer note: P10 responds — exact answer not clearly recorded]

OPEN INTERVIEW

48. **F:** Looking back at the whole session — what's your overall impression? What stood out most, positively and negatively?
49. **P:** The most positively surprising thing — what I found really special — was the design. Compared to Kicktipp it's genuinely, genuinely cool. Especially the use of club logos. And I also found it cool that you have to justify your prediction. But I could imagine it getting very tiring or overwhelming when you have to tip a full Bundesliga matchday with so many matches at once. Negatively — a few minor inconsistencies in the design, like starting with non-zero points. But that's obviously down to the prototype.
50. **F:** There were moments where the app deliberately slowed you down — asking you to justify a prediction or confirm a choice. Did those feel meaningful, or annoying? Did they change your interaction?
51. **P:** I actually found that genuinely cool — especially the idea of getting statistics about your own tipping patterns over time. What you've predicted in recent weeks or months. That I'd find really interesting. For a single match it works well. For many matches at once on a matchday, you'd have to think about how to handle that — I'm not sure what the right solution is there.
52. **F:** Did you feel in control the whole time, or did the app push you in directions you hadn't chosen?
53. **P:** No — there were no moments like that.
54. **F:** Did the match data and statistics make you feel more competent and better informed, or was it more like extra noise?
55. **P:** The statistics were really cool. Especially because I normally use other football apps — like Fotmob or SofaScore — to look up stats. So here I don't have to switch between apps. That's actually really nice.
56. **F:** Did the social features — the leaderboard and chat — feel particularly meaningful,

or more like functional extras?

57. **P:** Not strongly connected, to be honest. Maybe also because all the other profiles had the same default picture. But that's obviously a prototype thing. And because they weren't actually my friends. I think it would feel different in a real league with real friends and actual profile photos.
58. **F:** How does this app compare to Kicktipp or other tools you know?
59. **P:** Compared to Kicktipp — worlds apart. Kicktipp's design is really outdated. And it has so few statistics compared to this app. This feels modern and intuitive. With Kicktipp I feel like I'm using a website from 1998.
60. **F:** Is there anything else you'd like to share?
61. **P:** No, I think that covers everything.
62. **F:** Great — thank you so much for your time. All the best. Bye!
63. **P:** Ciao, ciao.

— *End of transcript* —

Note: Some passages (e.g. during questionnaire completion, betting app response, exact NPS score) were unclear in the original recording and have been summarised or marked accordingly.